BME205: Foundations of Biomedical Engineering

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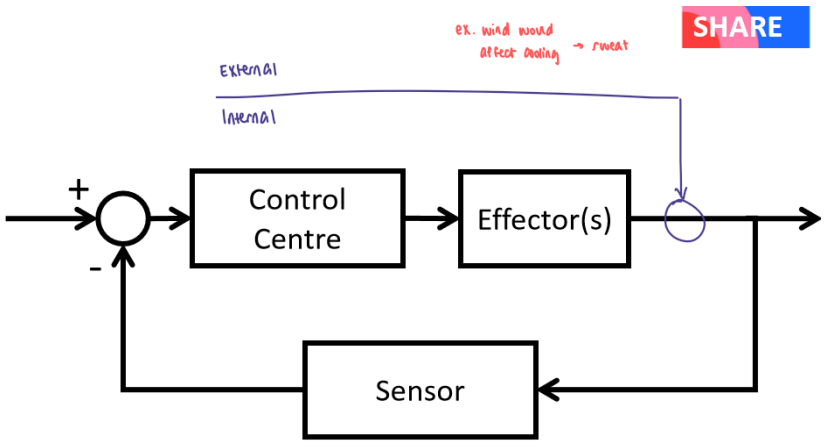
Homeostasis

Homeostasis: Maintenance of a near-constant condition in an organism.

- Important conditions that the body needs to maintain:
  - Temperature
  - Water
  - pH
  - O_2 vs CO_2
  - Ion concentration
  - Blood pressure
  - Energy (blood sugar, ATP)
  - Hormone

1. **Stimulus:** Something that causes detectable change in physical/chemical condition
   a. **External:** sweating, heart beat faster, hand pull back
   b. **Internal:** hungry, tired
2. **Receptors:** Detect stimuli (afferent)
   a. e.g. thermoreceptors (temperature), mechanoreceptors (pressure or distortion)
3. **Control center:** Nerve cells in central nervous system brain and spinal cord (hypothalamus)
   a. **Involuntary:** reflexes, sneeze, blood pressure
   b. **Voluntary:** jumping, lifting, chewing
4. **Effectors:** Perform a function that reacts to stimulus (efferent)
   a. e.g. muscle cells, secretion cells, sweat glands
5. **Response:** What the effector does
   a. e.g. cooling effect (sweating), increased heat loss

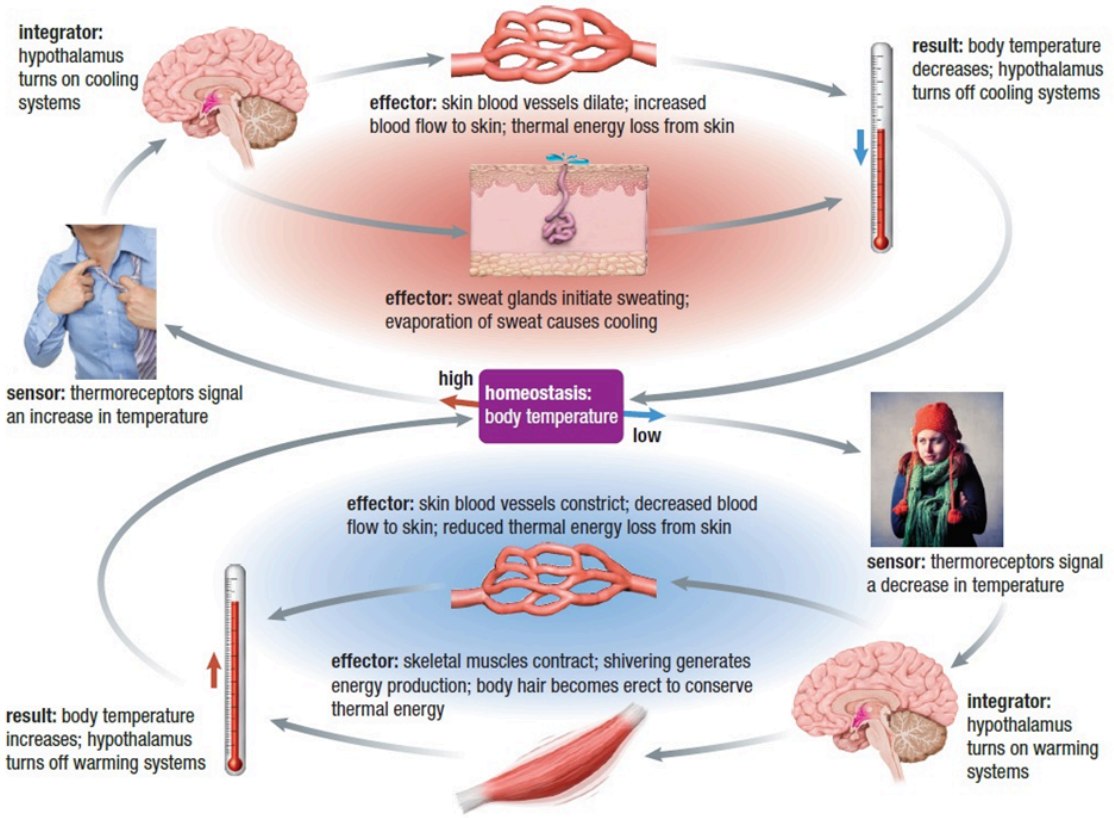


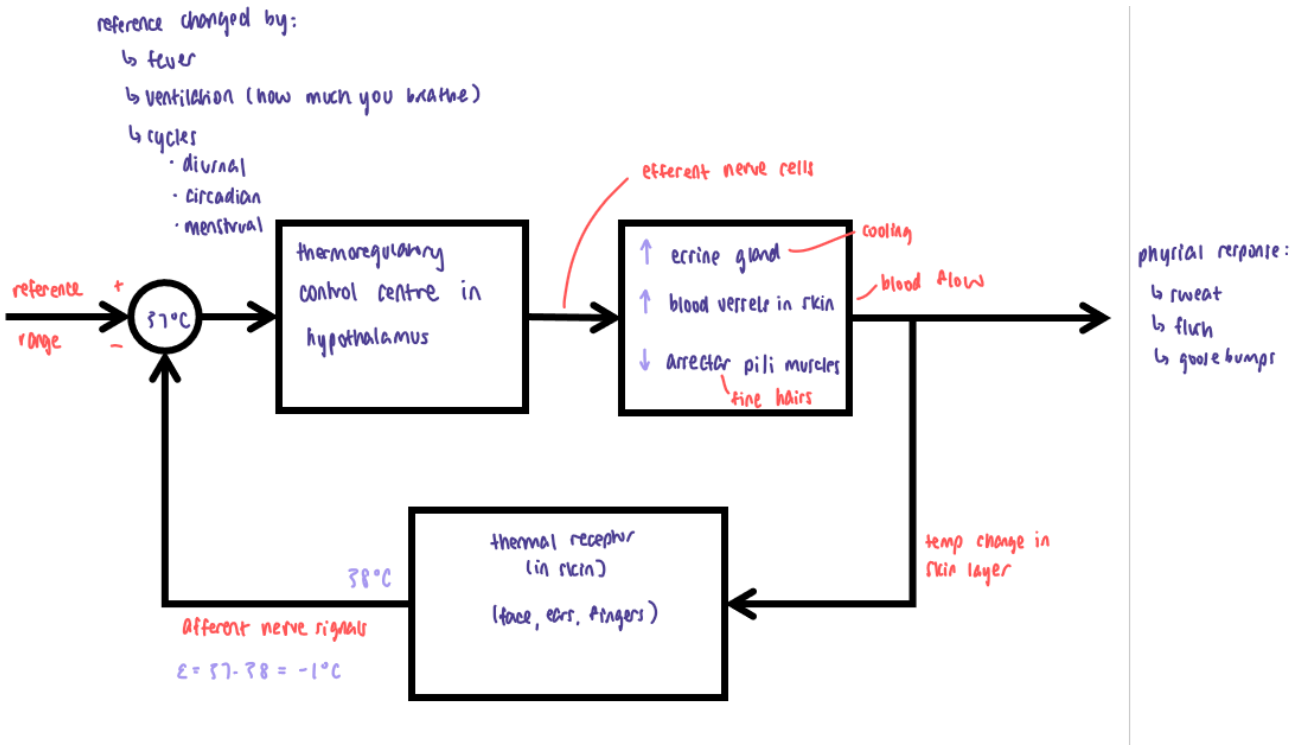
Positive Feedback Loop: A change in a direction causes additional change in the **same direction**

- Promotes the change
- **Amplifies** a small response

Negative Feedback Loop: A change in a direction causes additional change in the **opposite direction**

- Prevents the change
- Reduces an excessive response





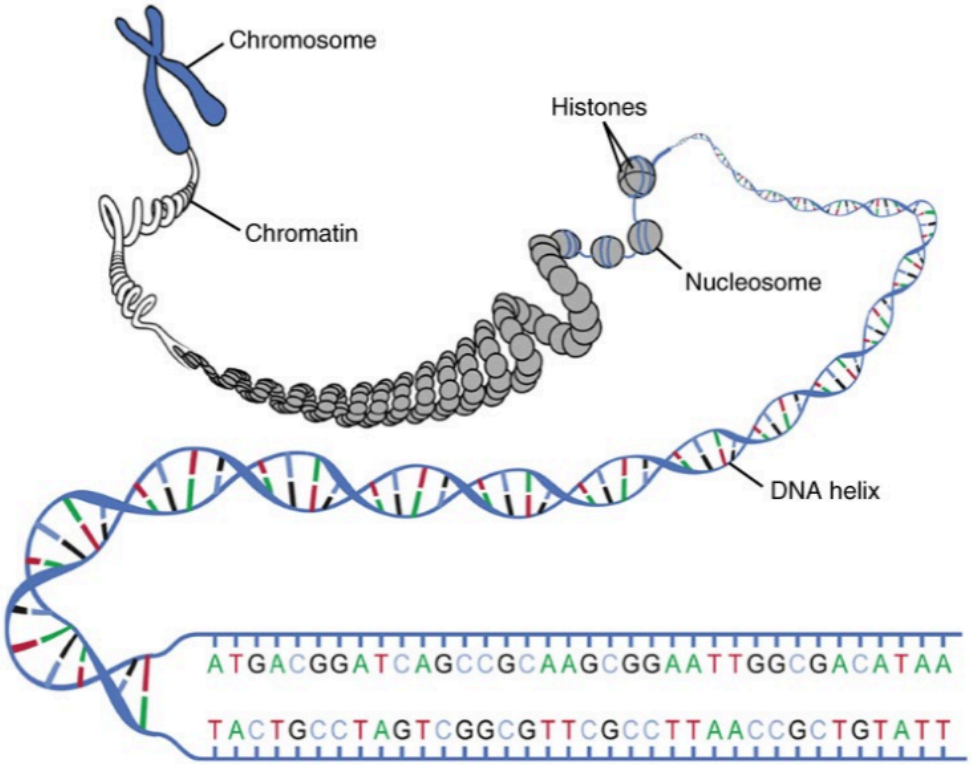
Cold Stress Response

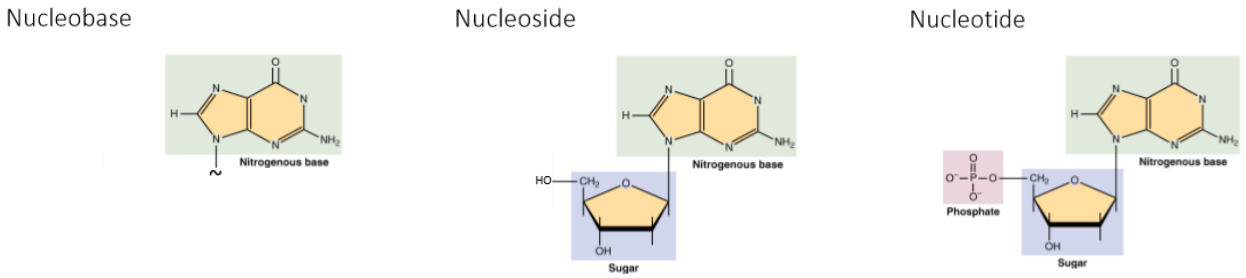
How does the body protect itself against excessive cold?

- Thermoreceptors detect drop in body temp → send signal to hypothalamus.

- Hypothalamus sends signal to effectors to initiate response:
    - **Blood vessels: vasoconstriction** to reduce blood flow to the surface (↓ heat loss)
    - Smooth muscles surrounding hair follicles: contraction causes goosebumps, hair stands up to “trap” warm air.

DNA

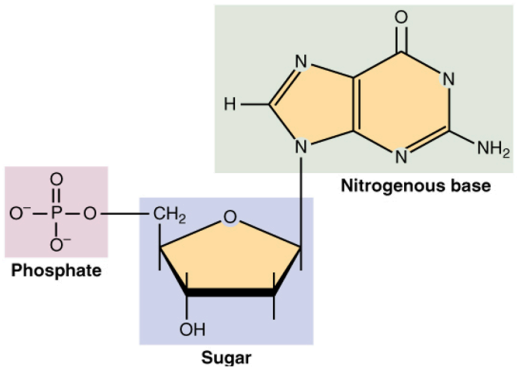




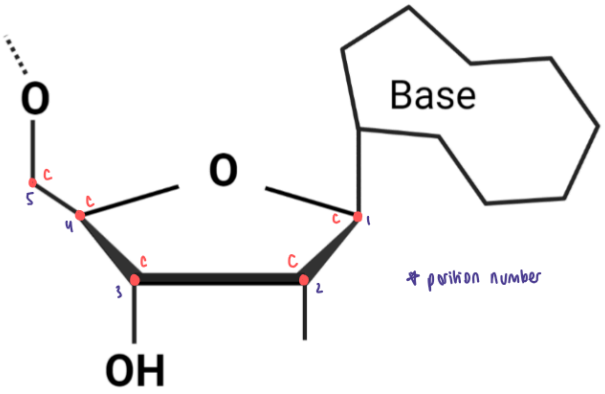
Nucleotide

- Composed of nucleotide monomers
    - Phosphate group
    - 5-Carbon sugar (ribose, deoxyribose)
    - Nitrogenous base (A, T, G, C)

e.g. DNA, RNA

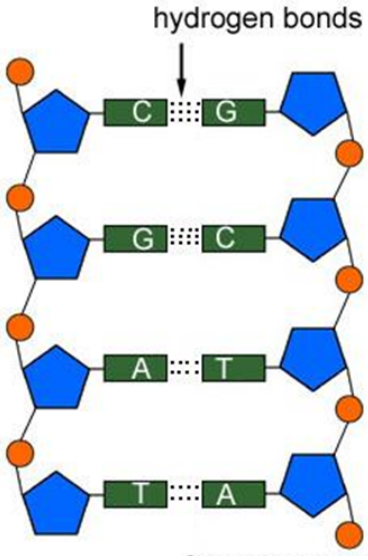


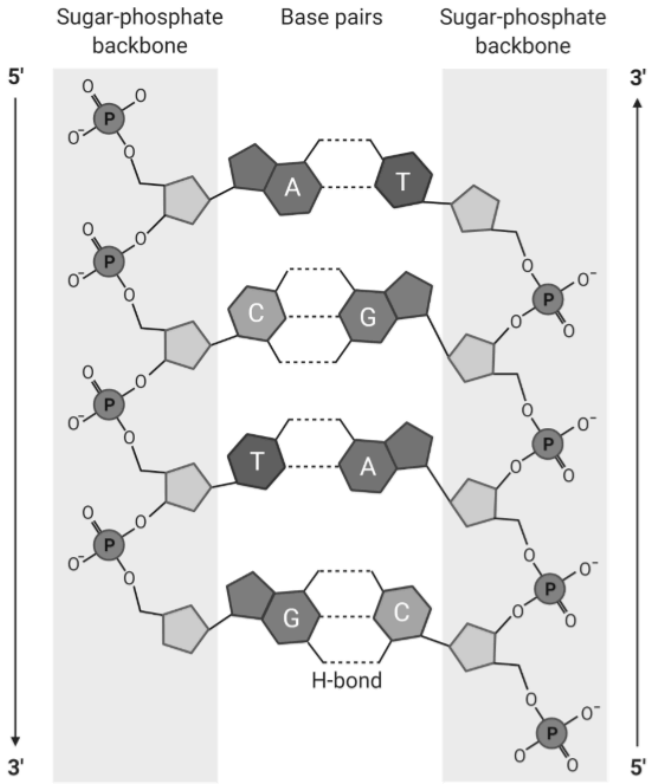
Numbering Convention of the Sugar Molecule (Deoxyribose is shown)



DNA

- **Two nucleic acid strands** twisted about each other to form a **'double helix'**
  - Backbone of alternating sugars and phosphates with the base pairs projecting away from the backbone
  - Hydrogen bonds between the bases hold the strands together





Nucleobases

Purines:

A Adenine

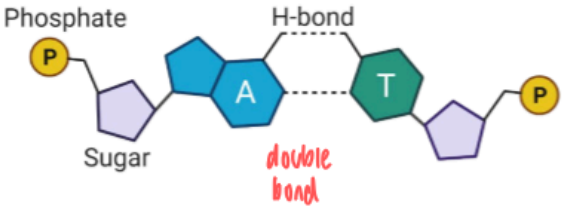
G Guanine

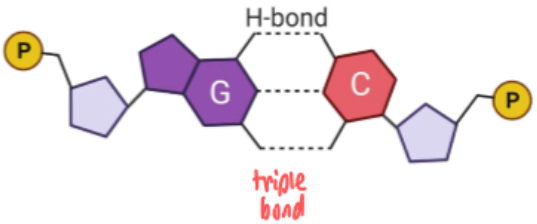
Pyrimidines:

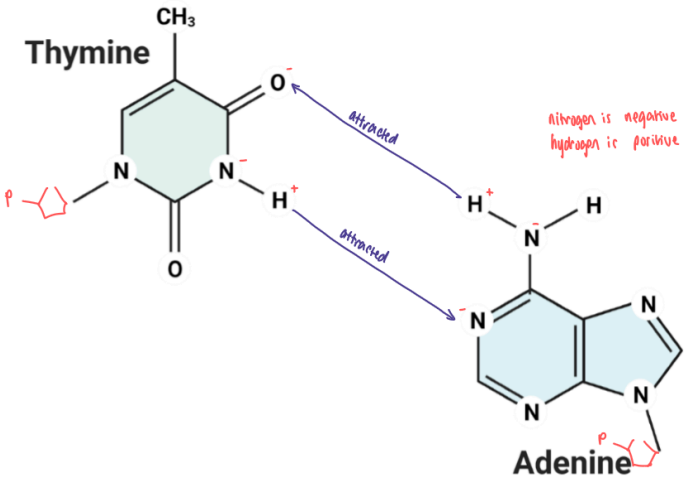
T Thymine

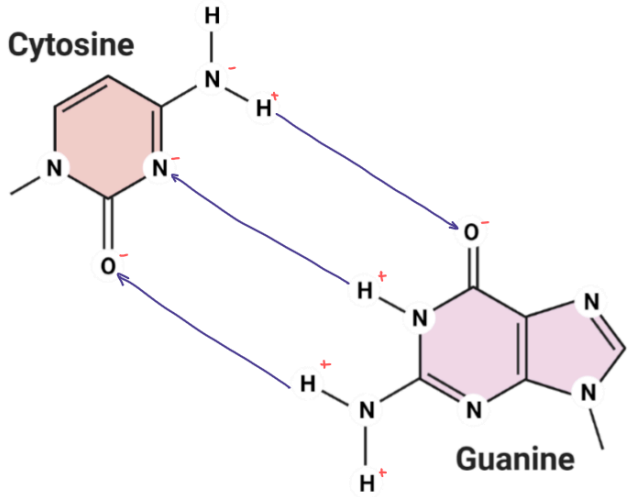
C Cytosine

Complementary nucleobase pairing

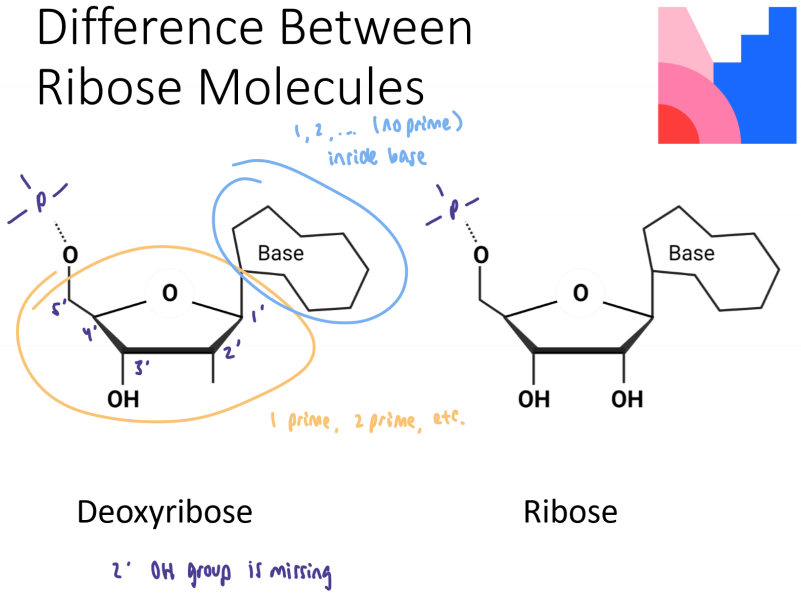








Ribose molecules



Deoxyribose

Ribose

2' OH group is missing

Transcription

The DNA strand being copied (complement) is known as the **template** strand (3' to 5').

The other DNA strand (not being copied, same) is the **coding** strand (5' to 3').

Initiation:

1. DNA transcription factor binds onto TATA box
2. RNA polymerase binds to the transcription factor complex
3. Binding unwinds the DNA for transcription to start.
4. Transcription starts at TAC

Elongation:

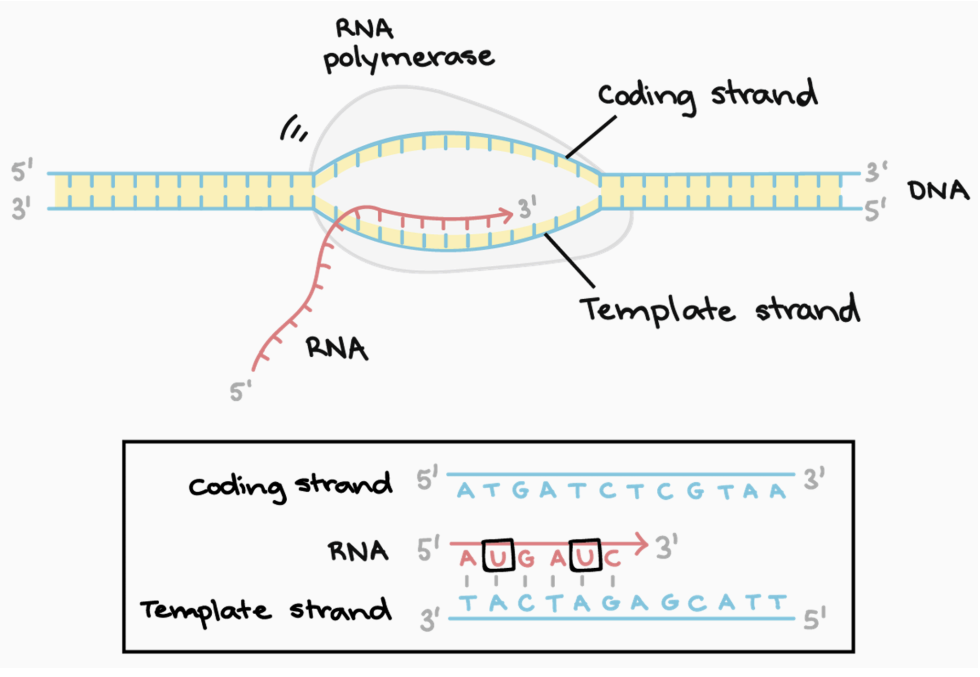
1. RNA polymerase unzips the helix
2. RNA polymerase starts building the single-stranded mRNA.

Termination:

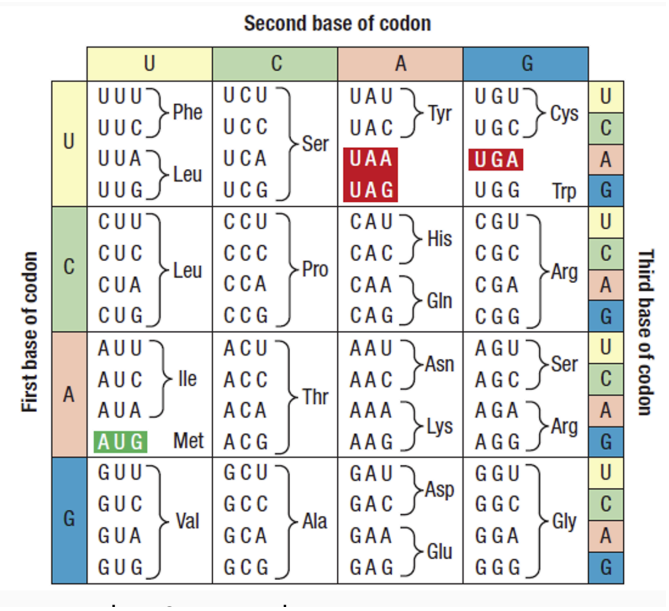
1. A terminator sequence tells RNA polymerase that it's reached the end of a gene.
    - Newly-made primary transcript dissociates from template strand
    - RNA polymerase is free to bind another promoter

pre-MRNA to RNA:

1. Adenylation
    - Poly-A tail: **AAUAAA** signals addition of 50-250 A's at 3' end to protect the mRNA from digestion by cytosolic enzymes
2. Splicing: Removal of Introns
    - In eukaryotes, DNA includes both coding (exons) & noncoding (introns) regions.
        - If introns are translated, the protein won't fold properly = nonfunctional!
        - Introns are removed by **spliceosomes** (enzyme-protein complex).



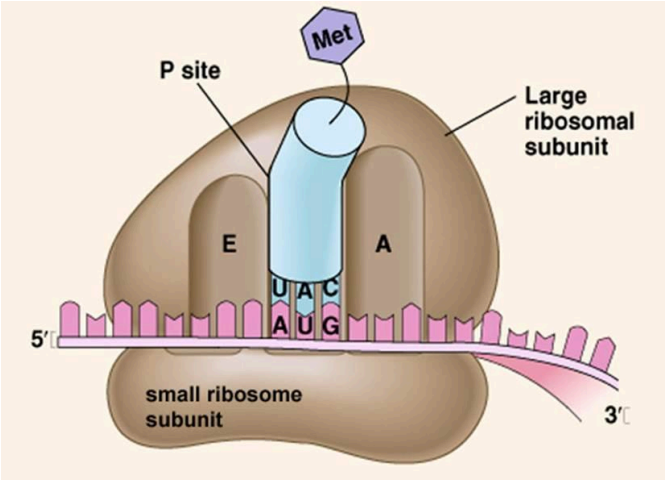
Translation



rRNA

Ribosomal RNA (rRNA), together with proteins, forms the ribosome.

- Each ribosome is made up of **2 subunits**:
  - **Large subunit (60S)**: consists of an A, P, and E site for a polypeptide - binds to codons
  - **Small subunit (40S)** - holds in position
- The 2 subunits clamp the mRNA between them.



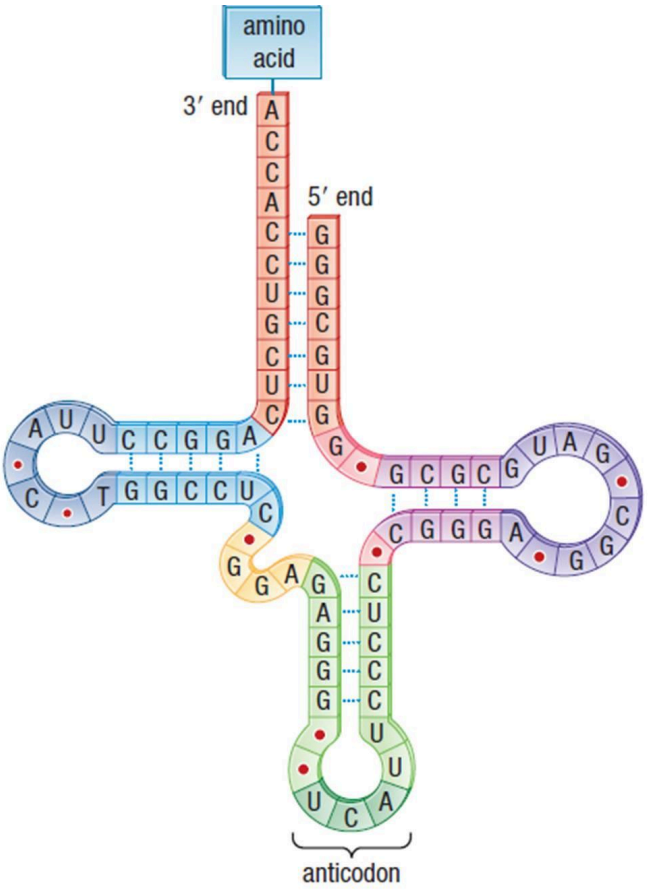
tRNA

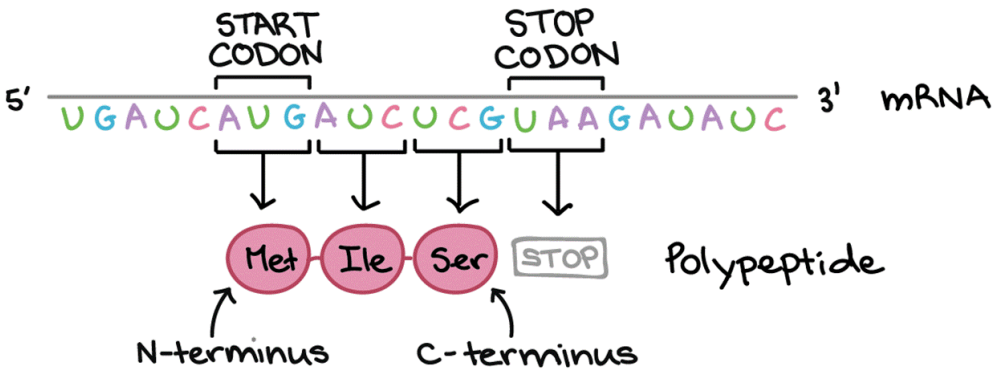
Transfer RNA (tRNA) carries amino acids to the ribosome.

- Small strand that folds in on itself to form a clover-leaf-like shape.
- Delivers the correct amino acid to the polypeptide-binding site of a ribosome.
- Every tRNA carries only one specific amino acid determined by its **anticodon**.
  - A tRNA carrying an amino acid is known as an **aminoacyl-tRNA**.

The **anticodon** of a tRNA will be complementary to an mRNA codon.

- Note: doesn’t have to be exactly complementary!
    - Wobble hypothesis
- Anticodon binds tRNA with the mRNA.





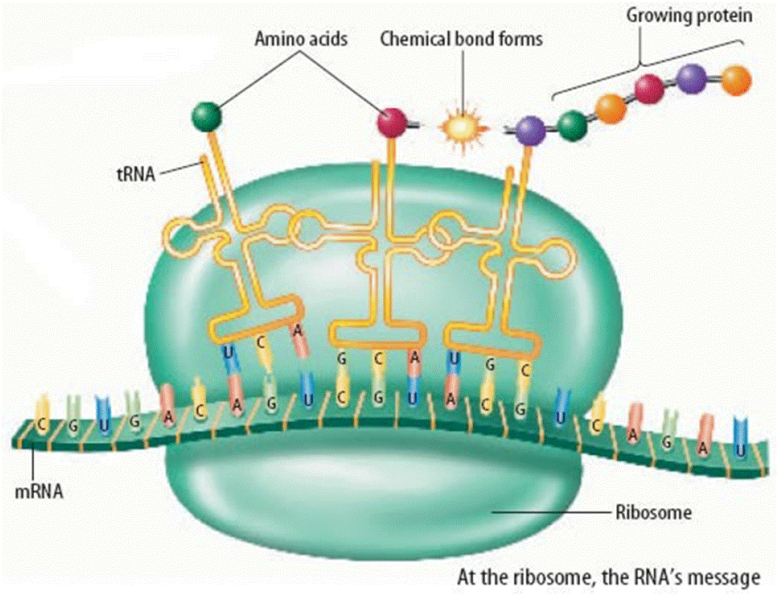
Start codon is AUG - corresponding anticodon is UAC

- If there was a mutation and something was added in middle of start codon, then the protein wouldn't be made.
- If there was a mutation and another G was added next to the G, then the process would start but it would be in wrong groupings (e.g. AU**G**GAUCUCGUAA -> AUG, GAU, CUC, GUA, ... -> wrong protein!)

Protein Synthesis Stage #2: Translation

Translation can be divided into 3 steps:

1. Initiation
2. Elongation
3. Termination

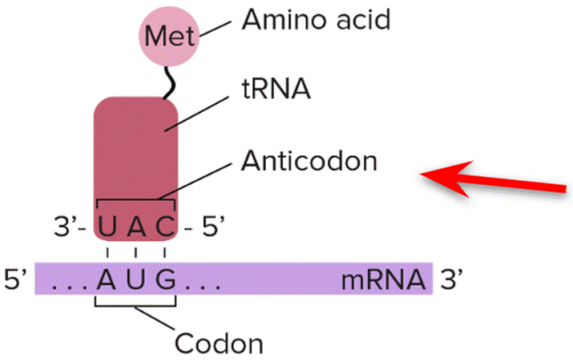


At the ribosome, the RNA's message is translated into a specific protein.

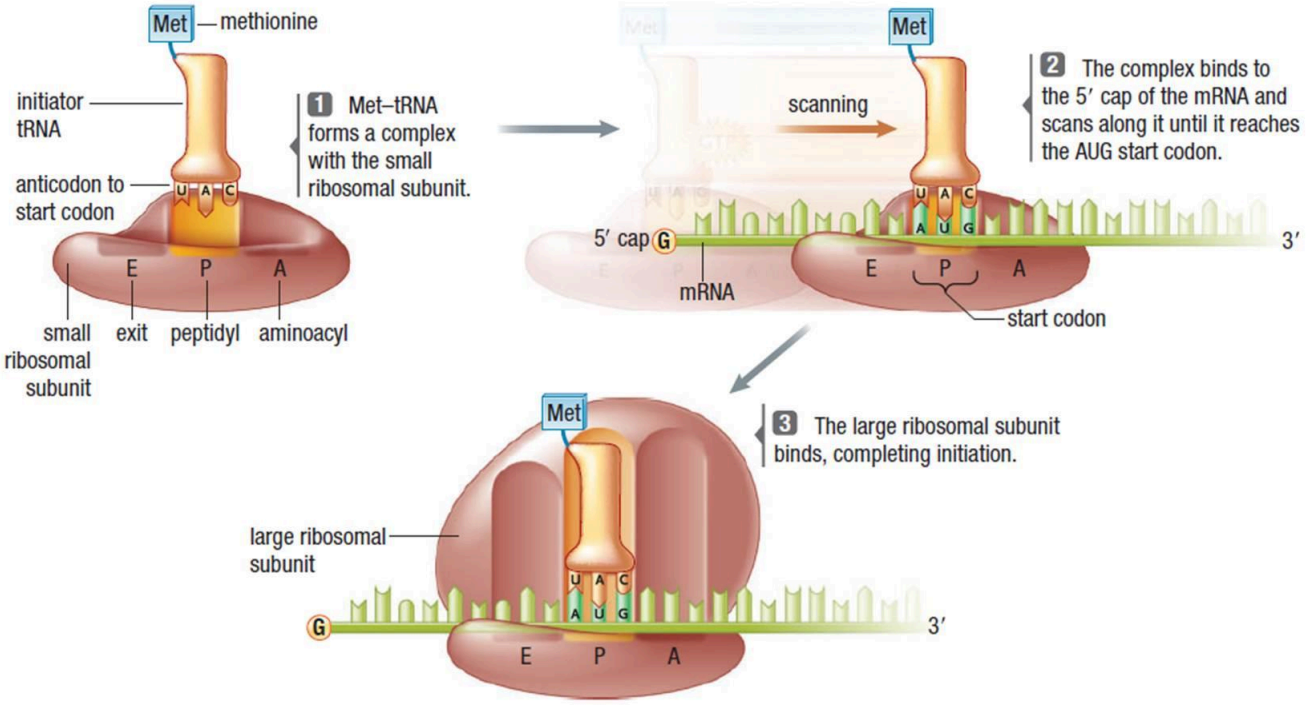
Step 1: Initiation

Translation can begin when mRNA enters the cytoplasm.

- **Initiator met-tRNA** binds to the small ribosomal subunit.
- Complex binds mRNA's 5' cap and scans until it finds the start codon.
- Once the start codon is found, the large ribosomal subunit binds as well.

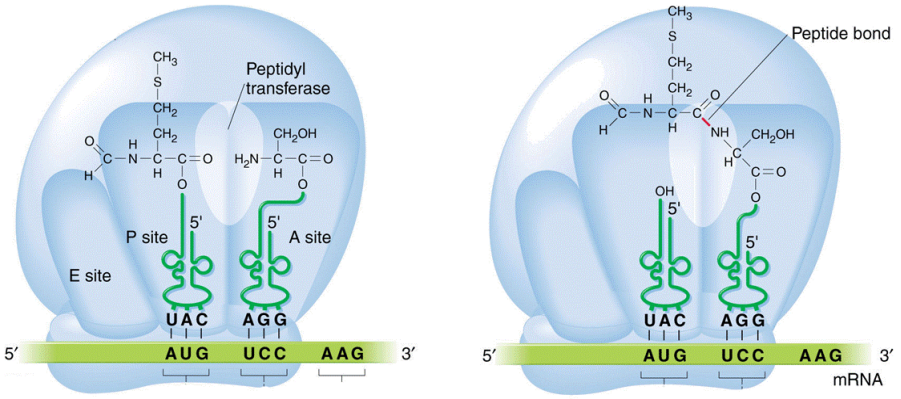


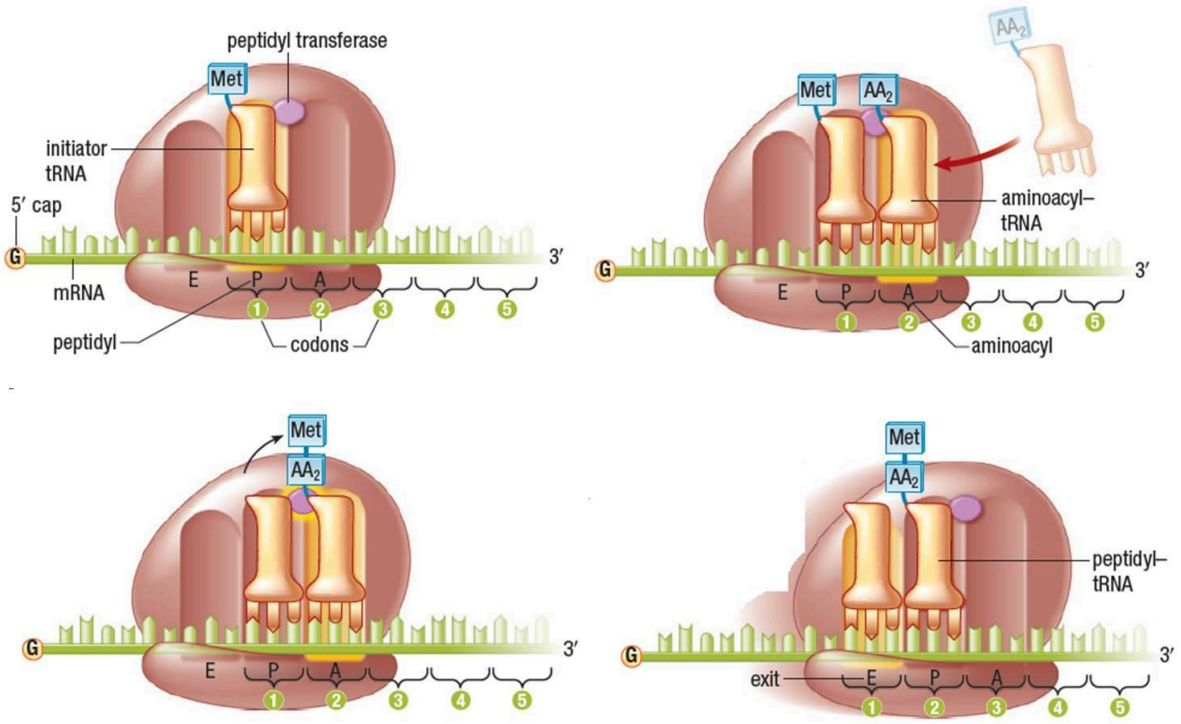
BINDING ESTABLISHES THE READING FRAME (HOW MRNA IS READ IN 3)



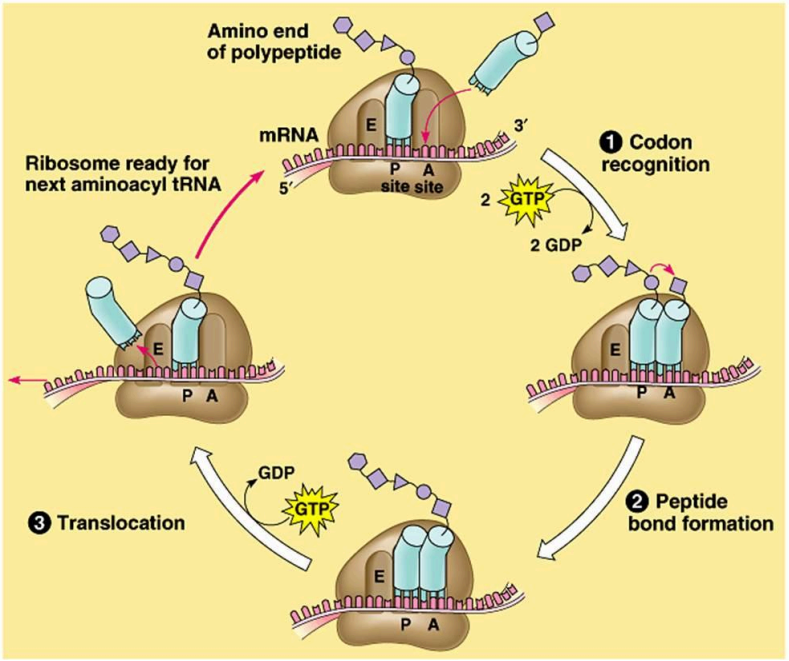
Step 2: Elongation

- An incoming aminoacyl-tRNA comes to the **A (aminoacyl) site**.
    - The first amino acid (methionine) is cleaved from the initiator tRNA.
    - Met is transferred onto the incoming amino acid.
    - Peptide bond is formed between the two amino acids
- Ribosome moves along the mRNA to the next codon.
    - The previous two tRNAs now occupy the P site and **E (exit) site**.
    - The empty tRNA at the E site is released.





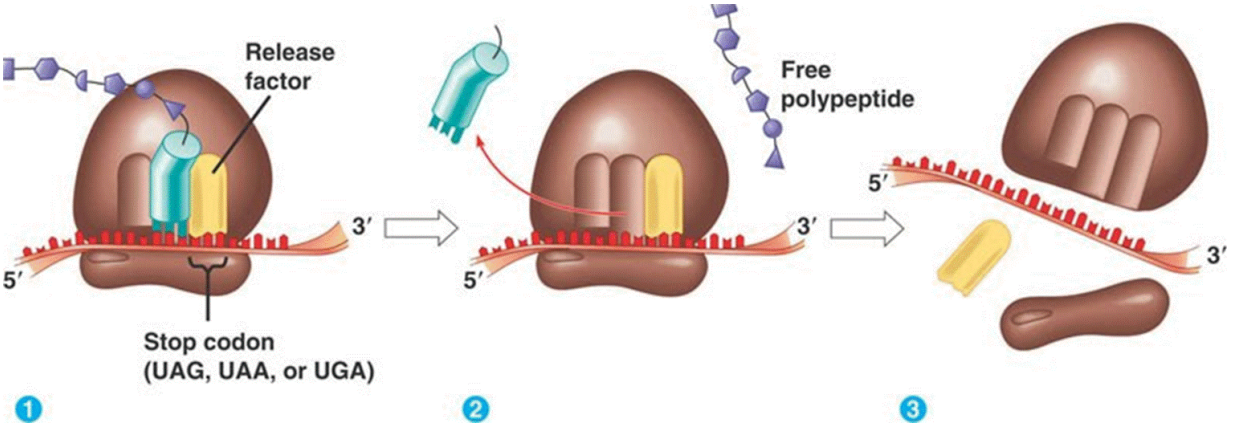
Energy for this step is given by GTP hydrolysis



Step 3: Termination

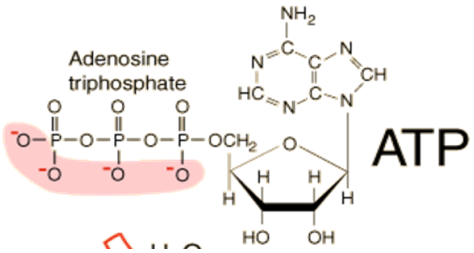
Termination occurs when the A site arrives at a stop codon…

- Here, a protein **release factor** binds to the A site instead of a tRNA.
- Polypeptide chain is released and ribosomal subunits separate.

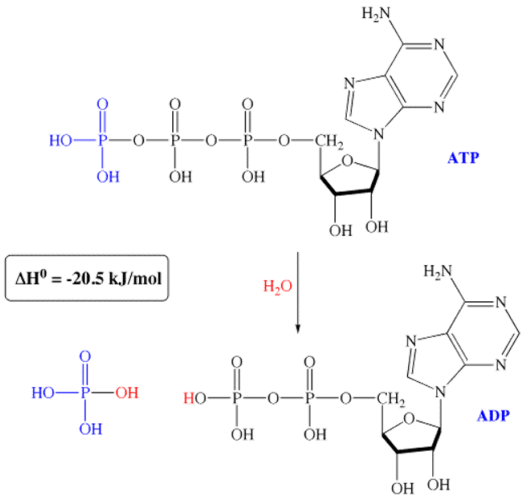

Release
factor
3′
5′
Stop codon
(UAG, UAA, or UGA)
1
Free
polypeptide
3′
5′
2
5′
3′
3

ATP Hydrolysis

- Hydrolysis of the end phosphate group from an ATP molecule releases energy
    - Concentration of negative charge in tail makes ATP very unstable

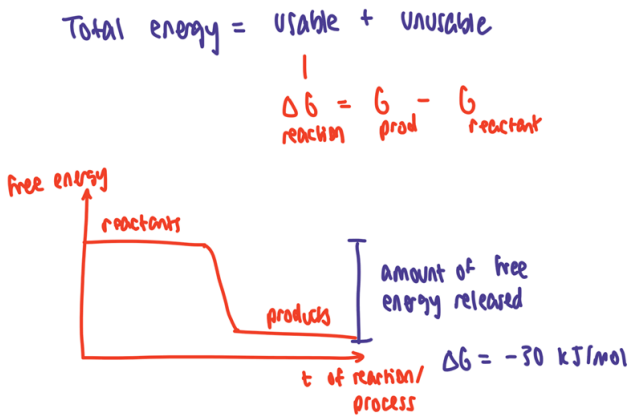


- When water is added, it breaks the relatively weak bond between the phosphate groups
    - Results in a (more stable) molecule of ADP and phosphate group
    - Releases energy

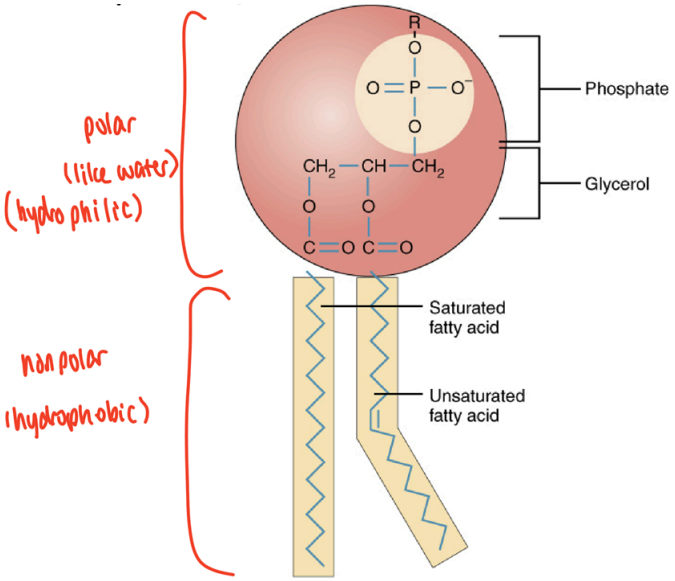


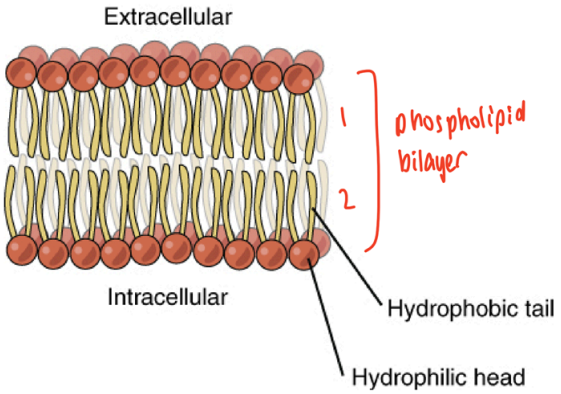
Free energy

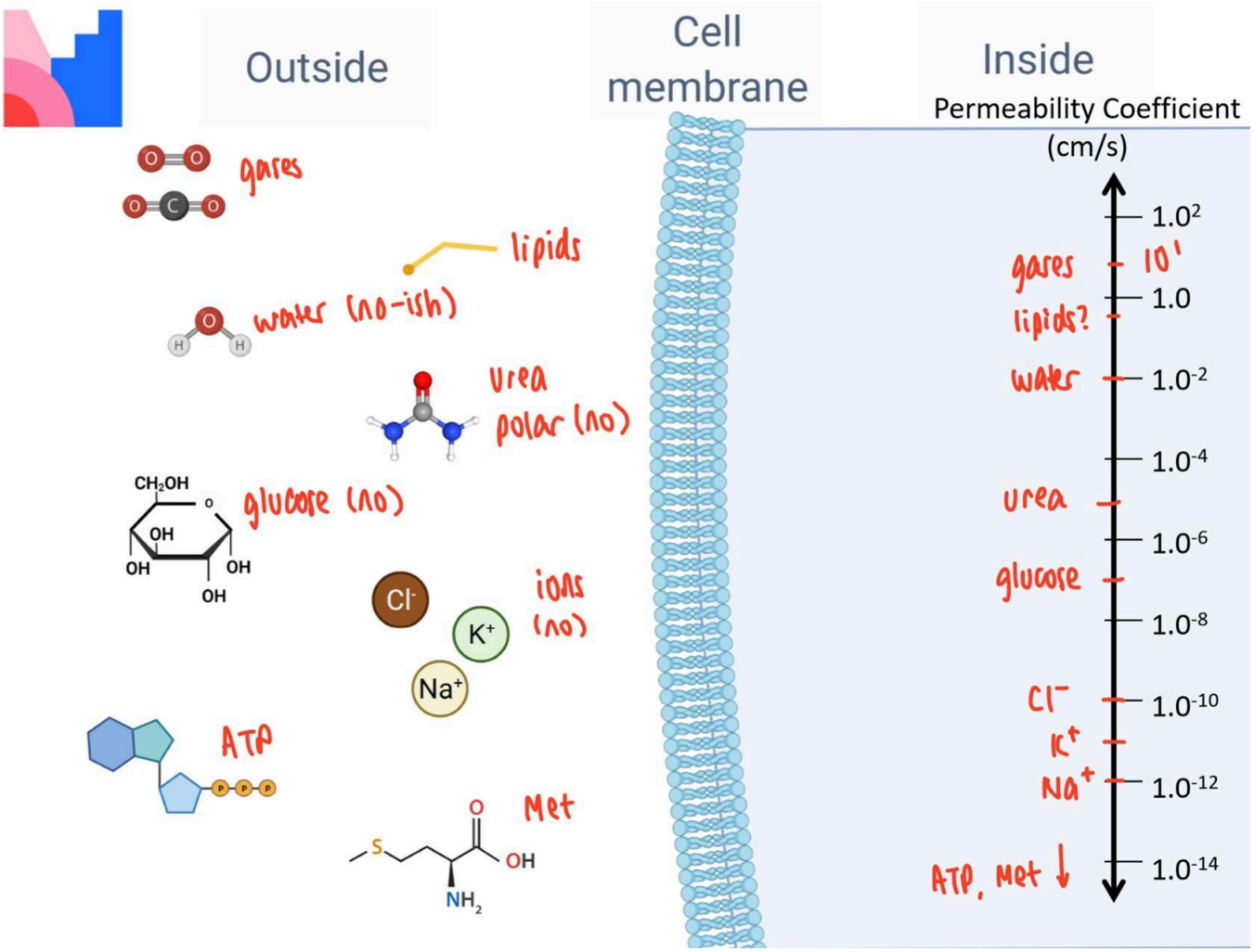
Total energy = usable + unusable



Cell Membrane







Types of Transport

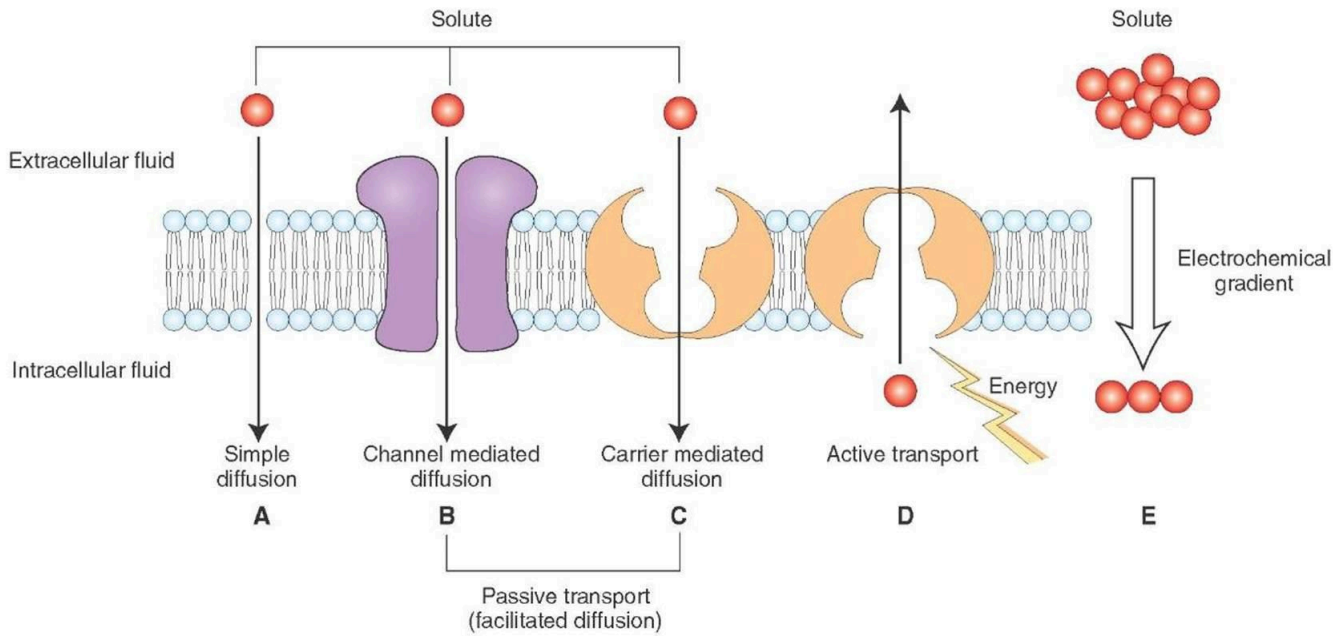
Passive transport

- High to low concentration, no ATP required

Active transport

- Movement of substances across a cell membrane against the concentration gradient using **pumps**
- Moving against concentration gradient = **requires energy (ATP)**

1. **Passive transport**
    - Simple diffusion
    - Facilitated diffusion
        - Channel proteins
        - Carrier proteins
2. **Active transport**
    - Pumps (Primary & secondary)
3. **Endocytosis, exocytosis**
    - Membrane-assisted transport



Passive transport

Simple Diffusion

- Molecules and ions are always moving around
- Random motion causes a net movement from regions of higher to lower concentration
- Dynamic equilibrium - net movement continues until there is no concentration gradient

Diffusion Across a Membrane

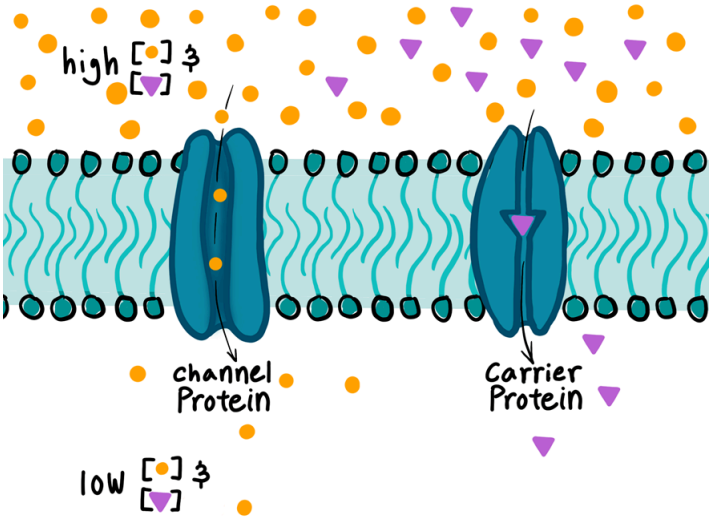
Affected by:

- Relative concentrations both inside and outside the cell
- How readily a molecule can cross the membrane

FACTOR	RATE OF DIFFUSION
Molecule Size	Rate of diffusion ⬇ as size increases.
Molecule Polarity	Rate of diffusion ⬇ as polarity increases.
Particle Charge	Charged particles **cannot diffuse** through the membrane.
Temperature	Rate of diffusion ⬆ as temperature increases.
Pressure	Rate of diffusion ⬆ as pressure increases.

Facilitated Diffusion

- Molecules must enter/exit using specific proteins embedded into the plasma membrane
- Still diffusion = molecules move with the concentration gradient
- 2 types of membrane proteins:
    - Channel proteins
    - Carrier proteins

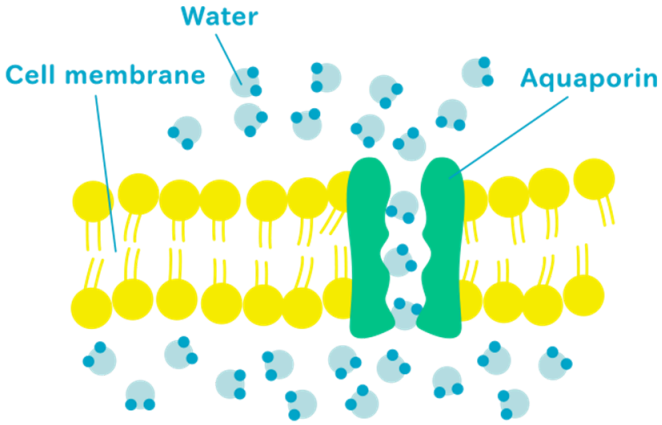


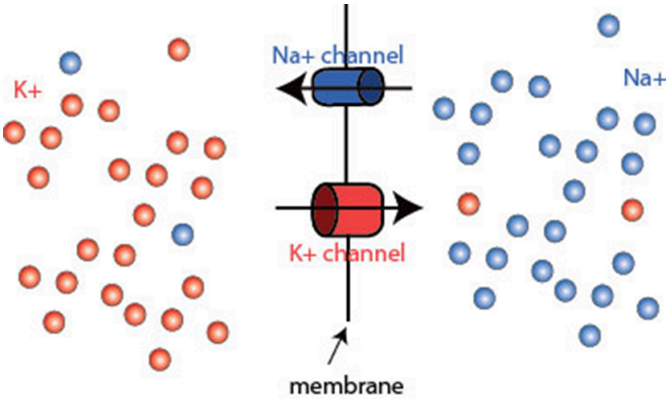
Channel Proteins

- Form tubular channels across the membrane
- When open, they are open on both sides
- Shape & size of its hole determine which particle it allows to pass
- Generally allow passage of water-soluble molecules (e.g. ions)

1. **Non-gated channel protein**

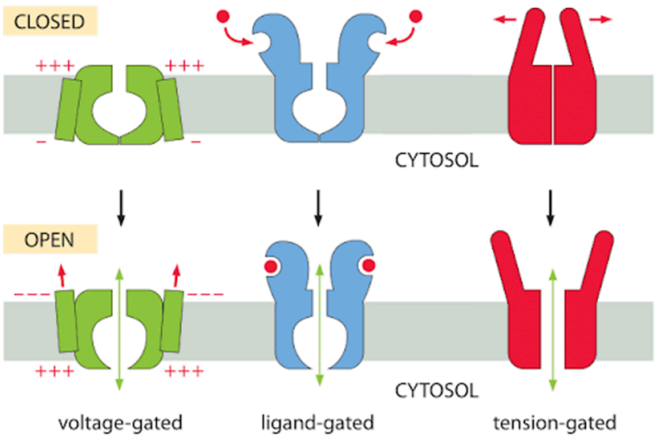
- Always open, selective, and bi-directional
    - *e.g.* ***leak channels:*** *allows diffusion of ions*
    - *e.g. aquaporins: allows diffusion of water*





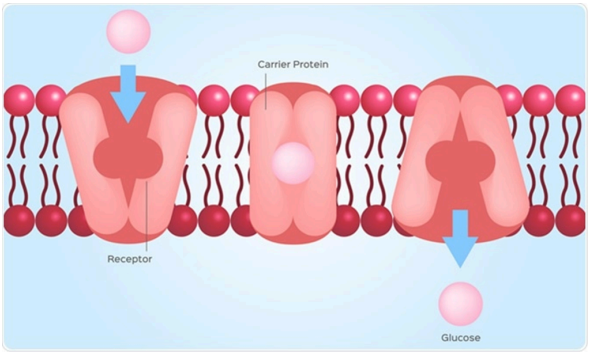
2. **Gated (ion) channel protein**

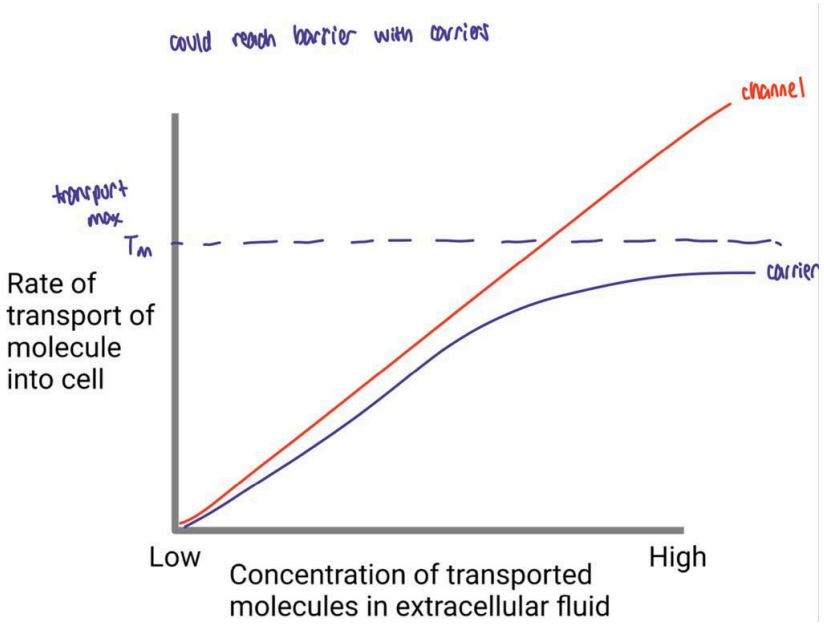
- Open & close in response to some stimulus
    - Voltage-gated
    - Ligand-gated
    - Tension-gated



Carrier Proteins

- Change shape to allow for the passage of specific substances
- Bind to the molecules and carry them to the other side
- Lower diffusion rates compared to channel proteins: bind a few at a time
- Can transport ions, small polar molecules, and also larger molecules



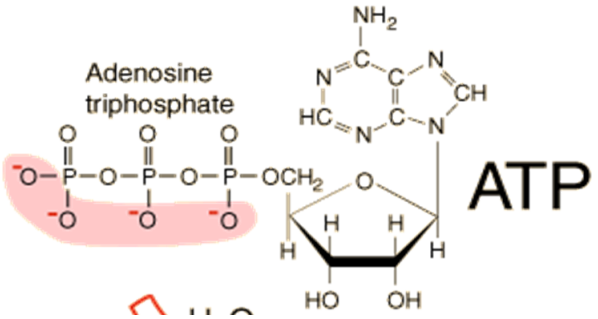


Active Transport (Primary & Secondary)

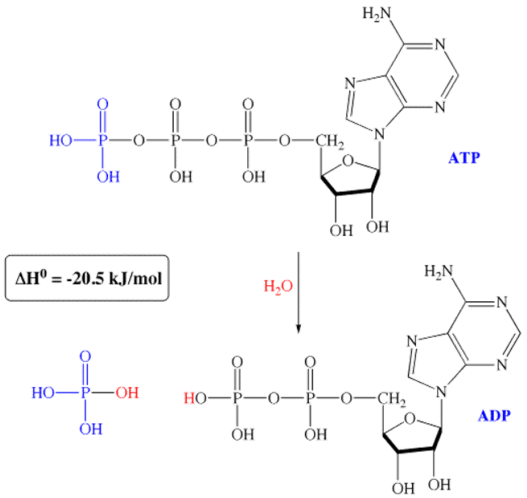
- **Primary** - transports Na^+, Ca^{2+}, K^+, and H^+
- **Secondary** - transports amino acids, monosaccharides, neurotransmitters

Pumps

- Hydrolysis of the end phosphate group from an ATP molecule releases energy
    - Concentration of negative charge in tail makes ATP very unstable

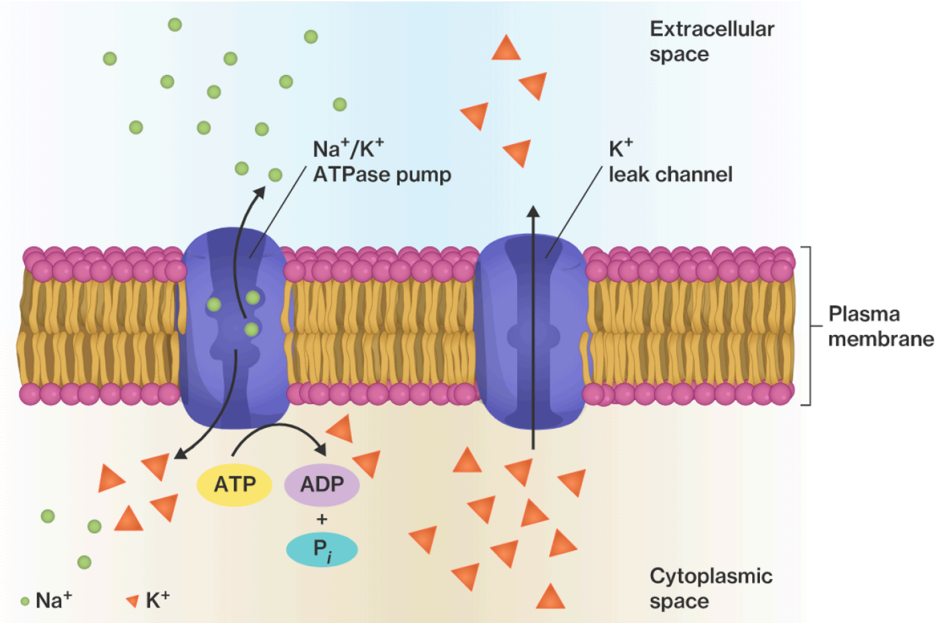


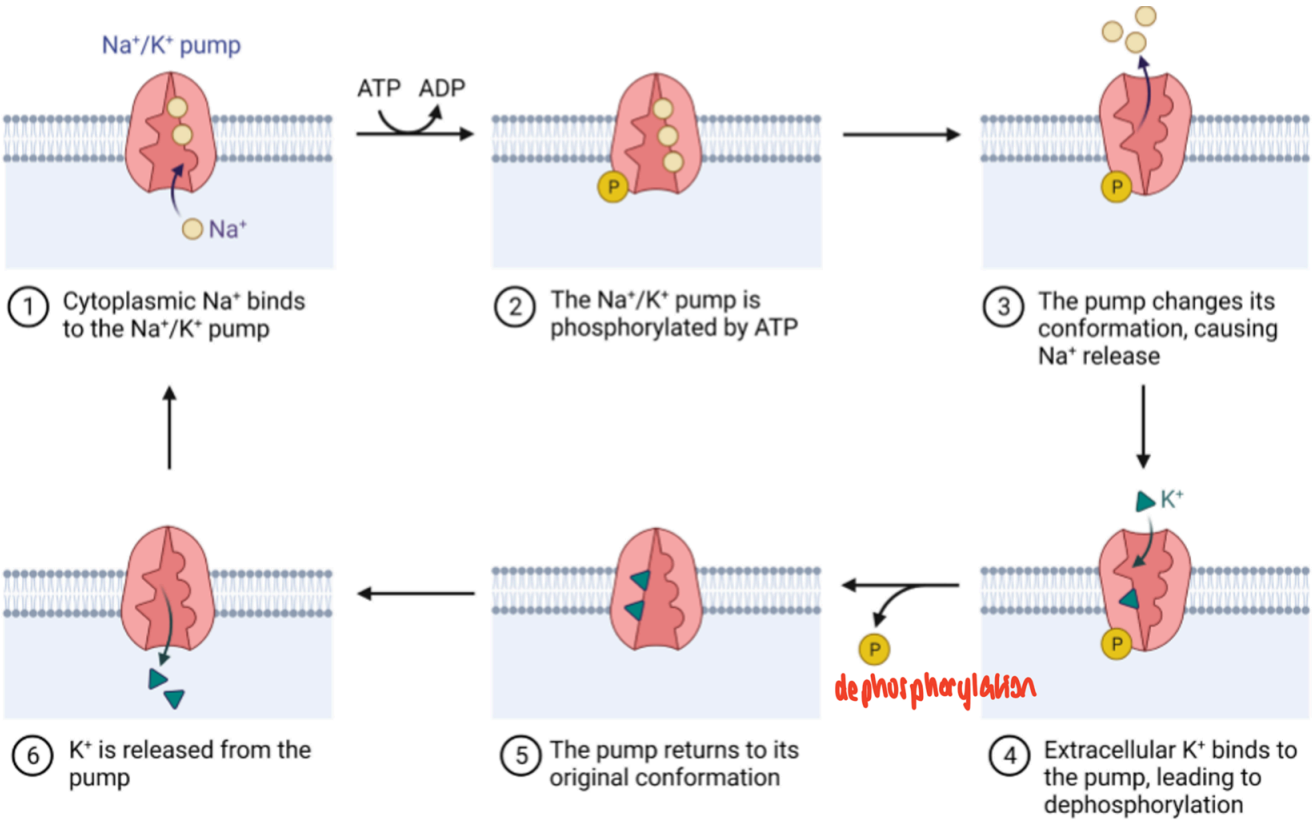
- When water is added, it breaks the relatively weak bond between the phosphate groups
    - Results in a (more stable) molecule of ADP and phosphate group



1. Primary

- ATP is **directly** used to transport substances
    - e.g. ion pumps use ATP to 'pump' ions against the conc. gradient
    - e.g. sodium-potassium pump: 3 Na+ out, 2 K+ in



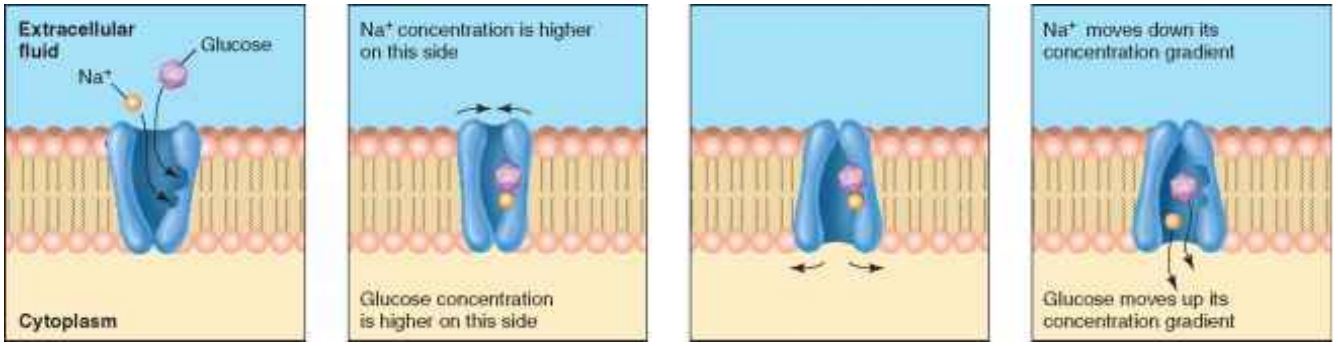


2. Secondary

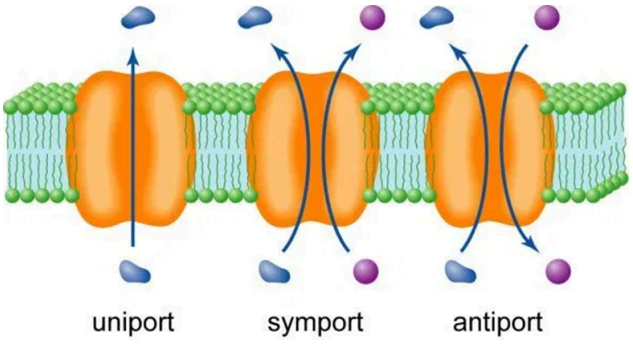
- ATP is **indirectly** used to transport substances
    - As an ion pump functions, a difference in ion concentration and thus charge (electric potential) builds up across the membrane

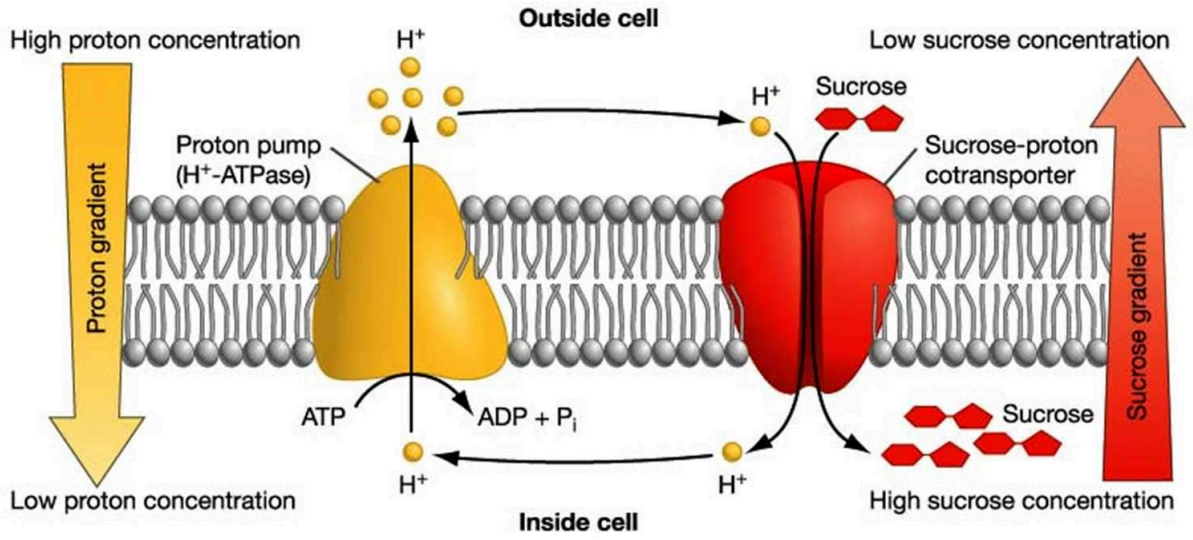
- Creates an **electrochemical gradient** used by secondary pumps to transport other molecules/ions against concentration gradient

- **Driving molecule** - the ion releasing energy as it moves down its concentration gradient (e.g. Na+)
- **Driven molecule** - the ion/molecule being pulled against its concentration gradient (e.g. glucose)



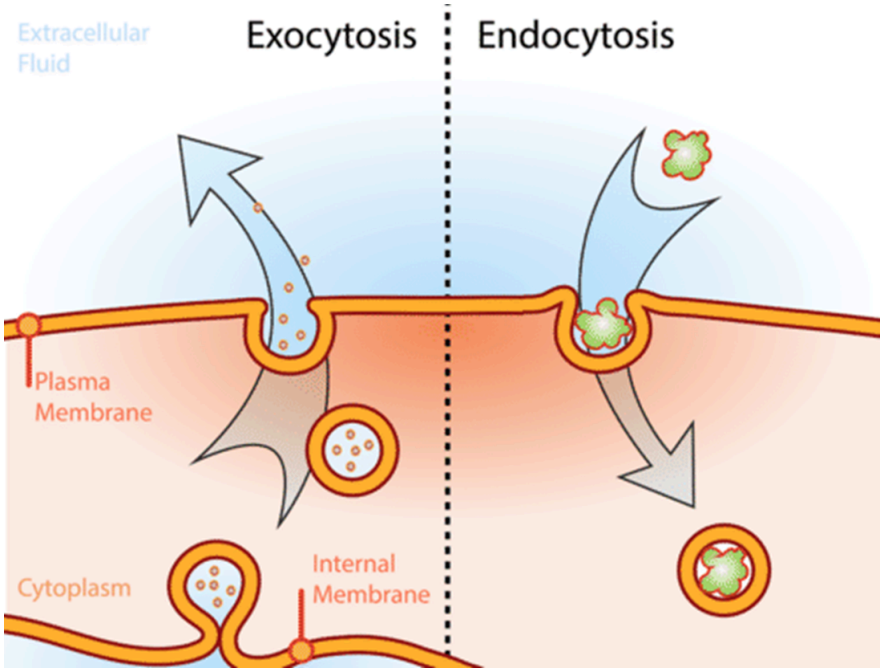
- **Coupling (cotransport) -** linking of energetically unfavourable reactions to energetically favourable reactions (which release energy) in order to 'power' them
  - **Symport** - both driving & driven molecule move in **same direction**
  - **Antiport** - driving & driven molecule move in **opposite directions**





Vesicular Transport

- For larger molecules that can't pass via diffusion or membrane proteins
- **Vesicles** are formed to surround and to move larger molecules across the cell membrane
- 2 types:
    - Endocytosis
    - Exocytosis

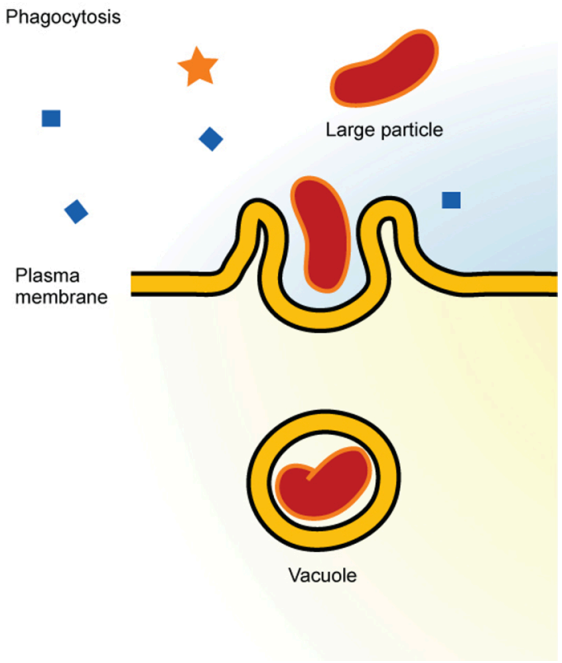


Endocytosis

- Cell engulfs extracellular material by folding the cell membrane around it and pinching off the membrane to form a vesicle

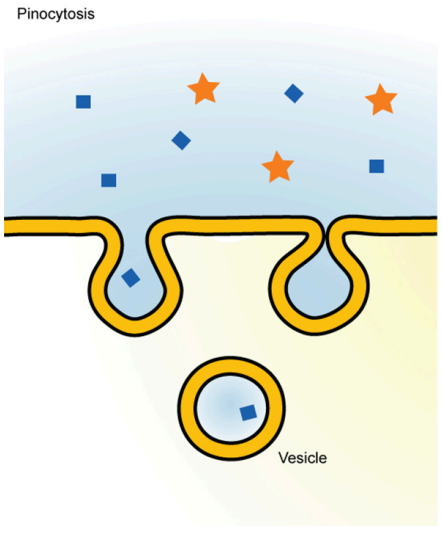
1. Phagocytosis ("cell eating")

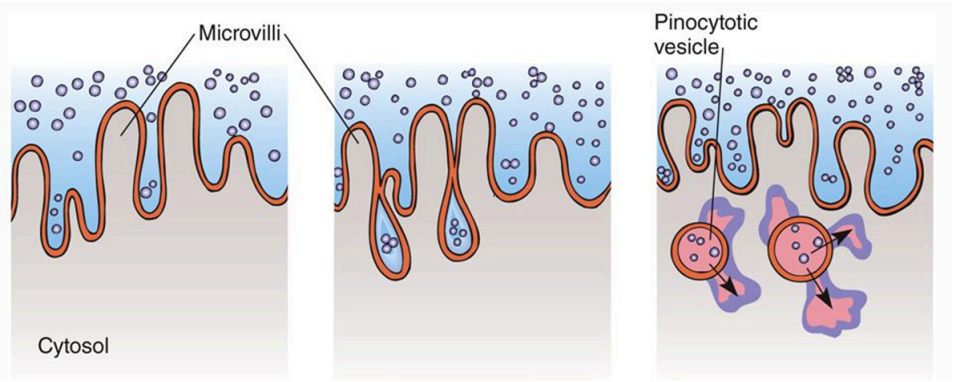
- Cell ingests other **cells or particles**
    - *e.g. how amoeba feed themselves*
    - *e.g. how WBCs kill bacteria, viruses, etc.*



2. Pinocytosis ("cell drinking")

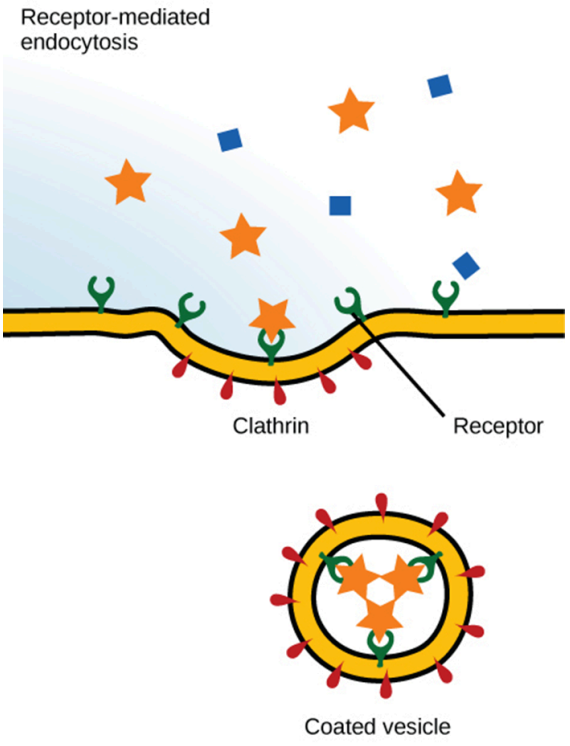
- Cell takes in some extracellular fluid
    - Takes in its dissolved solutes too
    - e.g. how microvill*i absorb nutrients*

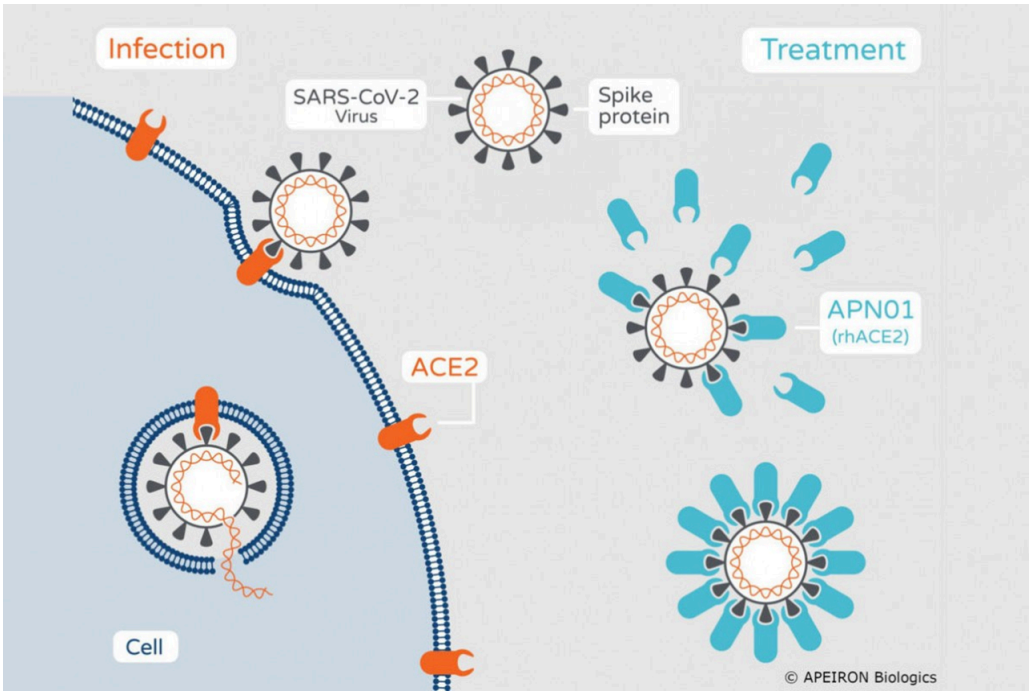




3. Receptor-mediated

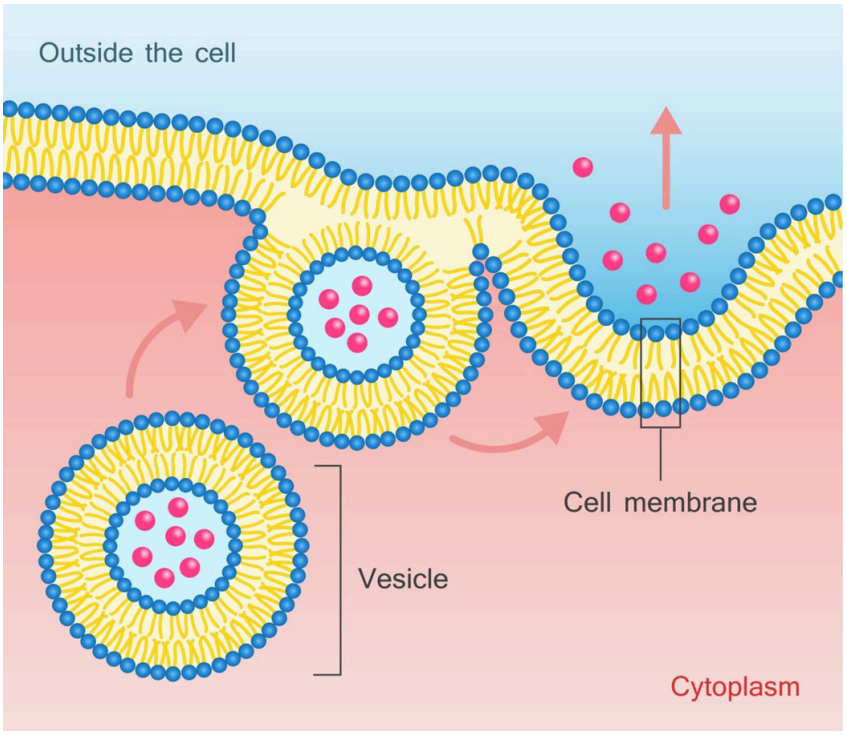
- Uses **receptor proteins** on cell membrane that bind with specific molecules outside of the cell
    - **Clathrin** coats this pit of receptors on the cytosol side of the membrane
    - Coated pit then pinches off into a vesicle
    - Can take up hormones, proteins, viruses, etc.
    - *e.g. COVID-19 is receptor mediated*





Exocytosis

- Vesicle fuses with the cell membrane to empty its contents into the extracellular environment
- Can carry out wastes, mucus, proteins, carbohydrates, neurotransmitters, etc.
- It provides a mechanism for secreting **large polar molecules**, such as protein hormones and enzymes that are unable to cross the plasma membrane.
- Vesicular contents are highly specific and are released only on receipt of appropriate signals.



Summary

	Simple diffusion	Facilitated diffusion	Primary active transport	Secondary active transport
Types of molecules transported	Nonpolar, any size (O2, CO2, fatty acids)	Channel: specific, small ions (Na+, K+, Ca2+, Cl-) Carrier: specific polar molecules (e.g. glucose)	Specific ions or polar molecules (Na+, K+, amino acids)	Specific polar molecules and ions for cotransport (e.g. glucose, amino acids, some ions)
Additional energy required	No	No	Yes	Yes
Against concentration/electrochemical gradient?	No (moves down conc. gradient)	No (moves down electrochemical gradient)	Yes (low -> high)	Yes

Limit	No more gradient	Channel: no more gradient Carrier: reaches max transport, saturated carrier	Transport max, saturated carrier	Transport max, saturated cotransport carrier

Ion Leak Channels

Due to active transport, Na+ is pushed out of the membrane and K+ is pushed in, against the concentration gradient (higher Na+ conc. outside, higher K+ conc. inside).

Ion leak channels is a type of passive transport (facilitated diffusion) that allows some Na+ ions to selectively leak in (following conc. gradient) and some K+ ions to selectively leak out (following conc. gradient)

Ion	Extracellular (mMol/L)	Intracellular (mMol/L)	Relative Permeability
Na+	150	15	$\frac{1}{15} - \frac{1}{75}$
K+	5	150	1
Cl-	110	20	$\frac{1}{2}$
Anions-	0	65	0

Resting Membrane Potential

Resting membrane potential = -70mV

Nernst Equation $E_x = \frac{61}{Z_x} log_{10} \frac{[C]_o}{[C]_i}$

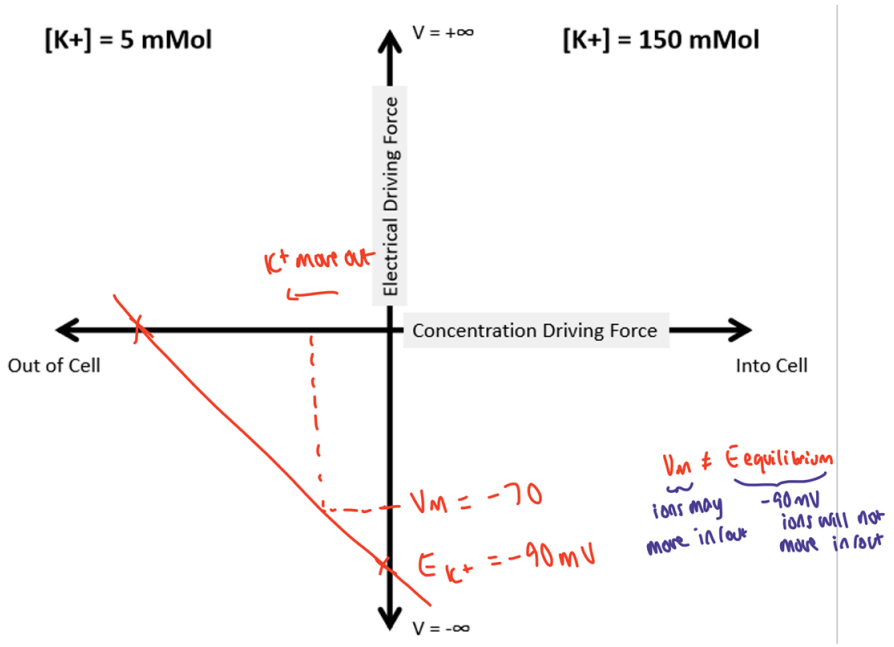
Where:

- E = Equilibrium potential (mV)
- x = Ion
- z = Valence
  (+1 for Na^+ | +2 for Ca^{2+} | -1 for Cl^- | etc.)
- $[C]_o$ & $[C]_i$ = Concentration outside/inside cell

Which direction will K+ go if a cell has a membrane potential of -70mV and E_k=-90mV?

Electrochemical Graphing Process:

1) Set up graph with the Electrical Driving Force (V_m) on y-axis and Concentration Driving Force on the x-axis
2) Write down the concentration inside & outside cell on the x-axis
3) Make a mark on the left or right side of the x-axis to show which direction the ion will move only considering the concentration differences
4) Make a mark on the y-axis according to whether the E_{ion} is positive or negative
5) Draw a straight line between the two marks
6) Determine whether the ion will move into or out of the cell at a given membrane potential (V_m)



3.1 Membrane Potential

Membrane potential: Difference in the electrical potential between inside and outside a cell

Resting membrane potential: Membrane potential when not producing electric signals

- Determined by multiple ions

	CONCENTRATION (millimoles/litre)		
Ion	Extracellular	Intracellular	Relative Permeability
Na^+	150	15	1
K^+	5	150	50–75
A^-	0	65	0

- Concentration gradient for K+ is always outward
- Concentration gradient for Na+ is always inward
- Electrical gradient for both itons is towards negatively charged side of membrane

Eventually, the net outward concentration gradient and the net inward electrical gradient become balanced, so there is no net movement of K+ across the membrane. The potential that would exist at this equilibrium is known as the K+ **equilibrium potential (E_k)**.

- Sign = the polarity of the excess charge **on the inside of the membrane**
- Equilibrium potential is a measure of the **membrane potential** (i.e., the magnitude of the electrical gradient) that exactly **counterbalances the concentration gradient** (i.e., the ratio between the ion's concentration outside and inside the cell) that exists **for the ion**.
    - Each ion has its own equilibrium potential

The greater the permeability of the plasma membrane for a given ion, the greater is the tendency for that ion to drive the membrane potential toward the ion's own equilibrium potential.

Nernst Equation: The equilibrium potential for a given ion of differing concentrations across a membrane.

$$E = 61 \log \frac{C_o}{C_i}$$

Example:

- Concentration gradient of K+ is outward (conc. higher inside)
- Electrical gradient of K+ is inward (more negative)

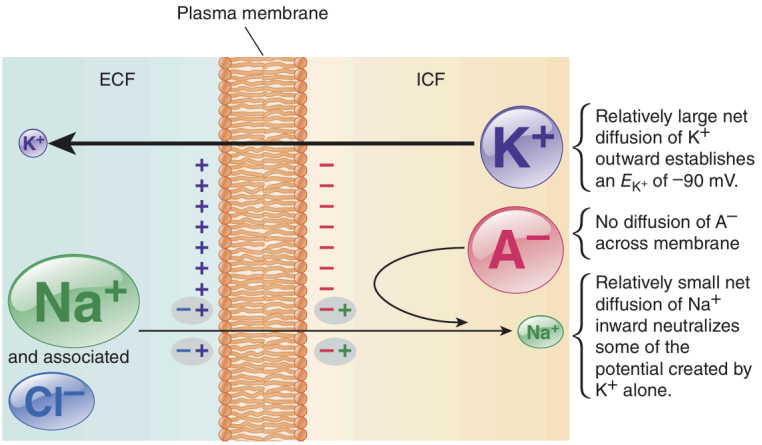
1 The Na^+–K^+ pump actively transports Na^+ out of and K^+ into the cell, keeping the concentration of Na^+ high in the ECF and the concentration of K^+ high in the ICF.

2 Given the concentration gradients that exist across the plasma membrane, K^+ tends to drive membrane potential to the equilibrium potential for K^+ (–90 mV), whereas Na^+ tends to drive membrane potential to the equilibrium potential for Na^+ (+60 mV).

3 However, K^+ exerts the dominant effect on resting membrane potential because the membrane is more permeable to K^+. As a result, resting potential (–70 mV) is much closer to E_{K^+} than to E_{Na^+}.

4 During the establishment of resting potential, the relatively large net diffusion of K^+ outward does not produce a potential of –90 mV because the resting membrane is slightly permeable to Na^+ and the relatively small net diffusion of Na^+ inward neutralizes (in *grey* shading) some of the potential that would be created by K^+ alone, bringing resting potential to –70 mV, slightly less than E_{K^+}.

5 The negatively charged intracellular proteins (A^-) that cannot cross the membrane remain unbalanced inside the cell during the net outward movement of the positively charged ions, so the inside of the cell is more negative than the outside.



Resting membrane potential = −70 mV

Changes in membrane potential:

1. **Polarization:** At resting potential, the membrane is polarized at -70 mV. The membrane is in polarization when the membrane potential is other than 0 mV.
2. **Depolarization:** Resting potential becomes less negative (e.g. -70 mV to -60 mV) - movement in positive direction.
3. **Repolarization:** Membrane returns to resting potential after being depolarizied. Movement in a negative direction.
4. **Hyperpolarization:** Membrane becomes more polarized than at resting potential. Membrane becomes more negative (e.g. -70 mV to -80 mV)

There are two basic forms of electrical signals:

(1) **Graded potentials:** serve as short-distance signals

(2) **Action potentials:** signal over long distances

Gated channels:

(1) **Voltage-gated channels:** open or close in response to changes in membrane potential

(2) **Chemically gated channels:** change conformation in response to the binding of a specific chemical messenger to a membrane receptor in close association with the channel

(3) **Mechanically gated channels:** respond to stretching or other mechanical deformation

(4) **Thermally gated channels:** respond to local changes in temperature (heat or cold)

Forms of electrical signals:

(1) Graded potential - short distance signals

(2) Action potential - long distance

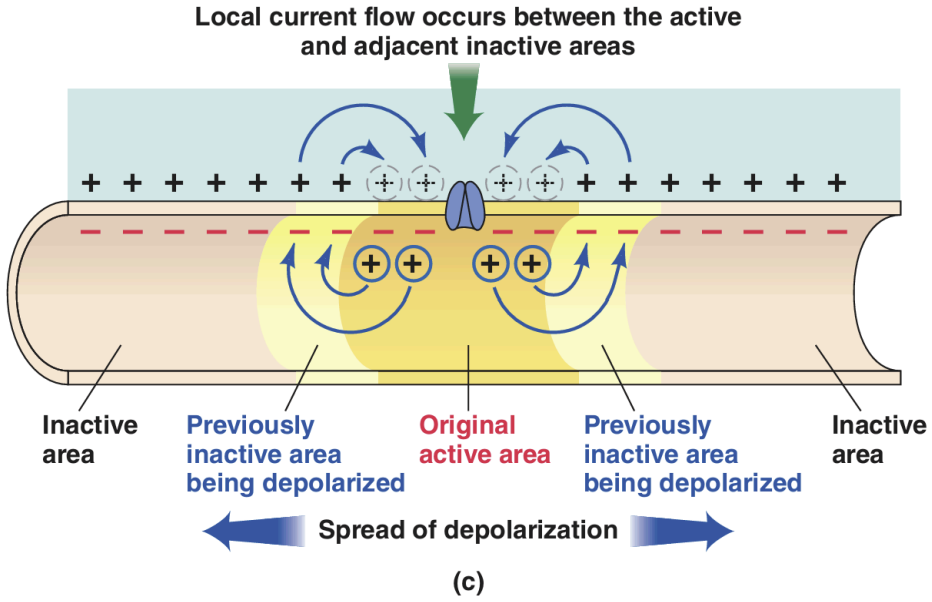
3.2 Graded potentials

Graded potentials are local changes in membrane potential that occur in varying grades or degrees of magnitude or strength.

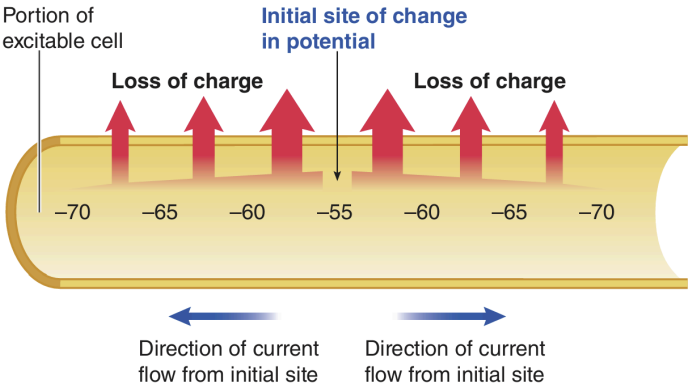
Triggering events:

- Graded potentials usually produced by **specific triggering event** that opens gated ion channels in specialized region of cell membrane (usually chemically or mechanically)
- Stronger the triggering event = more gated channels that open = greater the positive charge entering the cell = larger depolarizing graded potential at the point of origin.
- Longer duration of the triggering event = longer duration of the graded potential.

When a graded potential occurs locally in a nerve or muscle cell membrane, the remainder of the membrane is still at resting potential.



- Graded potentials can initiate action potentials, the long-distance signals, in an excitable cell.



› FIGURE 3-7 **Current loss across the plasma membrane.** Leakage of charge-carrying ions across the plasma membrane results in progressive loss of current with increasing distance from the initial site of potential change.

- To initiate an action potential, a triggering event causes the membrane to depolarize from the resting potential of −70 mV. **Depolarization** proceeds slowly at first, until it reaches a critical level known as **threshold potential**, typically between −50 and −55 mV.
- At threshold potential, an explosive depolarization takes place. A recording of the membrane potential at this time shows a sharp upward deflection to +30 mV; this coincides with the rapid reversal of the membrane potential, causing the inside of the cell to become positive compared to the outside.
- Just as rapidly, the membrane **repolarizes**, dropping back to resting potential.
- Often the forces that repolarize the membrane push the potential too far, causing a brief phase known as after **hyperpolarization**, during which the inside of the membrane briefly becomes even more negative than normal (e.g., −80 mV) before the resting potential is restored.

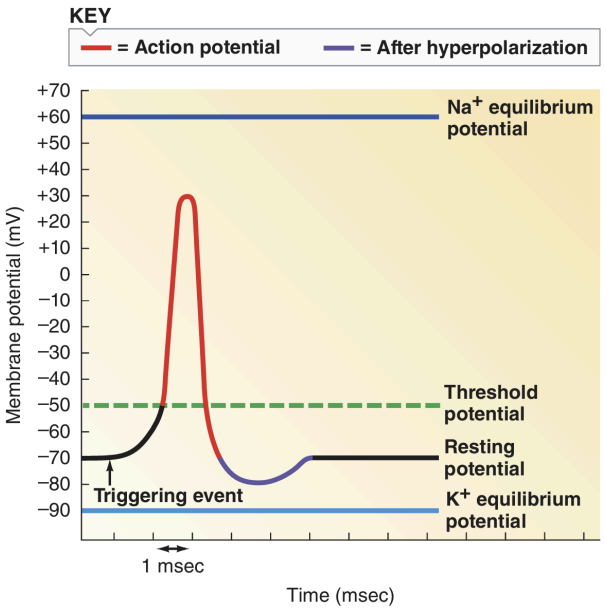
TABLE 3-2 Comparison of Graded Potentials and Action Potentials

Graded Potentials	Action Potentials
Graded potential change; magnitude varies with magnitude of triggering event	All-or-none membrane response; magnitude of triggering event coded in frequency rather than amplitude of action potentials
Duration varies with duration of triggering event	Constant duration
Decremental conduction; magnitude diminishes with distance from initial site	Propagated throughout membrane in undiminishing fashion
Passive spread to neighbouring inactive areas of membrane	Self-regeneration in neighbouring inactive areas of membrane
No refractory period	Refractory period
Can be summed	Summation impossible
Can be depolarization or hyperpolarization	Always depolarization and reversal of charges
Triggered by stimulus, by combination of neurotransmitter with receptor, or by spontaneous shifts in leak-pump cycle	Triggered by depolarization to threshold, usually through spread of graded potential
Occurs in specialized regions of membrane designed to respond to triggering event	Occurs in regions of membrane with abundance of voltage-gated Na^+ channels

3.3 Action Potential

Action Potentials

- **Nondecremental:** They do not lose strength as they travel (like a firecracker fuse).
- **All-or-None Law:** A stimulus either reaches a threshold (typically -50 to -55 mV) and triggers a full action potential, or it fails to reach it and nothing happens.
- **Reversal of Potential:** The membrane potential quickly swings from a resting state (-70 mV) to a positive peak (+30 mV), called the overshoot



Phase	Description	Ionic Mechanism
Rising Phase	Rapid depolarization toward +30 mV.	**Na+ influx:** Voltage-gated Na+ channels open rapidly.
Falling Phase	Rapid return toward resting potential.	**K+ efflux:** Na+ channels close (inactivate) and voltage-gated K+ channels open.
Hyperpolarization	Potential briefly becomes more negative than rest (e.g., -80 mV).	**Slow K+ closure:** K+ continues to leave the cell for a brief moment after reaching rest.

Voltage-gated Na+ channels have two gates:

1. **Activation gate:** Opens quickly at threshold.
2. **Inactivation gate:** A “ball” that blocks the channel shortly after it opens, stopping Na+ entry even if the stimulus is still present.

Propagation of Action Potential

Neuron consists of three basic parts: *cell body, dendrites, axon*

Cell body (soma): nucleus and organelles

Dendrites: *receive signals* from other nerve cells

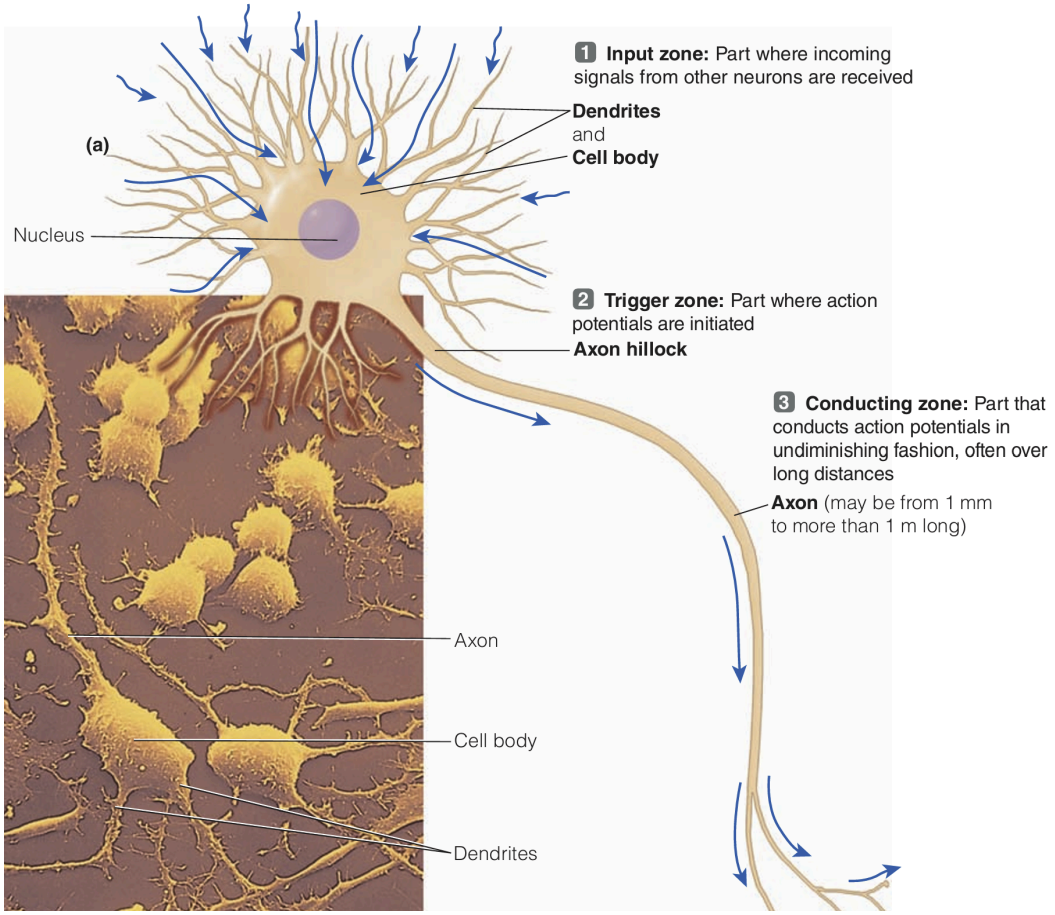
Axon: conduct action potentials *away from* cell body, terminates at other cells

Input Zone: Dendrites and cell body (soma) receive signals and produce **graded potentials**.

Conducting Zone: The axon transmits the action potential.

Output Zone: Axon terminals release chemical messengers to other cells.

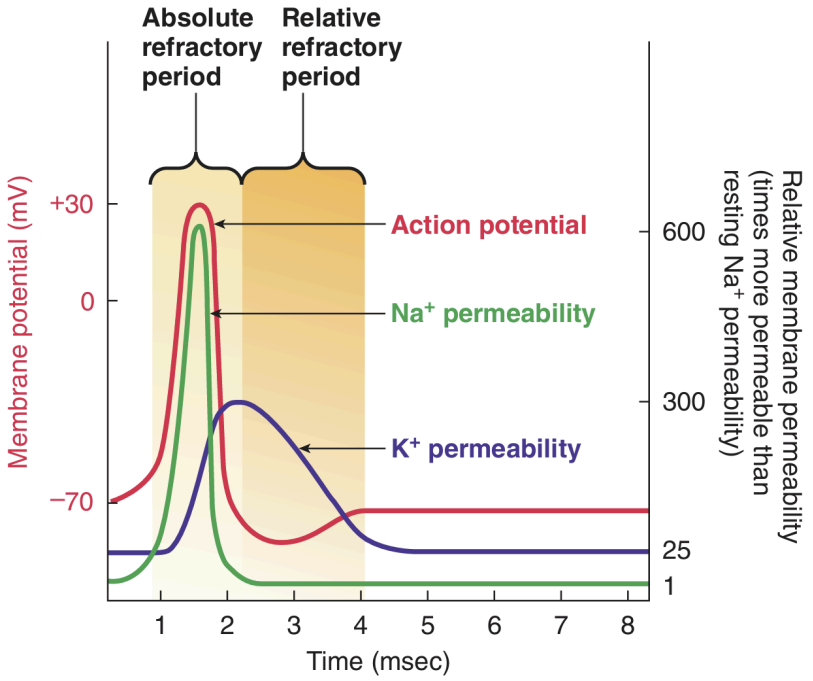
The Na+/K+ Pump: While only a tiny fraction of ions actually move during a single AP, the Na+/K+ pump is essential for maintaining the concentration gradients over time to prevent the cell from losing its ability to fire ("rundown").



Once an action potential starts at the **axon hillock** (the trigger zone), it travels down the axon via **contiguous conduction**.

- **Local Current Flow:** The positive charge inside an active patch of membrane flows to the adjacent negative (inactive) patch.

- **Self-Perpetuating:** This local flow brings the next patch to threshold, triggering an identical brand-new action potential. **The original action potential does not travel along the membrane.**
- **One-Way Travel:** The **refractory period** prevents the signal from moving backward. This period ensures that a recently active area cannot immediately fire again, forcing the impulse to move only toward the axon terminals.
    - **Absolute refractory period:** Time period when a recently activated patch of membrane is completely unresponsive to further stimulation
    - **Relative refractory period:** Second action potential can be produced only by a triggering event considerably stronger than usually necessary. This is due to (1) lingering inactivation of Na+ channels and (2) slow closing of K+ channels, so fewer Na+ channels are jolted open by triggering effect and K+ continues to leave.



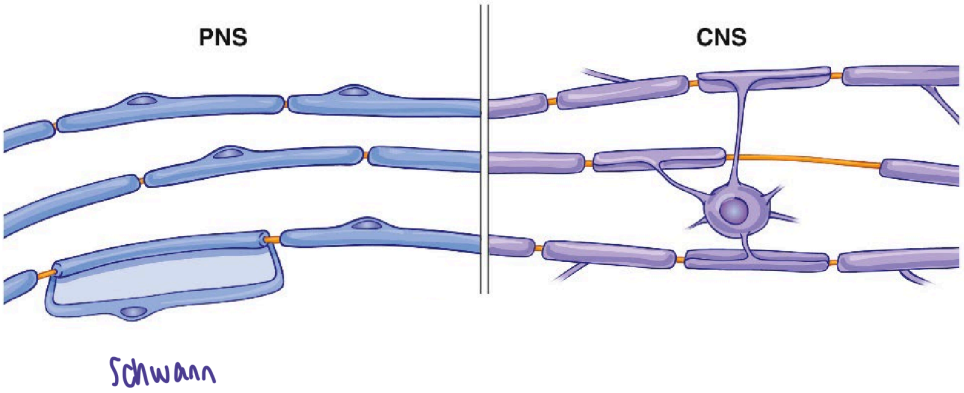
Strength of Stimulus and Frequency of Action Potential

- Stronger stimulus does not produce larger action potential, but it triggers a **greater frequency of action potentials per second**.
- Causes more neurons to reach threshold, increasing total information sent to CNS
- Once initiated, velocity of action potential travelling down axon depends on:
    (1) Whether fibre is myelinated
        - **Contiguous conduction** for unmyelinated
        - **Saltatory conduction** for myelinated
    (2) Diameter of fibre

Myelin

Myelin is composed of lipid-rich cells (**Schwann cells** in the peripheral nervous system and **oligodendrocytes** in the central nervous system) that wrap around the axon like electrical tape around a wire.

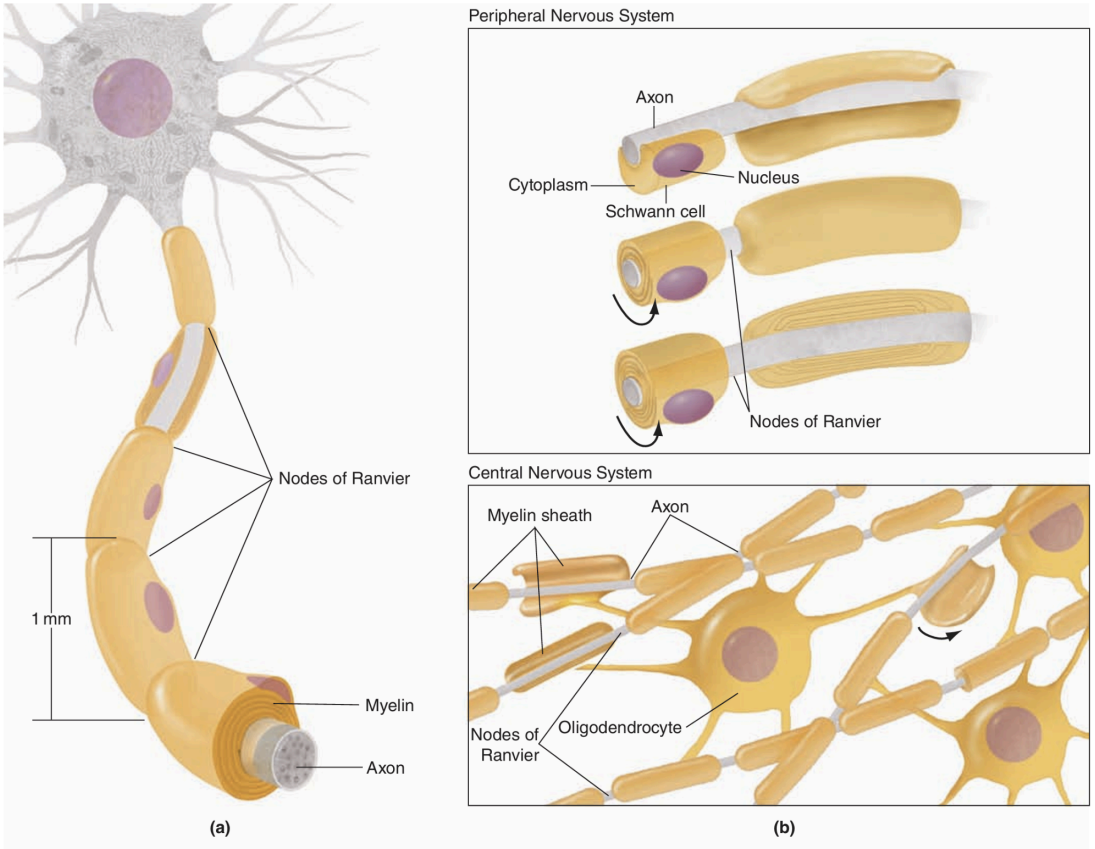
- **Insulation:** Myelin prevents “leakage” of current across the membrane.
- **Nodes of Ranvier:** Unmyelinated gaps between segments of myelin. They are the only places where the axonal membrane is exposed to the extracellular fluid and where **voltage-gated Na+ and K+ channels are concentrated**.



Saltatory Conduction

In myelinated fibers, the action potential does not have to be regenerated at every single micrometer of the membrane. Instead, it “jumps” from node to node.

1. **Fast Passive Spread:** When an action potential occurs at one node, the local current flows rapidly through the insulated (myelinated) portion of the axon.
2. **Regeneration at Nodes:** Because the myelin is so thick, the current doesn't lose much strength. When it reaches the next **Node of Ranvier**, it is still strong enough to depolarize the membrane to threshold.
3. **Result:** The action potential “skips” the myelinated sections, making conduction about **50 times faster** than in unmyelinated fibers of the same size.

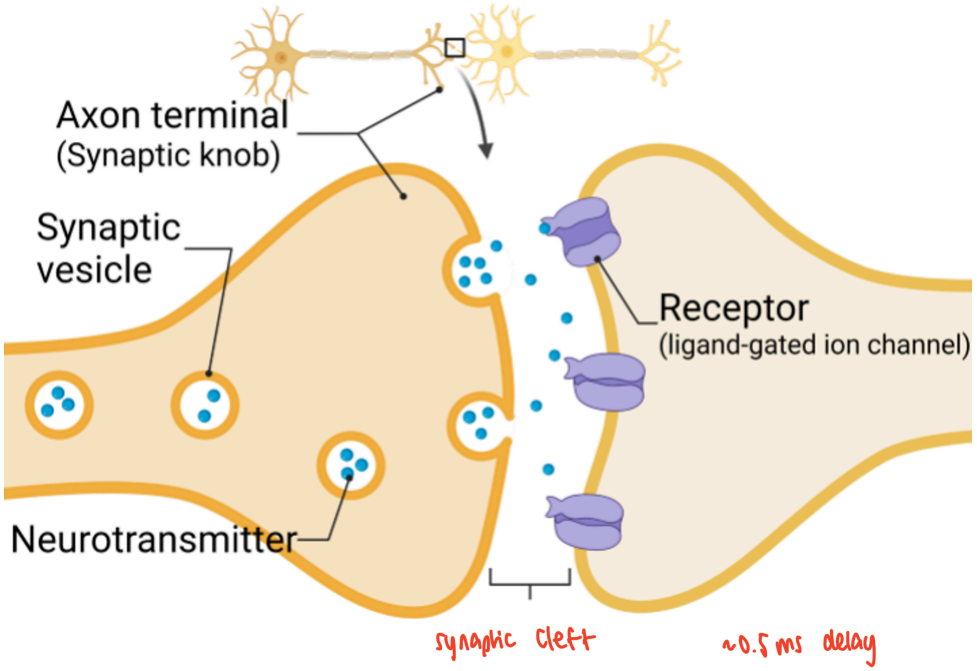

Nodes of Ranvier
1 mm
Myelin
Axon
(a)
Peripheral Nervous System
Axon
Cytoplasm
Nucleus
Schwann cell
Nodes of Ranvier
Central Nervous System
Myelin sheath
Axon
Nodes of Ranvier
Oligodendrocyte
(b)

3.5 Synapses and Neuronal Integration

The Anatomy of a Synapse

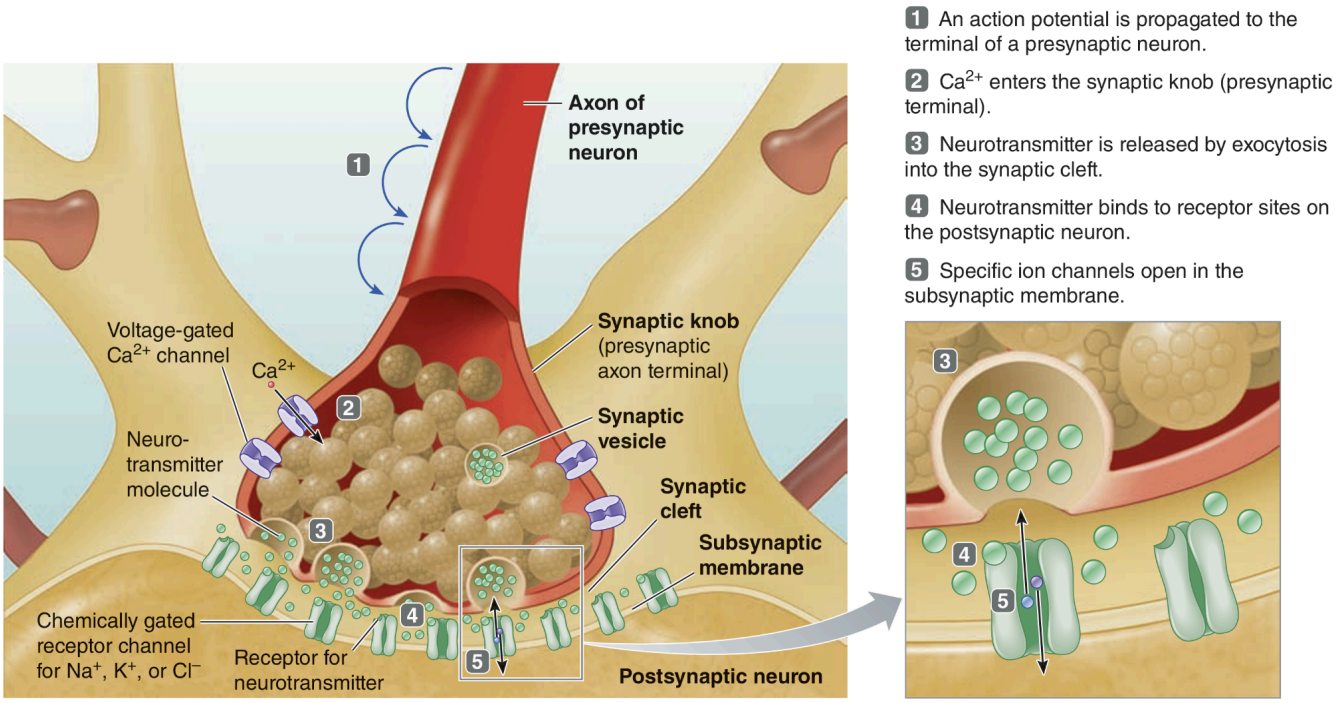
A neuron can terminate on a muscle, a gland, or another neuron. A junction between two neurons is a **synapse**.

- **Presynaptic Neuron:** The "sender" neuron that conducts action potentials toward the synapse. It ends in a **synaptic knob**.
- **Postsynaptic Neuron:** The "receiver" neuron whose potential is altered by the signal.
- **Synaptic Vesicles:** Sacs in the synaptic knob that store **neurotransmitters**.
- **Synaptic Cleft:** The narrow space between the two neurons. Because there is no physical contact, the electrical signal cannot jump the gap directly; it must be converted into a chemical signal.



Signal Transmission

1. **Action Potential Arrives:** The electrical signal reaches the synaptic knob.
2. **Ca^{2+} Entry:** Voltage-gated calcium channels open, and Ca^{2+} flows into the knob (since it is more concentrated in ECF)
3. **Exocytosis:** Ca^{2+} triggers synaptic vesicles to release neurotransmitters (stored in vesicles) into the cleft (synaptic delay of 1–2 msec to convert electrical to chemical).
4. **Binding:** Neurotransmitters diffuse across the cleft and bind to specific receptors on the **subsynaptic membrane** (post-synaptic membrane directly below knob).
5. **Ion Channels Open:** This binding opens chemically gated ion channels, changing the postsynaptic neuron's potential (excitatory or inhibitory).



Neurotransmitter Removal

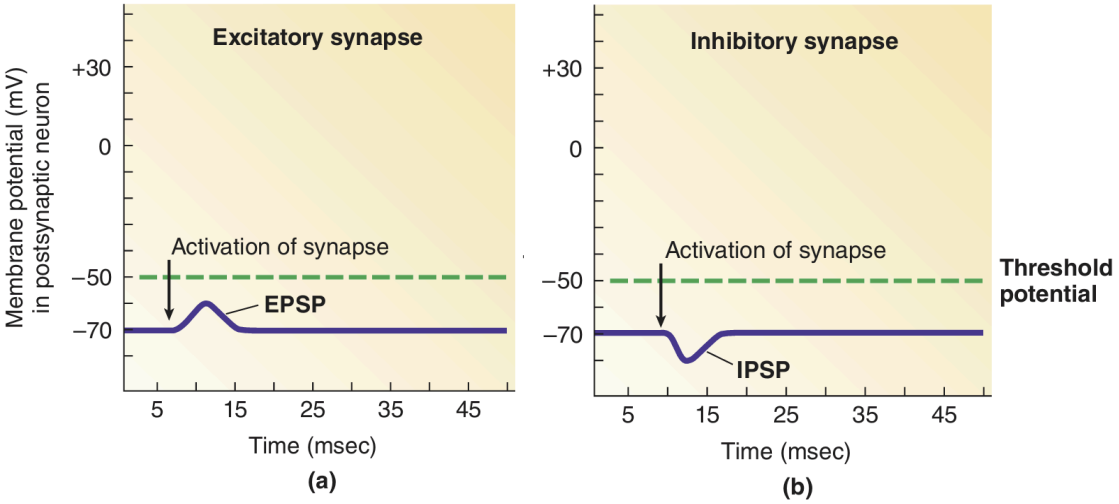
To keep the signal brief and "wipe the slate clean," the neurotransmitter must be removed via:

1. **Diffusion** away from the cleft.
2. **Enzymatic Inactivation** (destruction by enzymes).
3. **Reuptake** (transported back into the presynaptic knob to be recycled).

EPSP vs. IPSP

Depending on which ion channels open, the synapse is either excitatory or inhibitory. This **depends on the properties of the receptor, not the neurotransmitter**.

- **Excitatory Postsynaptic Potential (EPSP):** Opens nonspecific cation channels that permit both Na+ and K+. At resting potential, the drive for Na+ to enter is stronger than for K+ to leave, so the cell **depolarizes** (gets closer to threshold).
    - Example: glutamate
- **Inhibitory Postsynaptic Potential (IPSP):** Opens K+ or Cl- channels. K+ leaves or Cl- enters, making the inside more negative (hyperpolarization). This moves the cell farther from the threshold.
    - Example: GABA, glycine
- **Most** of the time, each axon terminal releases only one neurotransmitter (some exceptions though, like GABA + glycine)



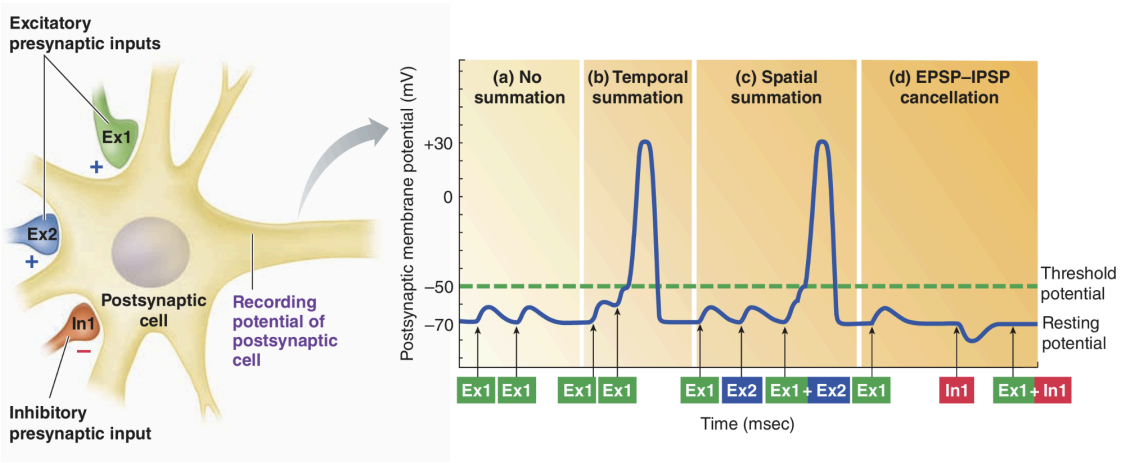
Grand Postsynaptic Potential (GPSP)

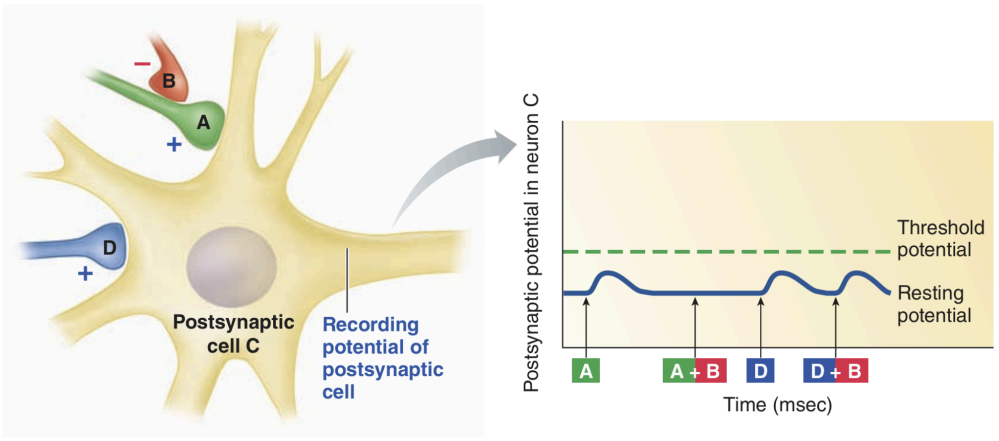
IPSPs and EPSPs are graded potentials, which can be of varying magnitude. A single EPSP is almost never strong enough to fire an action potential. The **GPSP** is the sum of all incoming signals (thousands of EPSPs and IPSPs) at any given moment.

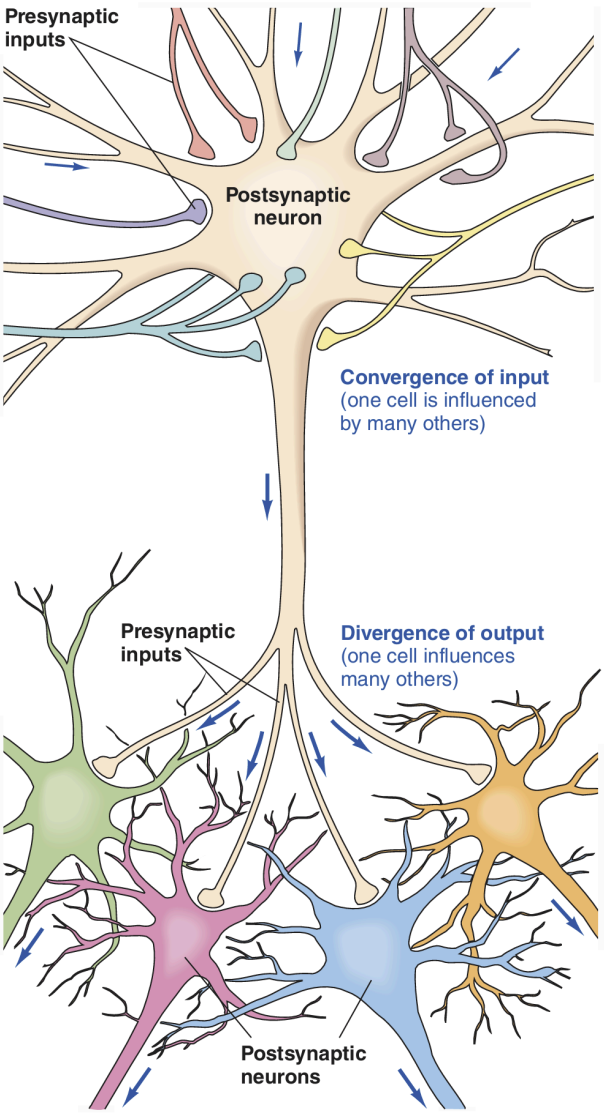
How Threshold is Reached (Summation):

- **Temporal Summation:** One presynaptic neuron fires many times in rapid succession.
- **Spatial Summation:** Several different presynaptic neurons fire at the same time.

Cancellation: If an excitatory and inhibitory signal fire simultaneously, they cancel each other out, and the potential stays near rest.


Excitatory presynaptic inputs
Ex1
+
Ex2
+
In1
−
Inhibitory presynaptic input
Postsynaptic cell
Recording potential of postsynaptic cell
Postsynaptic membrane potential (mV)
+30
0
−50
−70
(a) No summation
(b) Temporal summation
(c) Spatial summation
(d) EPSP–IPSP cancellation
Threshold potential
Resting potential
Ex1
Ex1
Ex1
Ex1
Ex1
Ex2
Ex1 + Ex2
Ex1
In1
Ex1 + In1
Time (msec)


−
B
A
+
D
+
Postsynaptic cell C
Recording potential of postsynaptic cell
Postsynaptic potential in neuron C
Threshold potential
Resting potential
A
A + B
D
D + B
Time (msec)

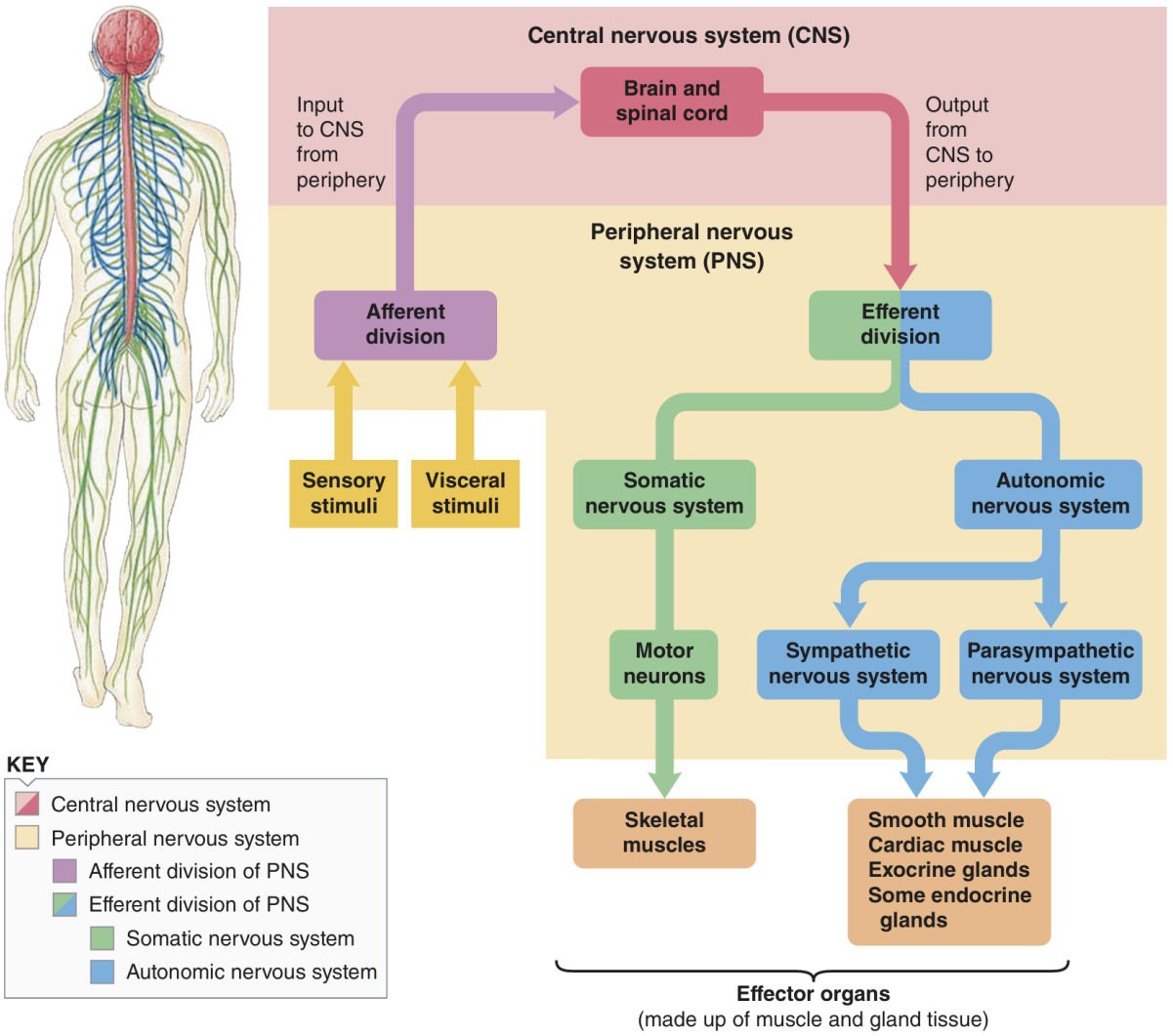

Presynaptic inputs
Postsynaptic neuron
Convergence of input
(one cell is influenced by many others)
Presynaptic inputs
Divergence of output
(one cell influences many others)
Postsynaptic neurons

4.1 Organization of the Nervous System

CNS vs. PNS

The nervous system is a single integrated unit, but it is divided into two primary parts:

- **Central Nervous System (CNS):** The "command center," consisting of the **brain** and **spinal cord**.
- **Peripheral Nervous System (PNS):** The "communication lines," consisting of nerve fibers that carry signals between the CNS and the rest of the body.



Subdivisions of PNS

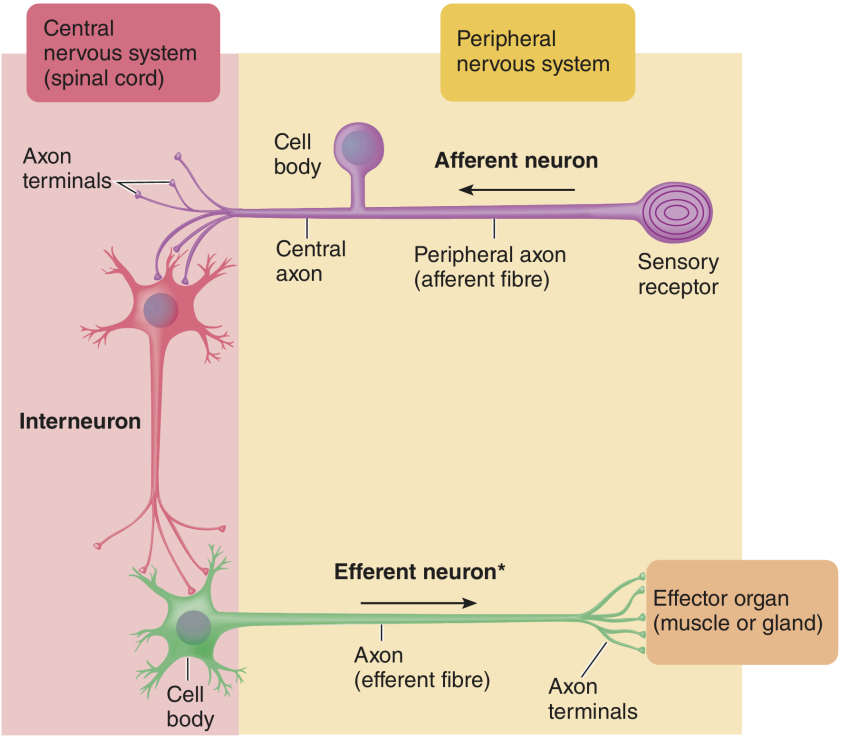
The PNS is broken down based on the *direction* of the information flow:

1. **Afferent Division (Input)**
    - Carries information **to the CNS**. It reports on the external environment (like temperature) and the internal environment (like hunger).
2. **Efferent Division (Output)**
    - Carries instructions from the CNS **to effector organs** (muscles or glands). It has two further branches:

- **Somatic Nervous System:** Supplies the **skeletal muscles** (voluntary movement).
- **Autonomic Nervous System (ANS):** Supplies **smooth muscle, cardiac muscle, and glands** (involuntary functions). It is further split into:
  - **Sympathetic Nervous System:** "fight or flight."
  - **Parasympathetic Nervous System:** "rest and digest."

Afferent, Efferent, and Interneurons

Neuron Type	Location	Role	Structure
Afferent	Primarily in the **PNS**.	Sensory: **Generates APs** in response to a stimulus and carries them toward the CNS.	Unique shape: No dendrites on the cell body; has a sensory receptor at the peripheral end.
Efferent	Primarily in the **PNS** (axons), but cell bodies are in the CNS (originate from CNS)	Motor: Carries **instructions away from the CNS** to effector organs.	Cell bodies receive many inputs within the CNS to integrate a "decision."
Interneurons	Entirely within the **CNS**.	Integrators: Connect afferent and efferent neurons; responsible for "mind" functions.	Make up 99% of all neurons (approx. 100 billion).



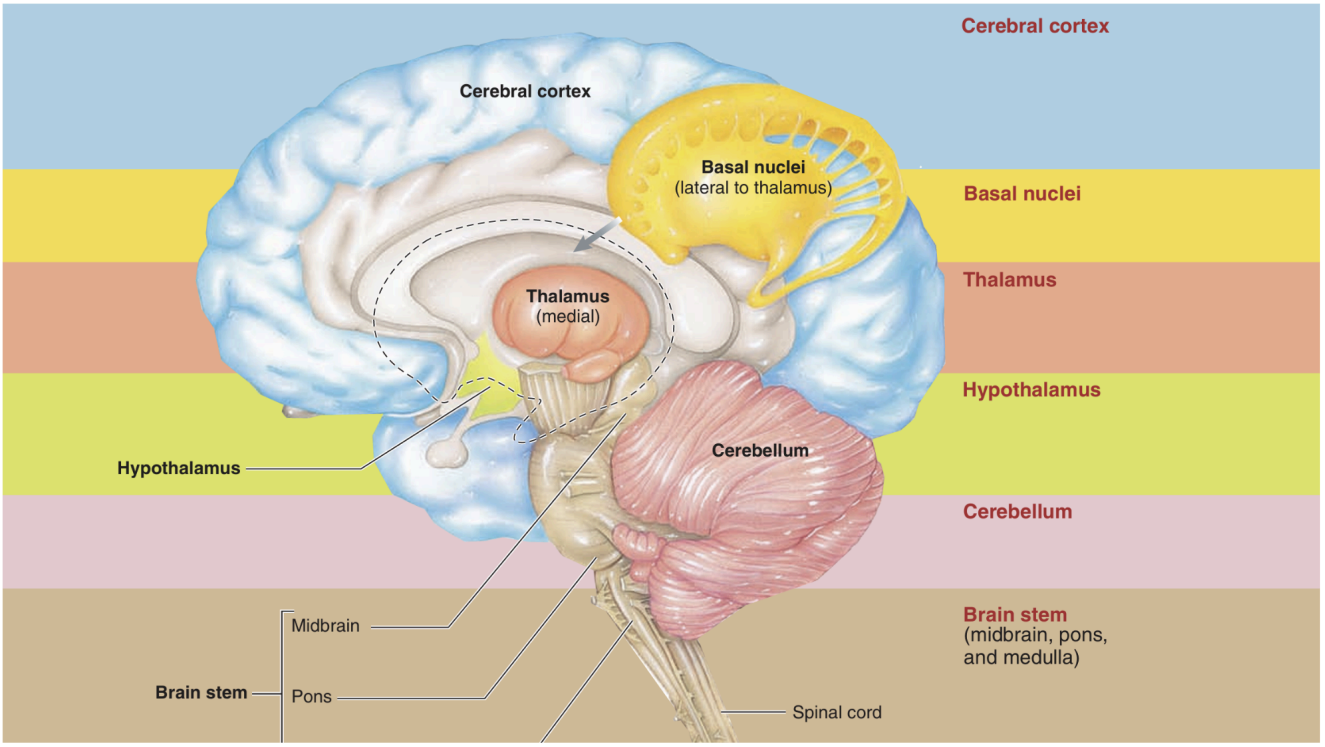
* Efferent autonomic nerve pathways consist of a two-neuron chain between the CNS and the effector organ.

Example: Hot object

1. **Afferent neurons** sense the heat and send a signal to the spinal cord.
2. **Interneurons** in the spinal cord receive that signal, "decide" what to do, and signal the efferent neurons.
3. **Efferent neurons** carry the "Pull away!" command to the arm muscles (the effectors).

Interneurons are also the source of everything we define as "human", including creativity, intellect, and motivation, by forming massive, complex networks between themselves.

4.2 Structure and Function of the Central Nervous System



1. **Brain stem**
    - **Components:** Midbrain, pons, and medulla.
    - **Role:** Controls **vegetative functions**: life-sustaining processes like breathing, circulation, and digestion (performed unconsciously)
    - This is why a person can remain alive in a "vegetative state" if their brain stem is intact, even if they have lost higher brain function.
2. **Cerebellum**
    - **Location:** Attached to the back of the brain stem.
    - **Role:** Coordinates **subconscious motor activity**. It handles balance, body position **(proprioception)**, and is essential for learning skilled tasks like dancing.
3. **Forebrain**
    - **Diencephalon**
        - **Hypothalamus: Homeostatic functions** for stability of internal environment
        - **Thalamus:** Primitive sensory processing
    - **Cerebrum:** Larger and highly convoluted, 80% of human brain weight
        - **Basal ganglia**
        - **Cerebral cortex:** Highly folded outer layer, **sophisticated neural functions** like voluntary initiation of movement, fine sensory perception, conscious thought, language, personality, mind/intellect traits

4.3 Protection of the Central Nervous System

Glial Cells

About 90% of CNS cells are glial cells. Unlike neurons, they do not conduct impulses but provide vital physical and metabolic support and communicate with neurons by chemical signals.

Astrocytes

- **Main connective tissue of CNS:** Support neurons in proper spatial relationships
- **Scaffold to guide neurons** during fetal brain development
- Induce formation of **blood–brain barrier**
- Repair of brain injury, form **neural scar tissue**
- Take up and **degrade GABA and released neurotransmitters** into raw materials for synthesis of more neurotransmitters by neurons
- **Take up excess K+** to help maintain proper brain–ECF ion concentration and normal neural excitability (elevation in brain K+ concentration can lead to seizures)
- Enhance synapse formation and strengthen synaptic transmission via chemical signalling with neurons

Oligodendrocytes

- Form myelin sheaths in CNS

Microglia

- **Immune cells:** Play a role in defence of brain as phagocytic scavengers

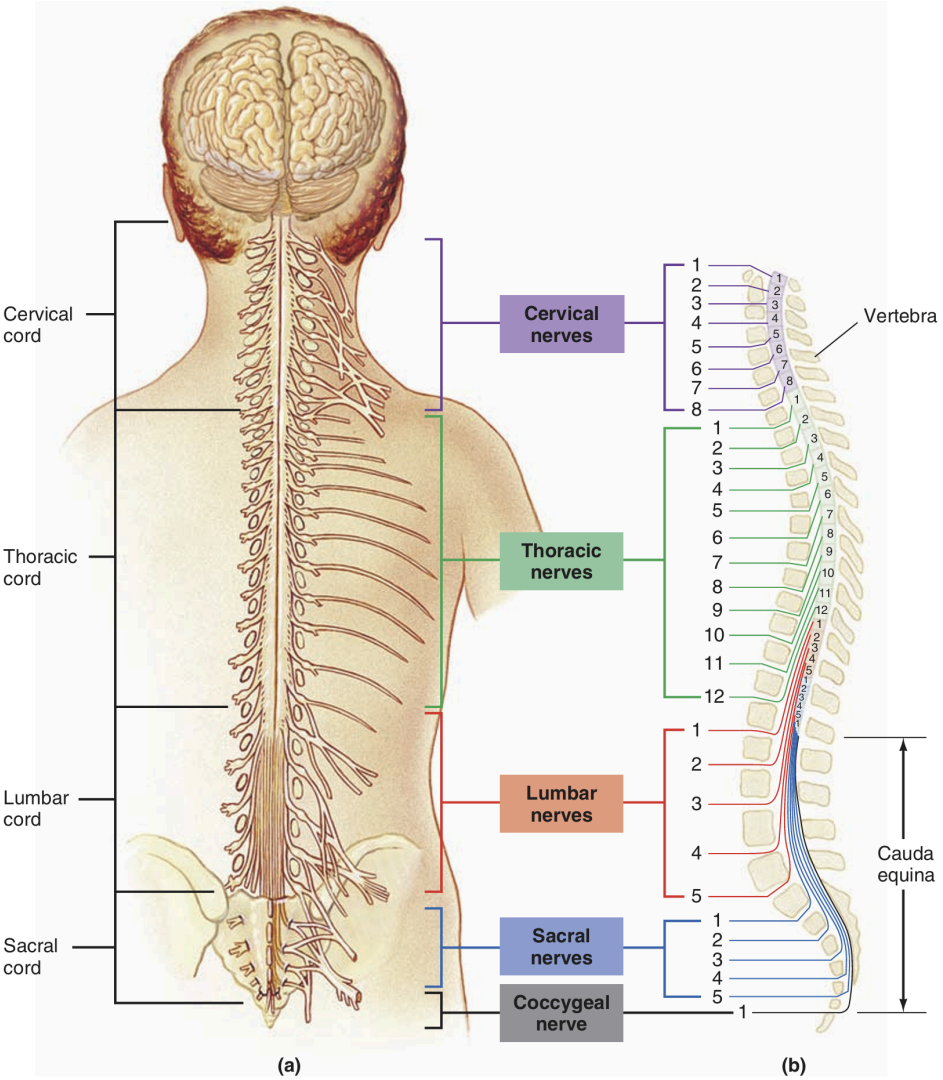
Ependymal cells

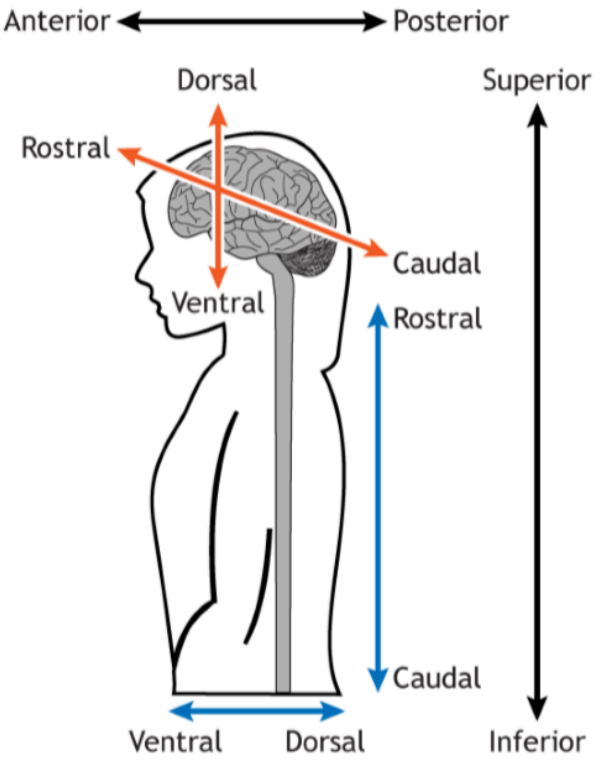
- Line internal cavities of brain and spinal cord
- Contribute to formation of cerebrospinal fluid
- Serve as neural stem cells with the potential to form new neurons and glial cells

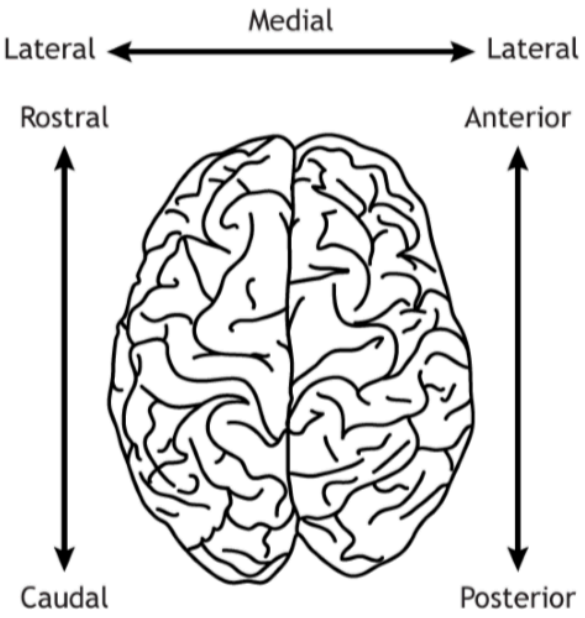
4.4 Spinal Cord

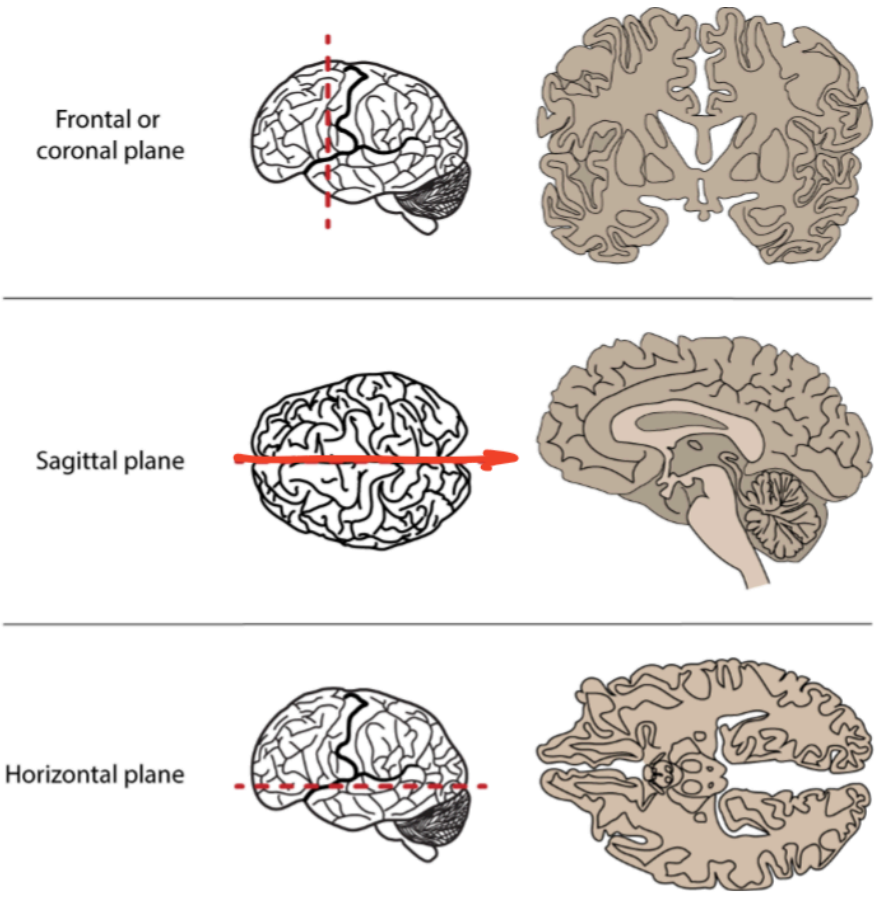
The spinal cord is a slender cylinder of nerve tissue protected by the **vertebral column**.

- **The Differential Growth:** The vertebral column grows faster and longer than the spinal cord. Consequently, the spinal cord ends around the **L1 or L2 lumbar vertebra** (waist level).
- **Cauda Equina ("Horse's Tail"):** Since the cord ends early, the remaining nerve roots must travel down the vertebral canal before exiting. This bundle of long nerve roots resembles a horse's tail.
- **Spinal Taps:** Because the cord ends at L2, doctors can safely insert a needle into the lower lumbar region to sample **Cerebrospinal Fluid (CSF)** without hitting the spinal cord.









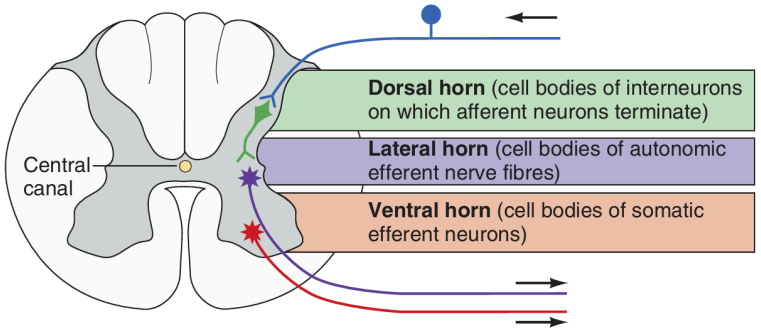
Grey Matter vs. White Matter

In the spinal cord, the layout is the opposite of the brain: **Grey is inside, White is outside.**

Grey Matter (The "Butterfly")

Contains cell bodies, dendrites, and glial cells. It is divided into "horns":

- **Dorsal (posterior) Horn:** Receives incoming **afferent** (sensory) signals.
- **Lateral Horn:** Contains cell bodies for autonomic (involuntary) efferent fibers.
- **Ventral (anterior) Horn:** Contains cell bodies for **efferent** (voluntary) motor neurons.

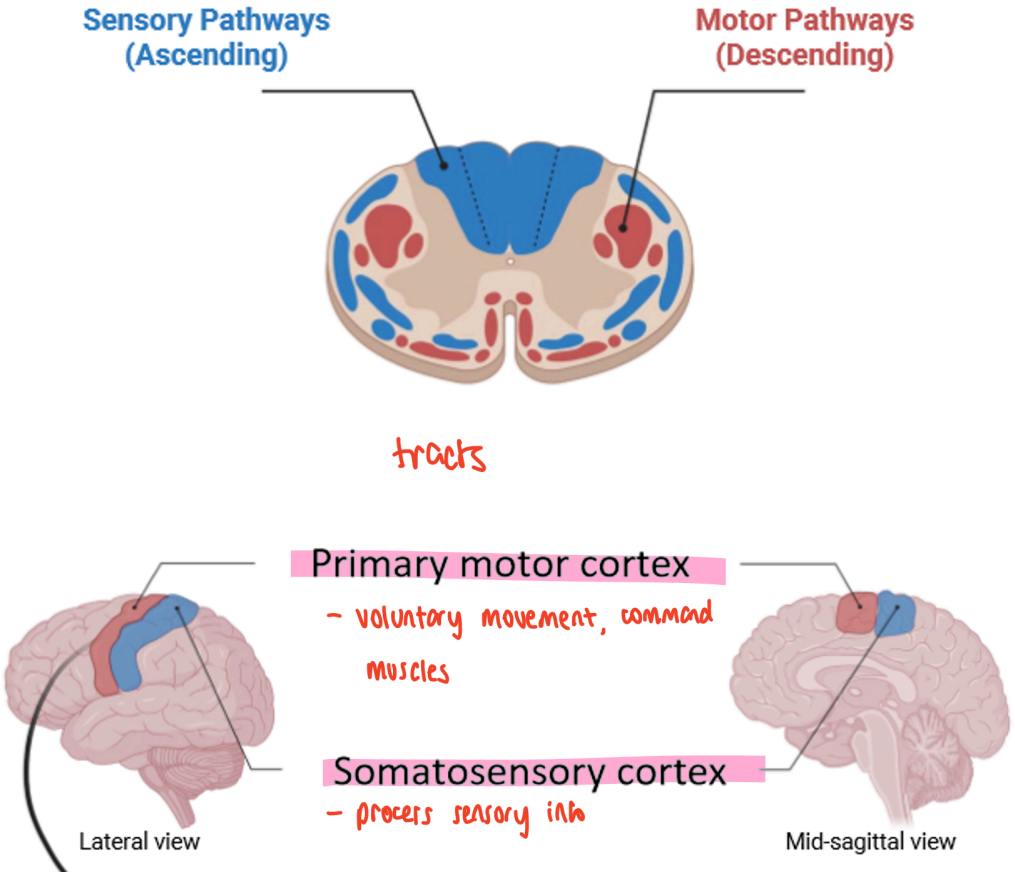


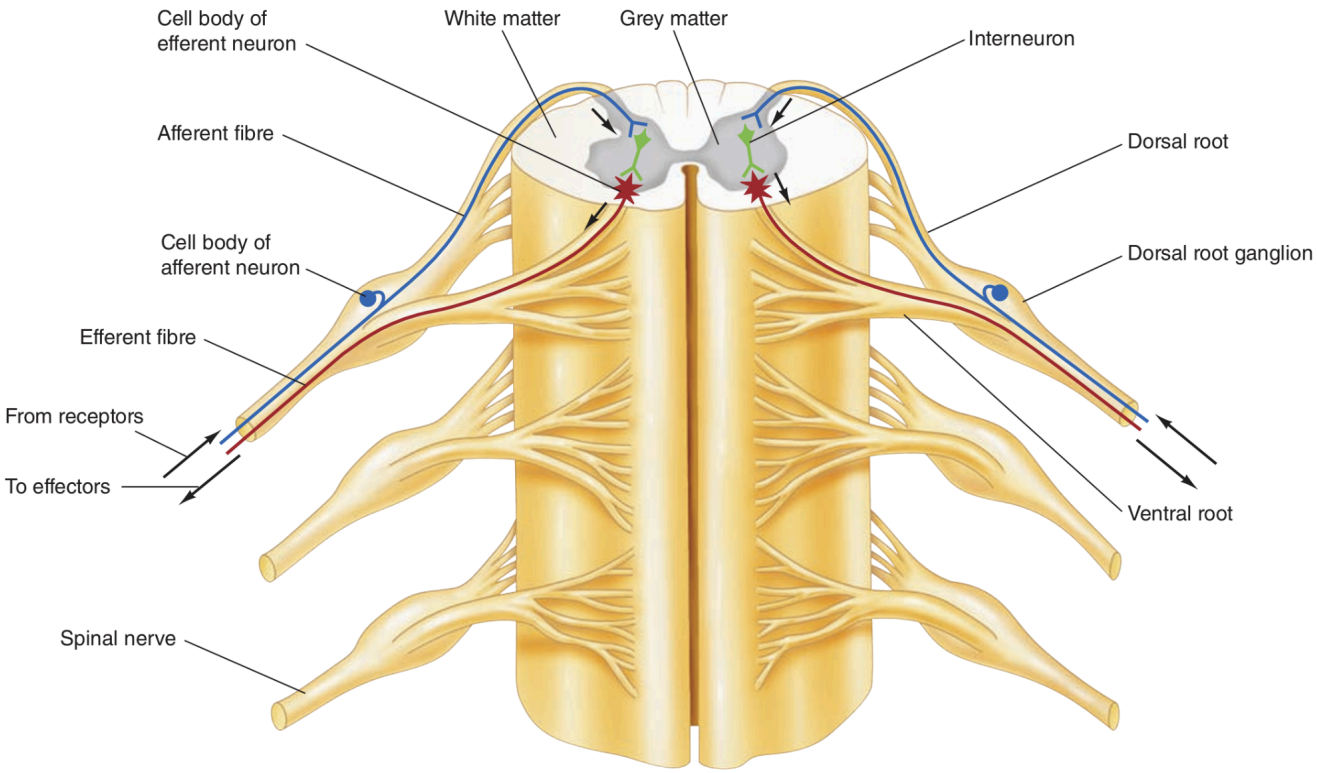
› FIGURE 4-12 Regions of the grey matter

White Matter (The "Tracts")

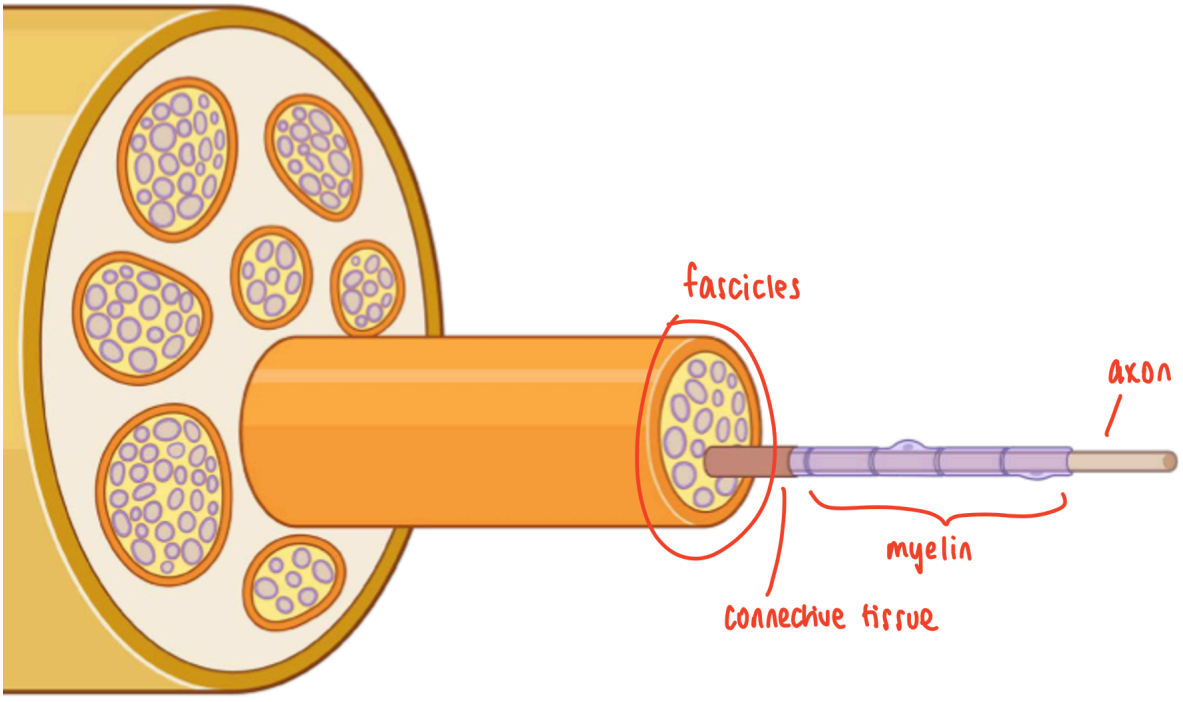
Contains bundles of axons (tracts) that carry signals up and down.

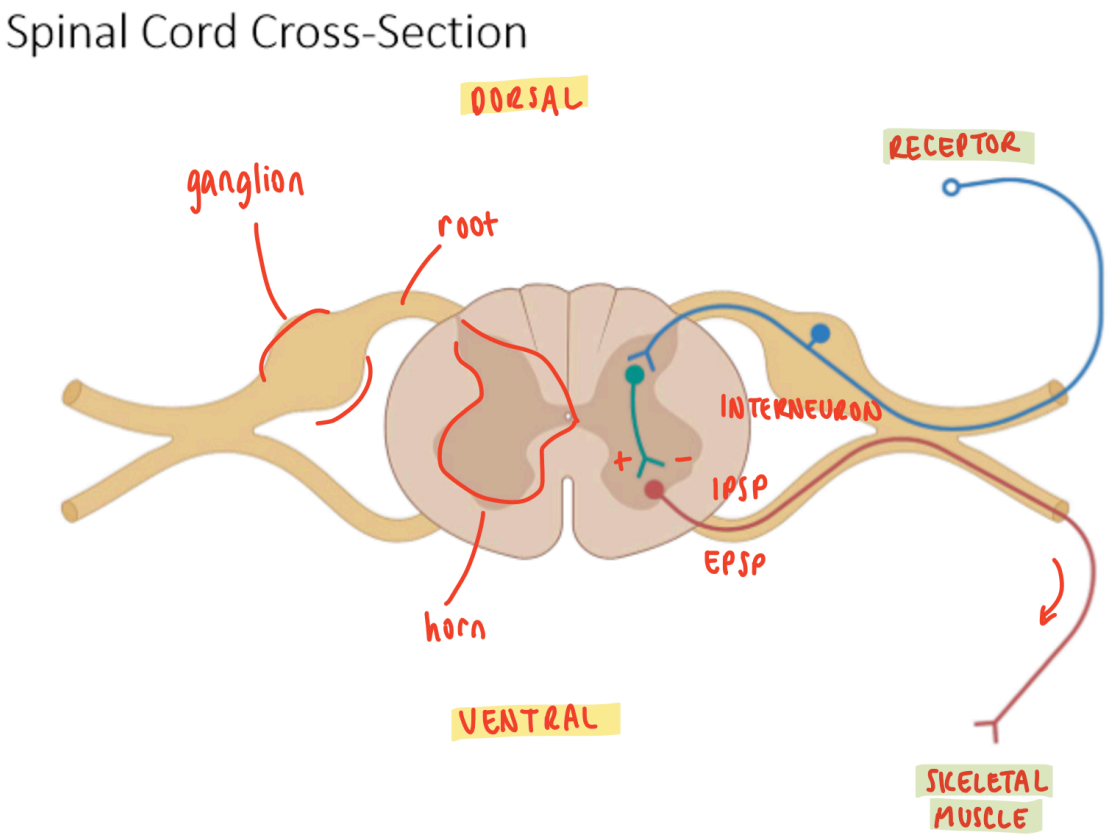
- **Ascending Tracts:** Carry sensory info **to the brain** (e.g., *spinocerebellar tract*).
- **Descending Tracts:** Carry motor commands **from the brain** (e.g., *corticospinal tract*).





Structure of Nerve





The Reflex Arc

A reflex is an automatic response to a stimulus. A **Reflex Arc** consists of five components:

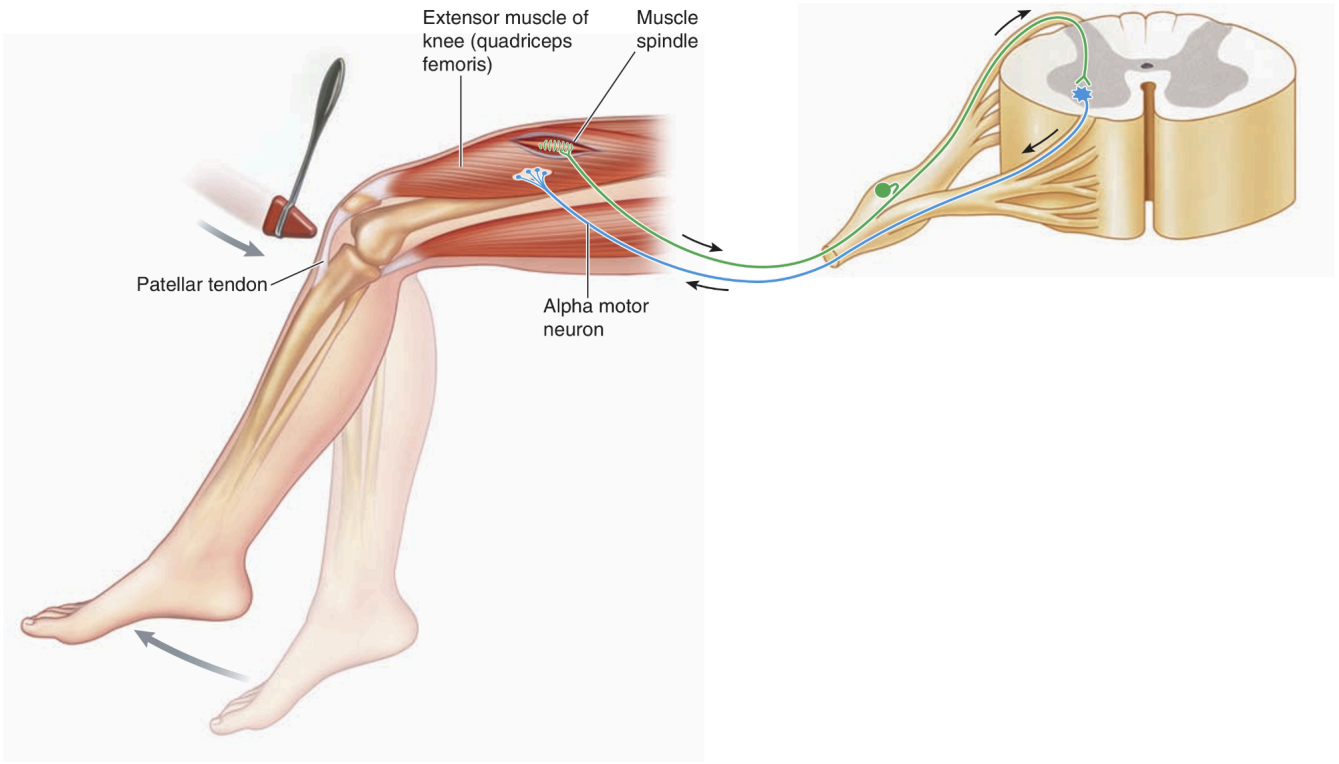
1. **Receptor:** Detects the stimulus.
2. **Afferent Pathway:** Sends the signal to the CNS.
3. **Integrating Centre:** Usually the spinal cord or brain stem; “decides” the response.
4. **Efferent Pathway:** Carries the command to the effector.
5. **Effector:** The muscle or gland that carries out the response.

Events that occur in the brain on arrival of signals via **ascending pathway**, such as awareness of pain, memory storage, and so on, are above and beyond reflex arc.

Types of Reflexes

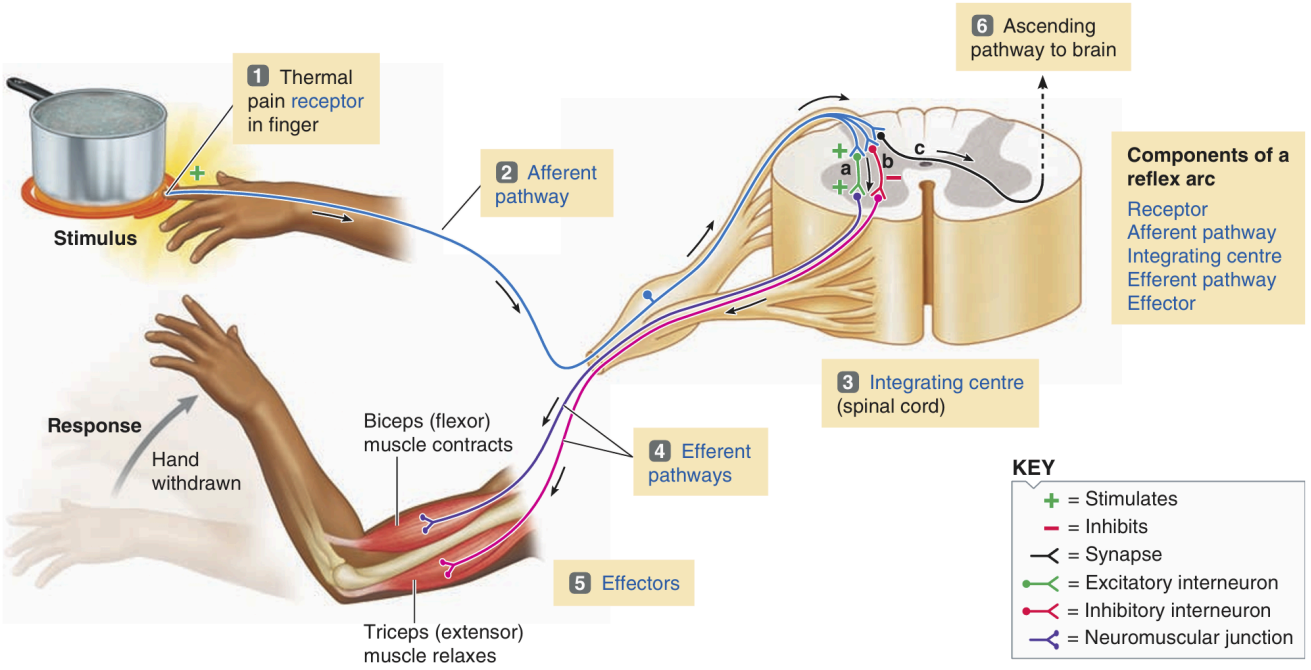
Stretch Reflex (Monosynaptic):

- **Knee-Jerk**: Tapping the patellar tendon stretches the muscle; the cord tells it to contract immediately to maintain posture.
- **EPSP**, stretch reflex



Withdrawal Reflex (Polysynaptic): Touching a hot stove.

- **Reciprocal innervation:** one muscle excited, one muscle inhibited
- **Biceps:** contracting/pulling muscle: excited
- **Triceps:** stretching/extending muscle: inhibited



1 Painful heat stimulus activates thermal pain receptor in finger.

2 Action potentials are generated in afferent pathway, which propagates impulses to the spinal cord.

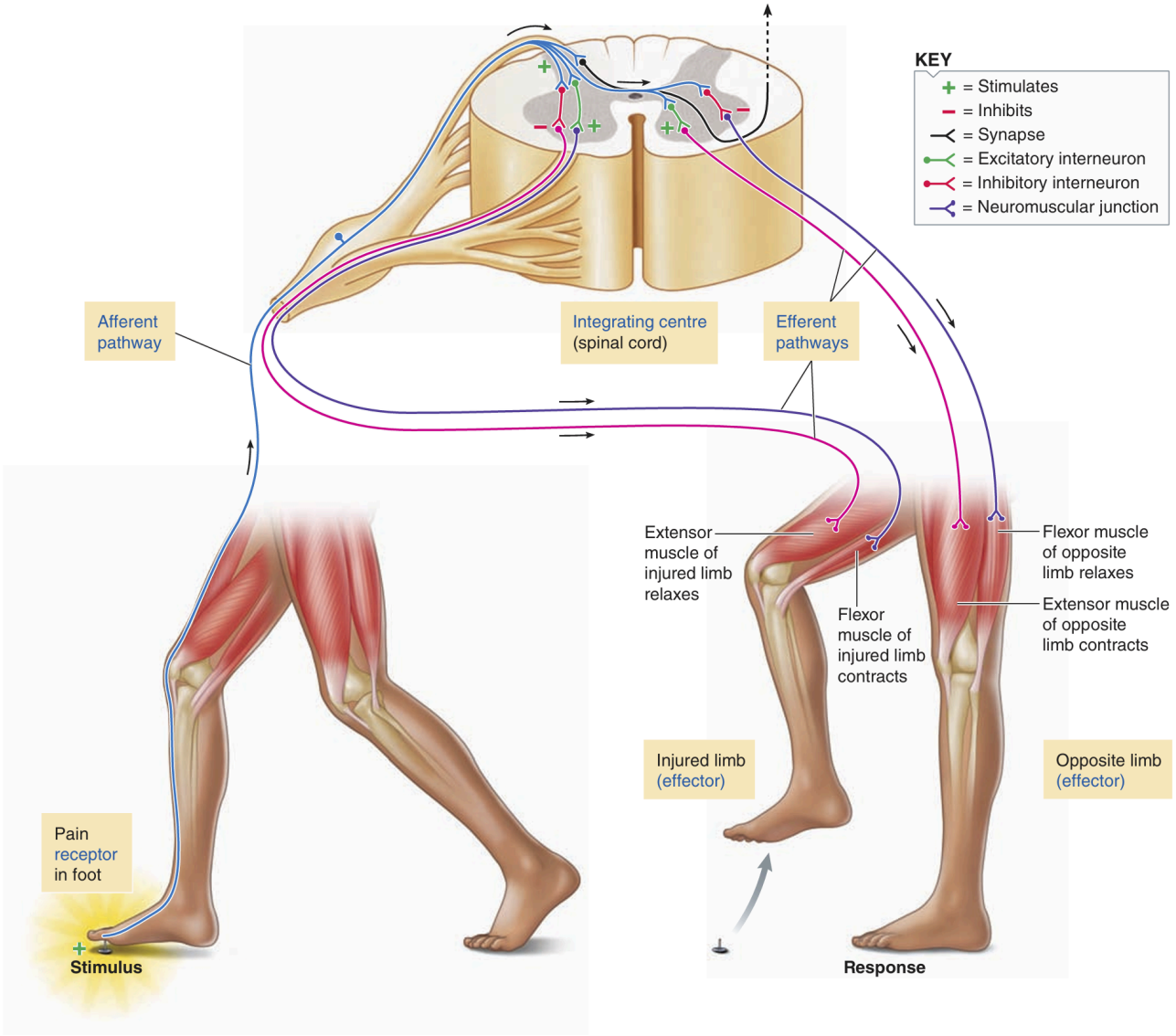
3 Spinal cord serves as the integrating centre. Here afferent neuron stimulates;
3a excitatory interneurons, which stimulate motor neurons to biceps;
3b inhibitory interneurons, which inhibit motor neurons to triceps;
3c interneurons that are part of ascending pathway to brain.

4 One efferent pathway stimulates biceps to contract. Other efferent pathway leads to relaxation of triceps by preventing counterproductive excitation and contraction of this antagonistic muscle.

5 The biceps and triceps are effectors. Resultant flexion of elbow joint pulls hand away from painful stimulus. This response completes the withdrawal reflex.

6 Events that occur in brain on arrival of signal via ascending pathway, such as awareness of pain, memory storage, and so on, are above and beyond reflex arc.

Crossed Extensor Reflex: If you step on a tack, one leg lifts (withdrawal) while the other leg automatically extends to bear your weight.



› FIGURE 4-17 **The crossed extensor reflex coupled with the withdrawal reflex.** The withdrawal reflex, which causes flexion of the injured extremity to withdraw from a painful stimulus. The crossed extensor reflex, which extends the opposite limb to support the full weight of the body

5.1 Introduction to the Peripheral Nervous System

This section defines the **Peripheral Nervous System (PNS)** as the essential communication bridge between the rest of your body (the periphery) and the **Central Nervous System (CNS)**. It categorizes how information flows in and out of the “command center.”

The **PNS** consists of nerve fibers (using both myelinated and unmyelinated neurons) that serve two primary roles:

- **Afferent Division:** Senses internal and external environmental changes through receptors and sends that information to the CNS.
- **Efferent Division:** Receives instructions from the CNS and carries them out through effector organs.

Efferent Subdivision

Somatic Nervous System

- Control: Conscious and voluntary.
- Effectors: Skeletal muscles.

Autonomic Nervous System

- Control: Subconscious and involuntary.
- Effectors: Various tissues and organs, such as smooth muscle in blood vessels.

5.2 The Peripheral Nervous System: Afferent Division

Afferent neurons (sensory neurons) carry action potentials from peripheral receptors toward the CNS.

- **Anatomy:** They feature a smooth, rounded cell body, a short axon, and a single long dendrite that functions similarly to an axon.
- **Organization:** Thousands of these cell bodies are clustered together outside the spine in a swelling called the **dorsal root ganglion**.
- **Processing:** Stimuli are perceived based on the specific location in the brain that receives the signal, with virtually no delay in transmission.

The Fight-or-Flight Response

This acute stress response demonstrates the "symphony" between nervous system divisions. When a threat (like a bear) is perceived:

- **Energy Diversion:** Non-essential functions slow down while vital survival functions accelerate.
- **Afferent Contributions:** The senses provide the data needed for decision-making (hearing rustling, seeing the bear).
- **Physiological Shifts:** Eyes use **accommodation** for focus, pupils dilate to let in light, and the brain processes distance to determine whether to fight or flee.

Somatosensation and Receptors

The somatosensory system consists of receptors (PNS) and processing centres (CNS) that create sensory modalities (e.g. pain, temperature), primarily within the **parietal lobes** of the cerebral cortex. It relies on four main receptor types:

1. **Thermoreceptors:** Detect temperature.
2. **Mechanoreceptors:** Detect mechanical energy/pressure.
3. **Photoreceptors:** Detect light.
4. **Chemoreceptors:** Detect specific chemicals.

Sensation: Internal vs. External

Information is categorized by whether it reaches our conscious mind based on the two pathways:

- **Visceral Afferents (Internal):** Monitor subconscious data like blood pressure or CO_2 levels. These are vital for homeostasis, though we may become aware of them during pain.
- **Sensory Afferents (External/Surface):** Reach conscious awareness and are split into:
    1. **Somatic Sensation:** "Body sense" from the skin (somaesthetic) and joints/muscles (proprioception).
    2. **Special Senses:** Vision, hearing, taste, smell, and equilibrium.

Perception: The Brain's Interpretation

Perception is the conscious interpretation of the world, but it is **not a direct replica of reality**. It is limited by:

- **Sensitivity:** We lack receptors for many energy forms (e.g., magnetic forces, X-rays).
- **High-Fidelity Gaps:** The brain accentuates some features while suppressing others.
- **Cognitive Manipulation:** The cerebral cortex sifts through data and compares it to memories. It often “completes the picture” or distorts info to make sense of it, as seen in optical illusions.

5.3 Receptor Physiology

Transduction: Turning Energy into Signals

Receptors are specialized structures at the peripheral endings of afferent neurons. Because the CNS only understands electrical language, receptors must perform **transduction**: converting various energy forms (light, heat, sound) into electrical signals.

- **Stimulus:** A detectable change that meets a minimum **threshold**.
- **Adequate Stimulus:** Each type of receptor is specialized to respond best to one type of energy, called the adequate stimulus (e.g. light for the eye in photoreceptors).
- **Receptor Potential:** A graded potential (depolarization) caused by a change in ion permeability (usually Na^+ influx).
- Sensation depends on **modality** of receptor stimulated - e.g. when you hit eye, you see stars because it stimulates photoreceptors

Types of Receptors

- **Photoreceptors** are responsive to visible wavelengths of light
- **Mechanoreceptors** are sensitive to mechanical energy.
    - Examples include **skeletal muscle receptors** sensitive to stretch; the receptors in the **ear** that contain fine hair cells, which bend as a result of sound waves; blood pressure—monitoring **baroreceptors**; and **osmoreceptors**, which respond to stretch. Other stimuli sensed are vibration and acceleration.
- **Thermoreceptors** are sensitive to varying amounts of heat.
- **Chemoreceptors** are sensitive to specific chemicals.
    - Include the receptors for smell and taste, the chemical content of the digestive tract, and oxygen and carbon dioxide in the blood.

Receptors can either be **(1) specialized ending of afferent neuron**, or **(2) separate receptor cell closely associated with peripheral ending of neuron**. Stimulation of the receptor alters membrane permeability by causing a nonselective opening of small ion channels. Electrochemical driving force is larger for sodium (Na^+) than other small ions at resting potential, for predominant effect is inward flux of Na^+ which depolarizes the receptor membrane.

The local depolarizing change in potential is called:

- **Generator Potential:** Occurs if the receptor is a specialized ending of the afferent neuron.
- **Receptor Potential:** Occurs in a separate receptor cell that releases a neurotransmitter to the neuron.

- **Initiation:** Unlike other neurons, afferent action potentials start at the **peripheral end** of the fiber, not the axon hillock.

Receptor potentials are graded potentials, so there is no refractory period.

Receptor Potentials and Action Potentials

The goal of any stimulus is to **open Na^+ channels** to depolarize the neuron. How this happens depends on the structure of the receptor:

Mechanism A: Separate Receptor Cells

- **Process:** The stimulus hits a separate cell (not the neuron itself), causing a receptor potential. This triggers the cell to release a **chemical messenger** (neurotransmitter).
- **Result:** The messenger diffuses across a space and binds to the afferent neuron, opening **chemically gated Na^+ channels**.

Mechanism B: Specialized Afferent Endings

- **Process:** Receptor is the "tip" of the neuron itself. The stimulus causes a generator potential directly at the tip.
- **Result: Local current flow** moves from the tip to the adjacent membrane, opening **voltage-gated Na^+ channels**.

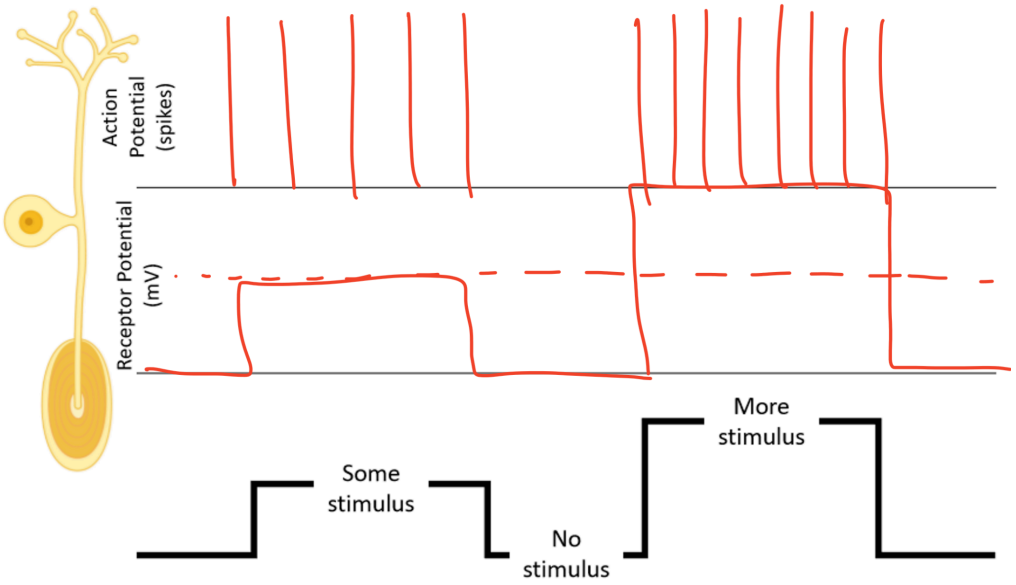
In both cases, if the Na^+ influx is strong enough to reach the **threshold**, an action potential is born.

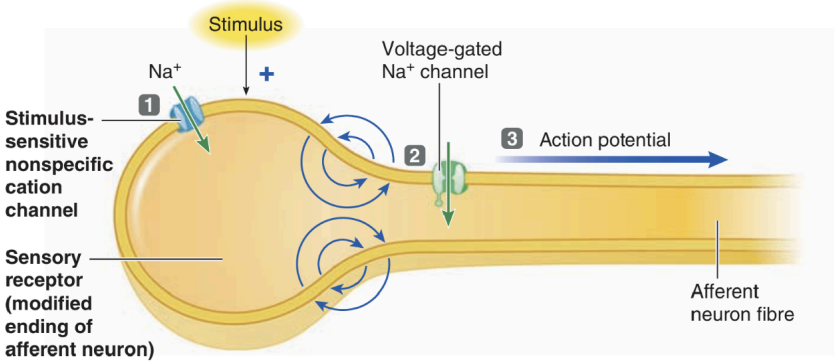
- **Efferent/Interneurons:** Action potentials start at the **axon hillock** near the cell body.
- **Afferent neurons:** Action potential starts at the **peripheral end** (the end farthest from the CNS), right next to the receptor. It then travels the long distance toward the cell body and spinal cord.

Intensity of stimulus is reflected by: **(1) magnitude of receptor potential** (higher frequency of action potentials) and **(2) area stimulated**.

Frequency Coding:

The larger the receptor potential, the greater the frequency of action potentials



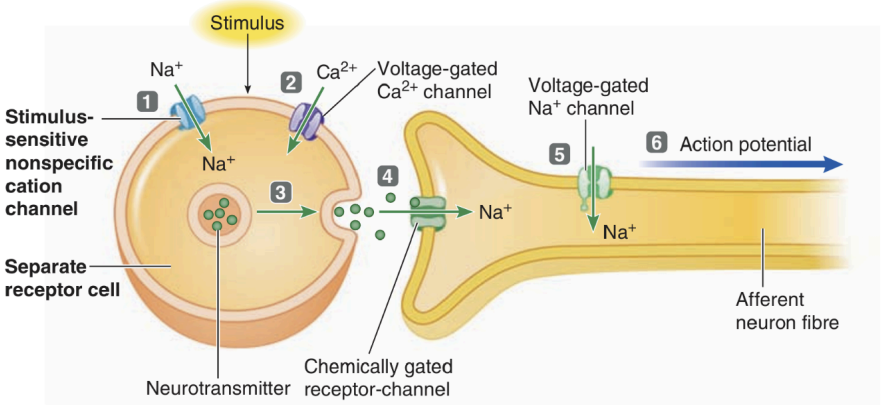


(a) Receptor potential in specialized afferent ending

1 In sensory receptors that are specialized afferent neuron endings, stimulus opens stimulus-sensitive channels, permitting net Na^+ entry that produces receptor potential.

2 Local current flow between depolarized receptor ending and adjacent region opens voltage-gated Na^+ channels.

3 Na^+ entry initiates action potential in afferent fibre that self-propagates to CNS.



(b) Receptor potential in separate receptor cell

1 In sensory receptors that are separate cells, stimulus opens stimulus-sensitive channels, permitting net Na^+ entry that produces receptor potential.

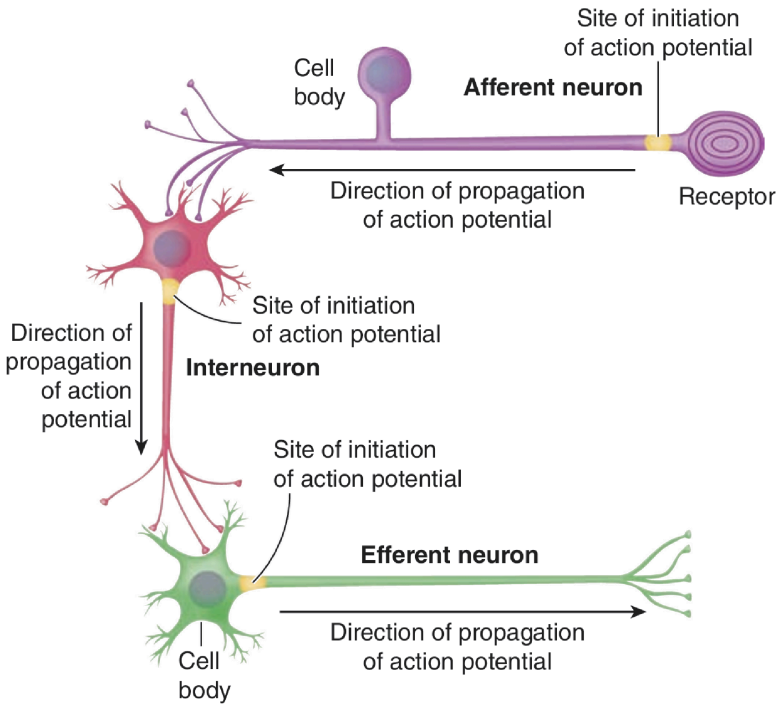
2 This local depolarization opens voltage-gated Ca^{2+} channels.

3 Ca^{2+} entry triggers exocytosis of neurotransmitter.

4 Neurotransmitter binding opens chemically gated receptor channels at afferent ending, permitting net Na^+ entry.

5 Resultant depolarization opens voltage-gated Na^+ channels in adjacent region.

6 Na^+ entry initiates action potential in afferent fibre that self-propagates to CNS.



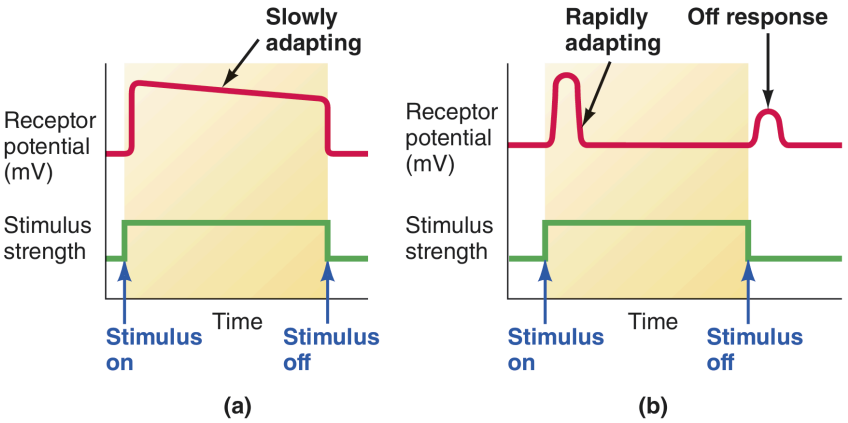
Tonic vs. Phasic

Tonic receptors

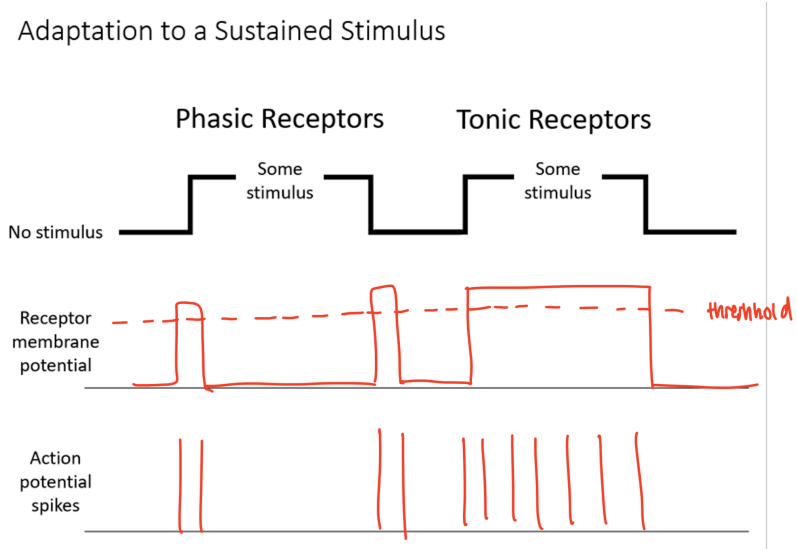
- Adapts slowly/does not adapt
- **Maintain information about stimulus**
- Examples: muscle stretch receptors (monitor muscle length), joint proprioceptors (degree of joint flexion)
- CNS must continually get information about these information -> receptors should not adapt to stimulus but continue to generate action potentials to relay information to CNS

Phasic receptors

- Rapidly adapting receptors
- When stimulus is removed, the receptor responds with slight depolarization called **off-response**
- Useful where it is important to signal a **change in stimulus intensity** rather than to relay status quo information
- Example: tactile (touch) receptors detect changes in pressure on skin surface -> you don't notice after you put clothes on that the clothes are still touching you



› **FIGURE 5-4 Tonic and phasic receptors.** (a) Tonic receptor. This receptor type does not adapt at all or adapts slowly to a sustained stimulus, and thereby provides continuous information about the stimulus. (b) Phasic receptor. This receptor type adapts rapidly to a sustained stimulus and frequently exhibits an off response when the stimulus is removed. Thus, the receptor signals changes in stimulus intensity rather than relaying status quo information.



Somatosensory pathways

On reaching the spinal cord, afferent information has two possible destinites:

(1) **Become part of a reflex arc**, bringing about an appropriate effector response

(2) **Relayed upward to the brain via ascending pathways** for further processing and possible conscious awareness

Somatosensory pathways: Each receptor is "hard-wired" to a specific path leading to a specific spot in the somatosensory cortex. The chains of neurons are called **labelled lines.**

- **Modality:** Because of these dedicated paths, the sensation depends on which line is activated, not how it was activated.
- If you trip a pain fiber with heat, chemicals, or a pinch, your brain always interprets the signal as "pain" because that specific "line" was used.

First-order sensory neuron: afferent neuron with peripheral receptor that first detects stimulus

Second-order sensory neuron: spinal cord or medullary, depending on which sensory pathway is involved

Third-order sensory neuron: in the thalamus, relays to cortex

TABLE 5-1 Coding of Sensory Information

Stimulus Property	Mechanism of Coding
Type of Stimulus (stimulus modality)	Distinguished by the type of receptor activated and the specific pathway over which this information is transmitted to a particular area of the cerebral cortex
Location of Stimulus	Distinguished by the location of the activated receptive field and the pathway that is subsequently activated to transmit this information to the area of the somatosensory cortex representing that particular location
Intensity of Stimulus (stimulus strength)	Distinguished by the frequency of action potentials initiated in an activated afferent neuron and the number of receptors (and afferent neurons) activated

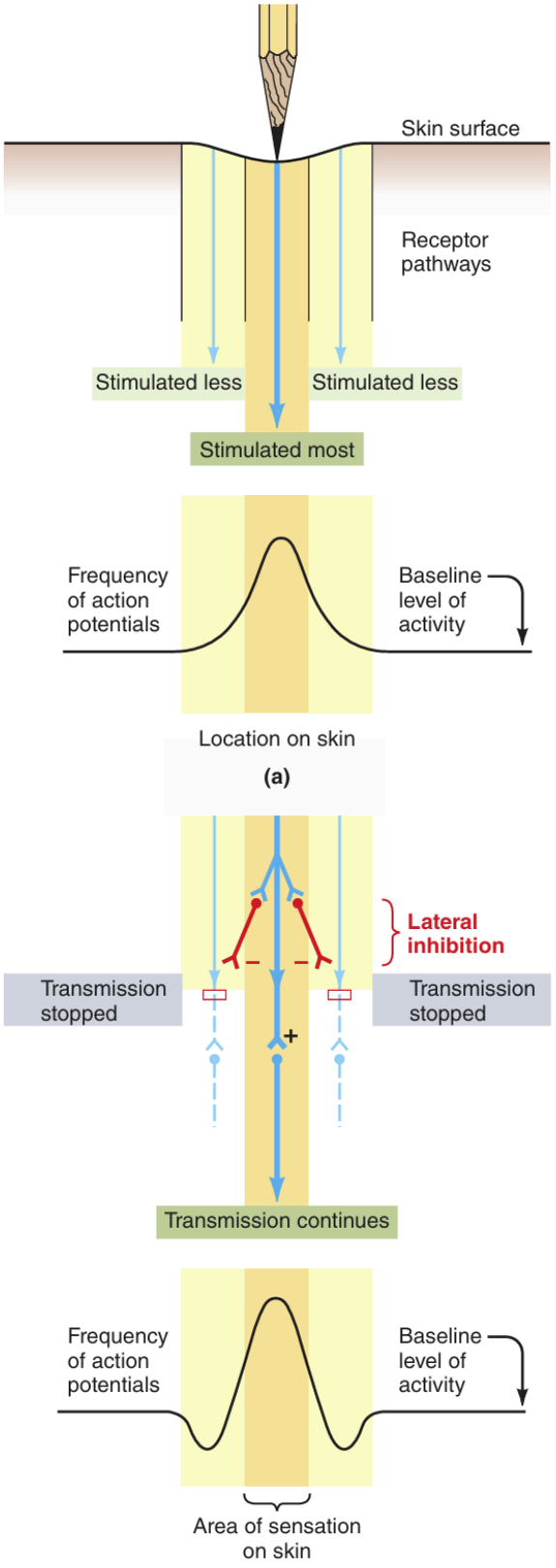
Acuity, receptive field size, and lateral inhibition

Receptive field: Each somatosensory neuron responds to stimulus information within this **circumscribed region of skin surface**.

Acuity: the **discriminative ability**

- Smaller receptive field = larger density = higher acuity
- **Lateral inhibition:** most strongly activated signal pathway originating from centre of stimulus area inhibits the less excited pathways from fringe area
    - Example: poke yourself with pencil, you only feel the most intense stimulus at the pencil point, not the surrounding
    - Caused by inhibitory interneurons that pass laterally between ascending fibres serving neighbouring receptive fields
    - Touch and vision has greatest lateral inhibition

Mechanoreceptors



5.9 Autonomic Nervous System

Somatic vs. Autonomic

- **Somatic** is **conscious**: effectors are skeletal muscles only
- **Autonomic** is **subconscious and involuntary**: effectors are smooth muscles, cardiac muscles, glands
    - Controlled by hypothalamus, brain stem, and spinal cord
    - Composed of the **sympathetic** and **parasympathetic** systems, which maintain a dynamic balance.
    - **Efferent nerve pathway** extending from CNS to innervated organ with a **two-neuron chain**

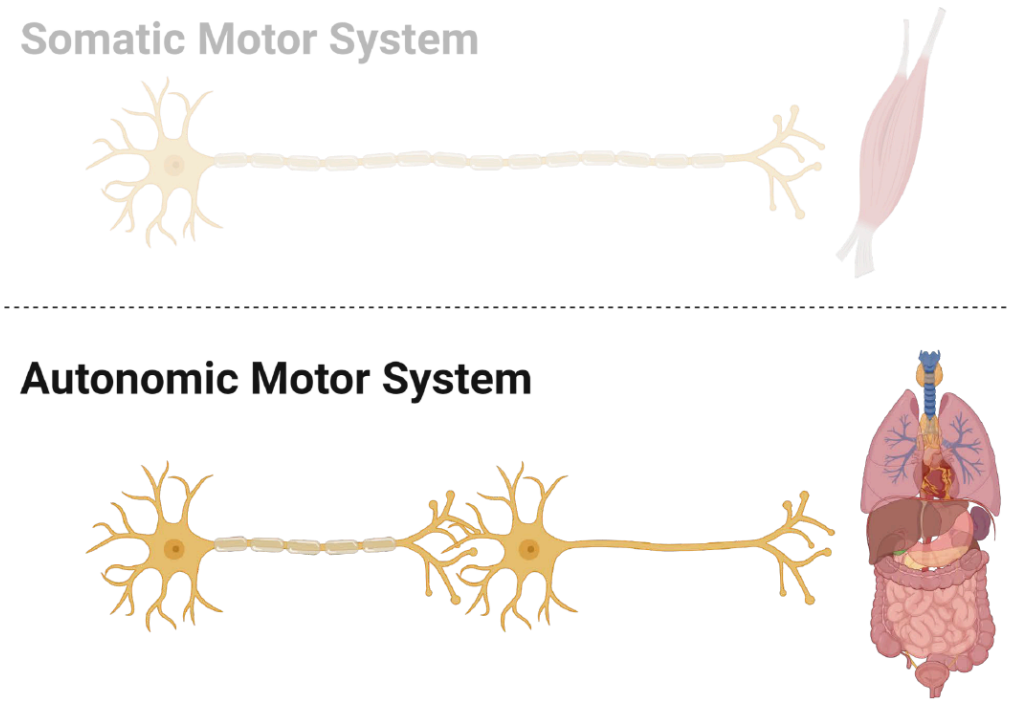
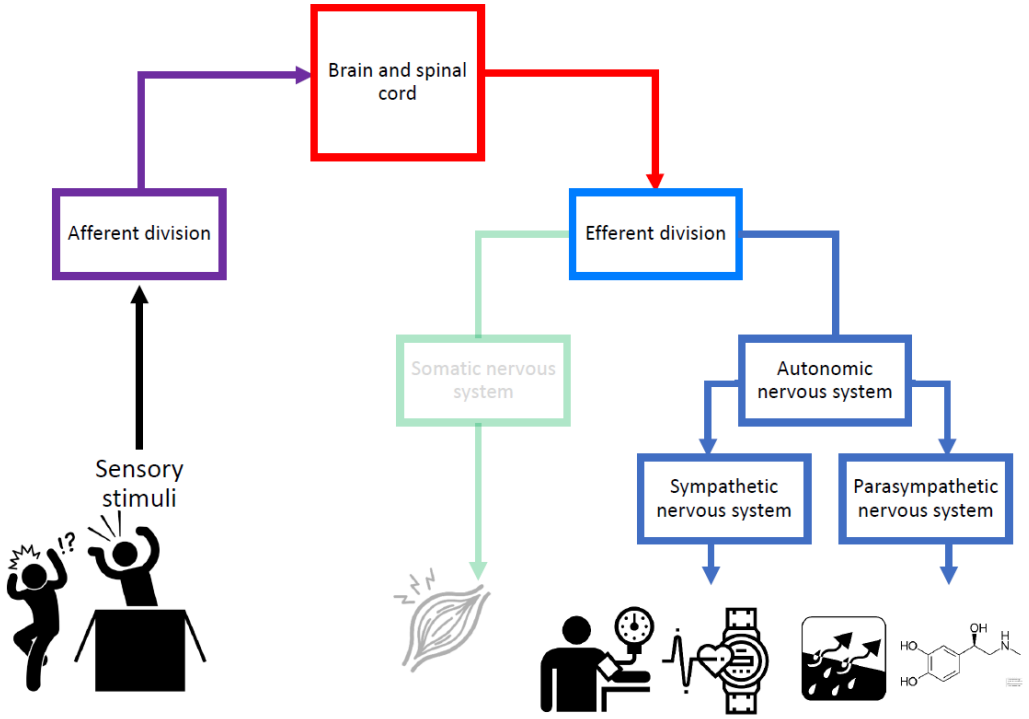


TABLE 5-10 Comparison of the Autonomic Nervous System and the Somatic Nervous System

Feature	Autonomic Nervous System	Somatic Nervous System
Site of Origin	Brain or lateral horn of spinal cord	Ventral horn of spinal cord for most; those supplying muscles in head originate in brain
Number of Neurons from Origin in CNS to Effector Organ	Two-neuron chain (preganglionic and postganglionic)	Single neuron (motor neuron)
Organs Innervated	Cardiac muscle, smooth muscle, exocrine and some endocrine glands	Skeletal muscle
Type of Innervation	Most effector organs dually innervated by the two antagonistic branches of this system (sympathetic and parasympathetic)	Effector organs innervated only by motor neurons
Neurotransmitter at Effector Organs	May be acetylcholine (parasympathetic terminals) or norepinephrine (sympathetic terminals)	Only acetylcholine
Effects on Effector Organs	Either stimulation or inhibition (antagonistic actions of two branches)	Stimulation only (inhibition possible only centrally through IPSPs on dendrites and cell body of motor neuron)
Types of Control	Under involuntary control	Subject to voluntary control; much activity subconsciously coordinated
Higher Centres Involved in Control	Spinal cord, medulla, hypothalamus, prefrontal association cortex	Spinal cord, motor cortex, basal nuclei, cerebellum, brain stem



Autonomic Nerve Pathways:

1. The axon, **preganglionic fibre (myelinated)**, located **in CNS**, synapses with the cell body of the second neuron, which lies within a **ganglion**.

a. Ganglion = cluster of neuronal cell bodies outside CNS

2. Axon of second neuron, **postganglionic fibre (unmyelinated)**, innervates effector organ (smooth muscle, heart)

TABLE 5-9 Distinguishing Features of Sympathetic Nervous System and Parasympathetic Nervous System

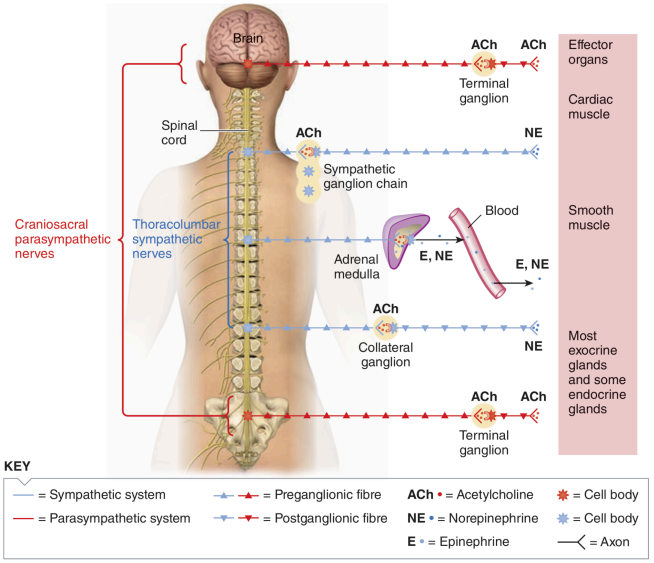
Feature	Sympathetic System	Parasympathetic System
Origin of Preganglionic Fibre	Thoracic and lumbar regions of spinal cord	Brain and sacral region of spinal cord
Origin of Postganglionic Fibre (location of ganglion)	Sympathetic ganglion chain (near spinal cord) or collateral ganglia (about halfway between spinal cord and effector organs)	Terminal ganglia (in or near effector organs)
Length and Type of Fibre	Short cholinergic preganglionic fibres	Long cholinergic preganglionic fibres
	Long adrenergic postganglionic fibres	Short cholinergic postganglionic fibres
Effector Organs Innervated	Cardiac muscle, almost all smooth muscle, most exocrine glands, and some endocrine glands	Cardiac muscle, most smooth muscle, most exocrine glands, and some endocrine glands
Types of Receptors for Neurotransmitters	For preganglionic neurotransmitter: nicotinic	For preganglionic neurotransmitter: nicotinic
	For postganglionic neurotransmitter: α_1, α_2, β_1, β_2	For postganglionic neurotransmitter: muscarinic
Dominance	Dominates in emergency fight-or-flight situations; prepares body for strenuous physical activity	Dominates in quiet, relaxed situations; promotes "general housekeeping" activities, such as digestion

Sympathetic: short preganglionic, long postganglionic

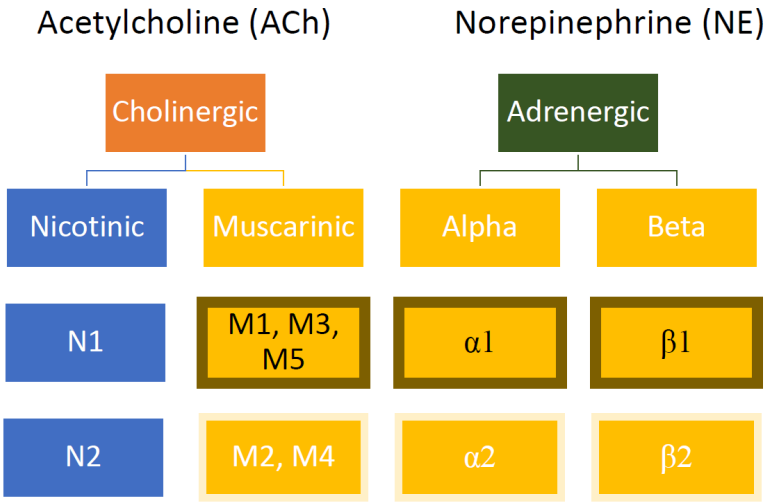
- Most synapse with postganglionic neurons in **sympathetic ganglion chain** on side of spinal cord
- Some pass through ganglion chain without synapsing, later synapse in **collateral ganglia** halfway between CNS and organs

Parasympathetic: long preganglionic, short postganglionic

- Preganglionic fibres from cranial (brain) and sacral (lower spinal cord) of CNS
- Do not end until reaching **terminal ganglia** near effector organs

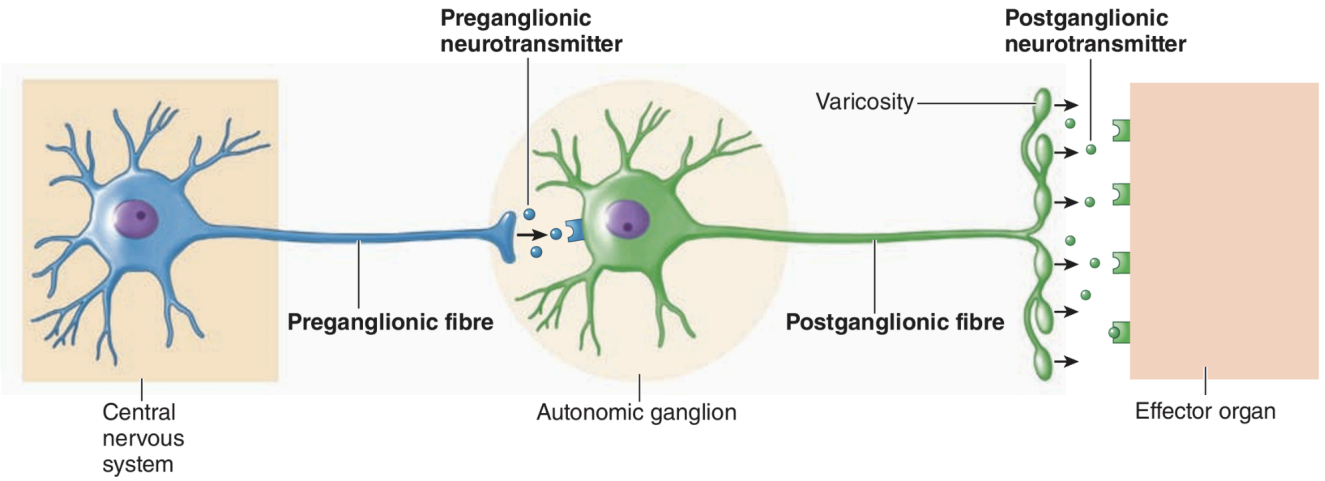


Neurotransmitters



The chemical signals used by the ANS vary by branch and stage of the pathway:

- **Preganglionic**: Both systems release **ACh (Cholinergic fibers)**.
- **Postganglionic**:
  - Parasympathetic: Releases **ACh (Cholinergic fibers)**.
  - Sympathetic: Most release **Norepinephrine (Adrenergic fibers)**.
- Postganglionic autonomic fibers do not end in a single terminal. Instead, they have numerous swellings called **varicosities** that release neurotransmitters **over a wide area** of the organ rather than single cells
  - Allows the signal to influence an entire organ



Dual Innervations

Most internal organs are dually innervated, meaning they **receive fibers from both systems**. Sympathetic and parasympathetic usually has opposite effects (e.g. sympathetic increases heart rate, parasympathetic decreases; sympathetic slows down movement in digestive tract, parasypathetic enhances it)

- **Autonomic Tone:** Under normal conditions, both systems are partially active, providing a "baseline" level of stimulation.
- **Sympathetic Dominance:** Occurs when the sympathetic rate increases and parasympathetic decreases (e.g. increased heart rate during stress).
- **Parasympathetic Dominance:** Occurs when the parasympathetic rate increases and sympathetic decreases (e.g. enhanced digestion during rest).

Types of Receptors

1. **Cholingeric**

   Nicotinic:

   - Found on postganglionic cell bodies in all autonomic ganglia
   - **Respond to acetylcholine** from both sympathetic and parasymathetic **preganglionic** fibres

   Muscarinic:

   - Found on effector cell membranes (smooth, cardiac muscle, glands)
   - Bind with acetylcholine released from parasympathetic postganglionic fibres

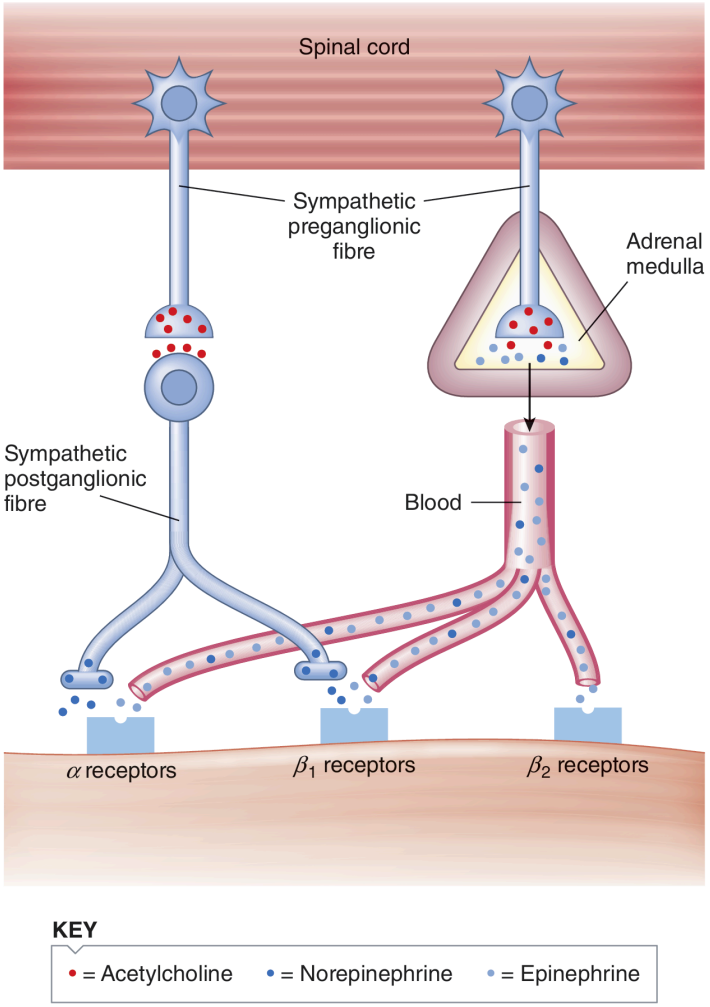
2. **Adrenergic**

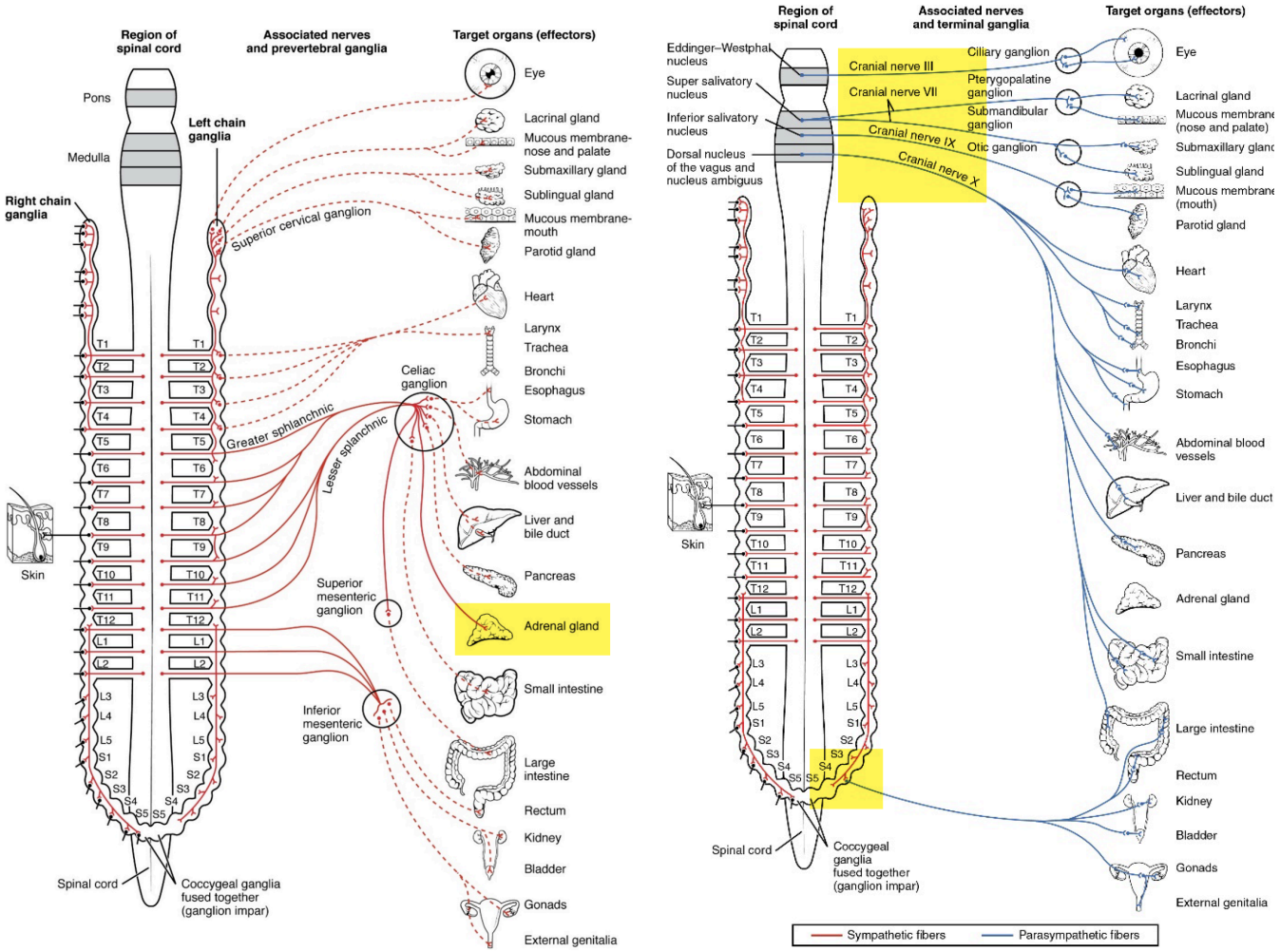
   Beta:

   - **B2:** bind primarily with epinephrine, **inhibitory** (e.g. arteriorlar/bronchiolar dilation caused by relaxation of smooth muscle)
   - **B1:** norepinephrine and epinephrine, **excitatory** (e.g. increased rate and force of cardiac contraction)

   Alpha:

   - Greater sensitivity to norepinephrine than to epinephrine
   - **A2: inhibitory** response in effector organ (e.g. decreased smooth muscle contraction in digestive tract)
   - **A1: excitatory** response in effector organ, **most sympathetic** (e.g. artieriolar constriction caused by increased contraction of smooth muscle)


Spinal cord
Sympathetic preganglionic fibre
Adrenal medulla
Sympathetic postganglionic fibre
Blood
α receptors
β1 receptors
β2 receptors
KEY
= Acetylcholine
= Norepinephrine
= Epinephrine



Only sympathetic nerves to:

1. Adrenal gland
2. Sweat (ecrine glands)
3. Blood vessels

Parasympathetic nerves from:

1. Cranial
2. Vagus (cranial nerve X)
3. Sacral

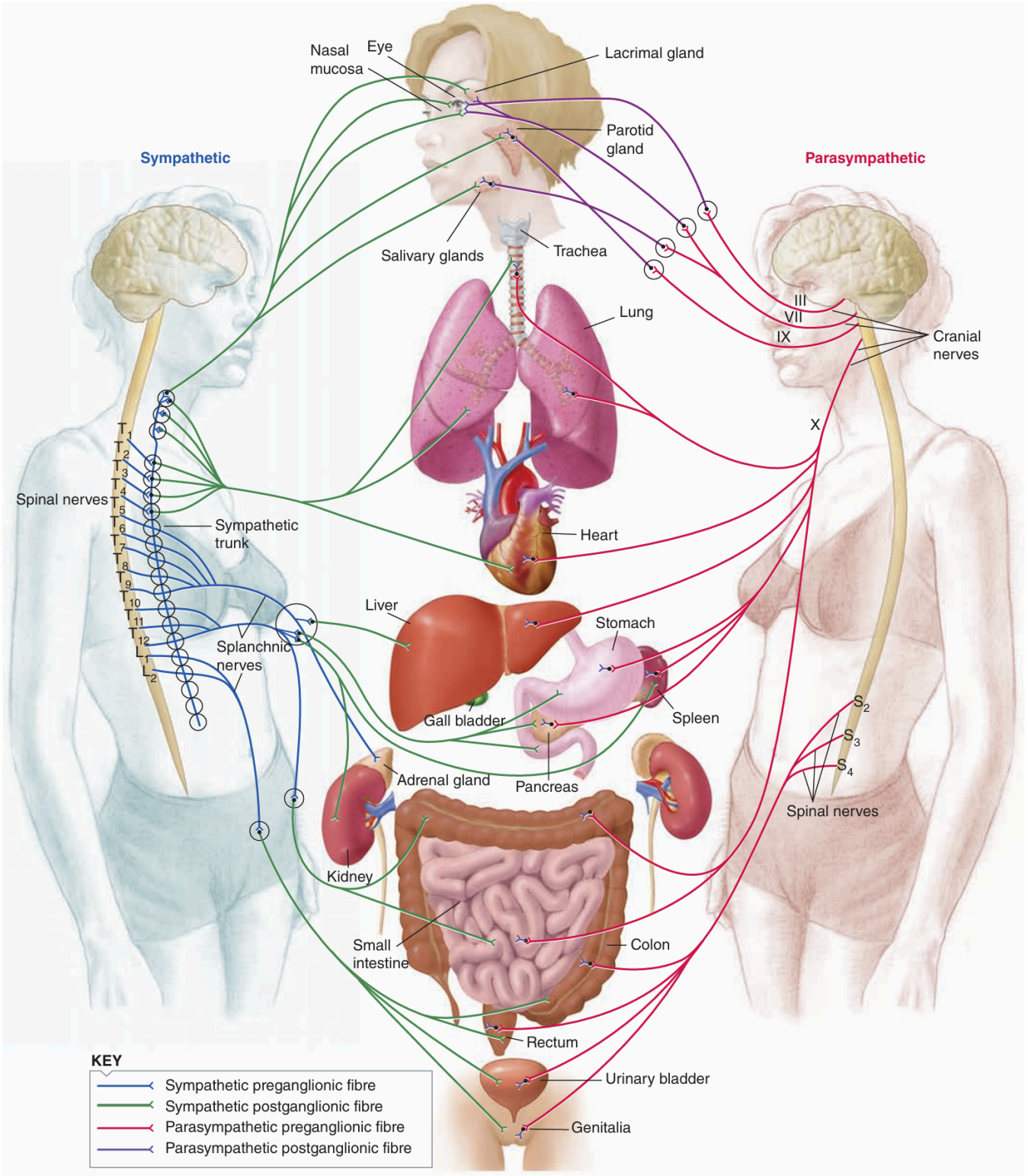

Sympathetic
Parasympathetic
Nasal mucosa
Eye
Lacrimal gland
Parotid gland
Salivary glands
Trachea
Lung
III
VII
IX
Cranial nerves
X
Spinal nerves
Sympathetic trunk
Heart
Liver
Stomach
Splanchnic nerves
Gall bladder
Spleen
Adrenal gland
Pancreas
Kidney
Small intestine
Colon
Rectum
Urinary bladder
Genitalia
S2
S3
S4
Spinal nerves
KEY
Sympathetic preganglionic fibre
Sympathetic postganglionic fibre
Parasympathetic preganglionic fibre
Parasympathetic postganglionic fibre

TABLE 5-8 Effects of Autonomic Nervous System on Various Organs

Organ	Effect of Sympathetic Stimulation	Effect of Parasympathetic Stimulation
Heart	Increased rate, increased force of contraction (of whole heart)	Decreased rate, decreased force of contraction (of atria only)
Blood Vessels	Constriction	Dilation of vessels supplying the penis and clitoris only
Lungs	Dilation of bronchioles (airways)	Constriction of bronchioles
	Inhibition of mucous secretion	Stimulation of mucous secretion
Digestive Tract	Decreased motility (movement)	Increased motility
	Contraction of sphincters (to prevent forward movement of contents)	Relaxation of sphincters (to permit forward movement of contents)
	Inhibition of digestive secretions	Stimulation of digestive secretions
Urinary Bladder	Relaxation	Contraction (emptying)
Eye	Dilation of pupil	Constriction of pupil
	Adjustment of eye for far vision	Adjustment of eye for near vision
Liver (glycogen stores)	Glycogenolysis (glucose released)	None
Adipose Cells (fat stores)	Lipolysis (fatty acids released)	None
Exocrine Glands		
Exocrine pancreas	Inhibition of pancreatic exocrine secretion	Stimulation of pancreatic exocrine secretion (important for digestion)
Sweat glands	Stimulation of secretion by most sweat glands	Stimulation of secretion by some sweat glands
Salivary glands	Stimulation of small volume of thick saliva rich in mucus	Stimulation of large volume of watery saliva rich in enzymes
Endocrine Glands		
Adrenal medulla	Stimulation of epinephrine and norepinephrine secretion	None
Endocrine pancreas	Inhibition of insulin secretion; stimulation of glucagon secretion	Stimulation of insulin and glucagon secretion
Genitals	Ejaculation and orgasmic contractions (males); orgasmic contractions (females)	Erection (caused by dilation of blood vessels in penis [male] and clitoris [female])
Brain Activity	Increased alertness	None

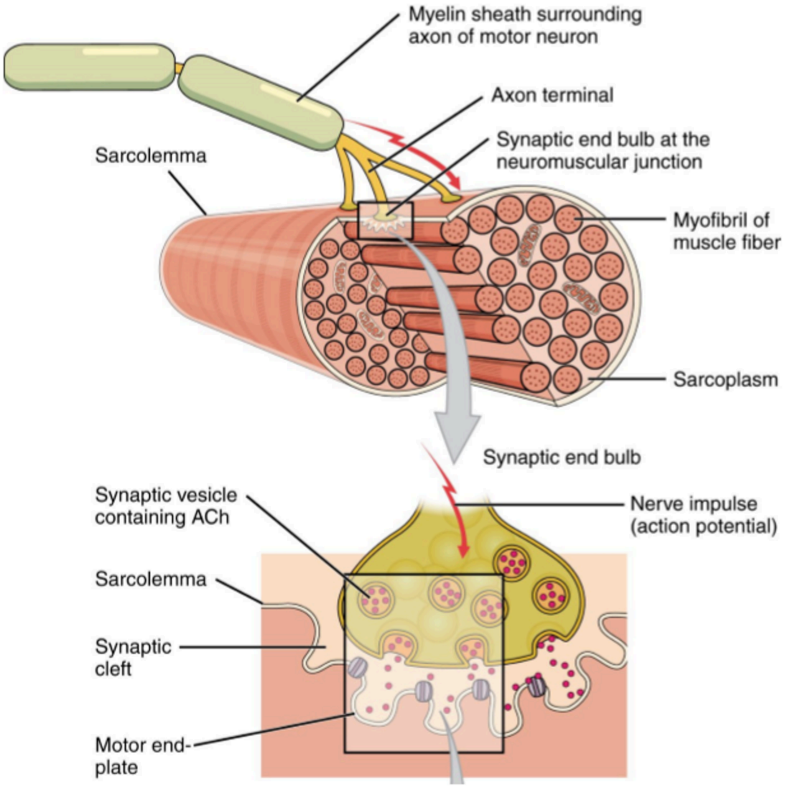
5.10 Somatic Nervous System

Motor neurons and skeletal muscle

- Motor neuron axon terminals release **acetylchline** -> excitation and contraction of muscle cells
- Motor neurons **only stimulate** skeletal muscle, not inhibit (unlike autonomic fibres and effector organs)
    - Inhibition of skeletal muscle only done through CNS inhibitory synaptic input to dendrites and cell bodies of motor neurons to that particular muscle
- Motor neurons are the **final common pathway**
    - Only way other parts of nervous system can influence skeletal muscle activity is by acting on these motor neurons
    - Level of activity depends on relative balance of EPSPs and IPSPs by presynaptic inputs
- Skeletal muscle activity is mostly subconsciously controlled
- **Axon is continuous from CNS to skeletal muscle** (not two-neuron chain)

5.11 Neuromuscular Junction

- Action potential of a motor neuron is rapidly propagated from the cell body within CNS to skeletal muscle along the myelinated axon (efferent fibre) of neuron.
- As it approaches muscle, it divides into terminal branches and loses myelin sheath -> each forms a **neuromuscular junction** with a muscle cell
- Axon terminal is enlarged into knoblike structure, the **terminal button**, which fits into a cleft in the underlying muscle fibre



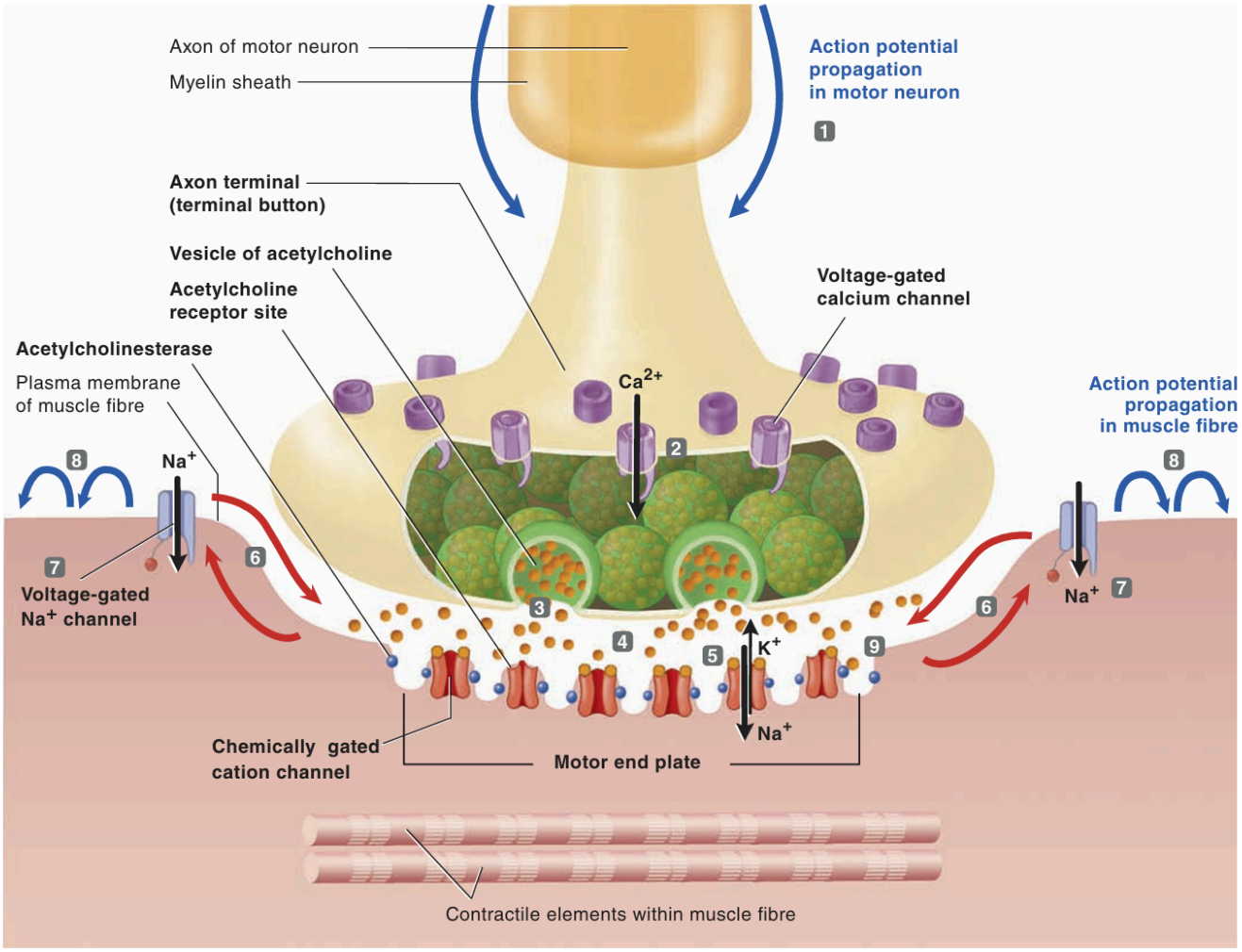
Acetylcholine

- Synaptic cleft between nerve and muscle cells are too large for electrical transmission of impulse
- Neurotransmitter **acetylcholine (ACh)** is used to carry signal between neuron terminal and muscle fibre

Each terminal button contains thousands of vesicles with ACh

1. Propagation of action potential ot axon terminal triggers opening of voltage-gated **Ca^{2+} channels** in terminal button
2. Permits Ca^{2+} to diffuse into terminal button down concentration gradient
3. Releases ACh by exocytosis from vesicles in the cleft
4. ACh (nicotinic) diffuses across cleft, **binds with specific receptor sites** with specialized membrane proteins unique to motor end-plate portion of muscle fibre membrane
5. Binding **opens nonspecfic cation channels** -> larger movement of Na^+ into muscle cell compared to smaller movement of K^+ outward

6. Results in **end-plate potential (EPP)** -> local current flow between depolarized end plate and adjacent membrane
    a. Similar to EPSP (but larger), also a graded potential
7. Opens voltage-gated Na^+ channels in adjacent membrane
    a. **Motor end-plate region itself does not have a threshold potential**, so action potential cannot generate at this site -> must go to adjacent membrane
8. Reduces potential to threshold -> action potential propagated
9. ACh destroyed by acetylcholinesterase, enzyme on motor end-plate membrane, terminating response



1 An action potential in a motor neuron is propagated to the axon terminal (terminal button).

2 This local action potential triggers the opening of voltage-gated Ca^{2+} channels and the subsequent entry of Ca^{2+} into the terminal button.

3 Ca^{2+} triggers the release of acetylcholine (ACh) by exocytosis from a portion of the vesicles.

4 ACh diffuses across the space separating the nerve and muscle cells and binds with receptor channels specific for it on the motor end plate of the muscle cell membrane.

5 This binding brings about the opening of these nonspecific cation channels, leading to a relatively large movement of Na^+ into the muscle cell compared to a smaller movement of K^+ outward.

6 The result is an end-plate potential. Local current flow occurs between the depolarized end plate and the adjacent membrane.

7 This local current flow opens voltage-gated Na^+ channels in the adjacent membrane.

8 The resultant Na^+ entry reduces the potential to threshold, initiating an action potential, which is propagated throughout the muscle fibre.

9 ACh is subsequently destroyed by acetylcholinesterase, an enzyme located on the motor end-plate membrane, terminating the muscle cell's response.

Synapse vs. Neuromuscular Junction

Similarities	Differences
• 2 excitable cells separated by narrow cleft with no direct transmission of electrical activity • Axon terminals store neurotransmitters released by Ca^{2+}-induced exocytosis • Binding of neurotransmitter opens specific channels in membrane, allowing ionic movements that alter membrane potential • Resultant change is graded potential	• Synapse is junction between two neurons; neuromuscular junction is between **motor neuron and skeleton muscle fibre** • 1-to-1 transmission of action potential between motor neuron and muscle fibre, while action potential in postsynaptic neuron usually needs sum of EPSPs • Neuromuscular is always excitatory (EPP), synapse can be excitatory (EPSP) or inhibitory (IPSP) ○ Inhibition of skeletal muscles can only occur in CNS through IPSPs at dendrites and cell body of motor neuron

8.1 Muscle Physiology Introduction

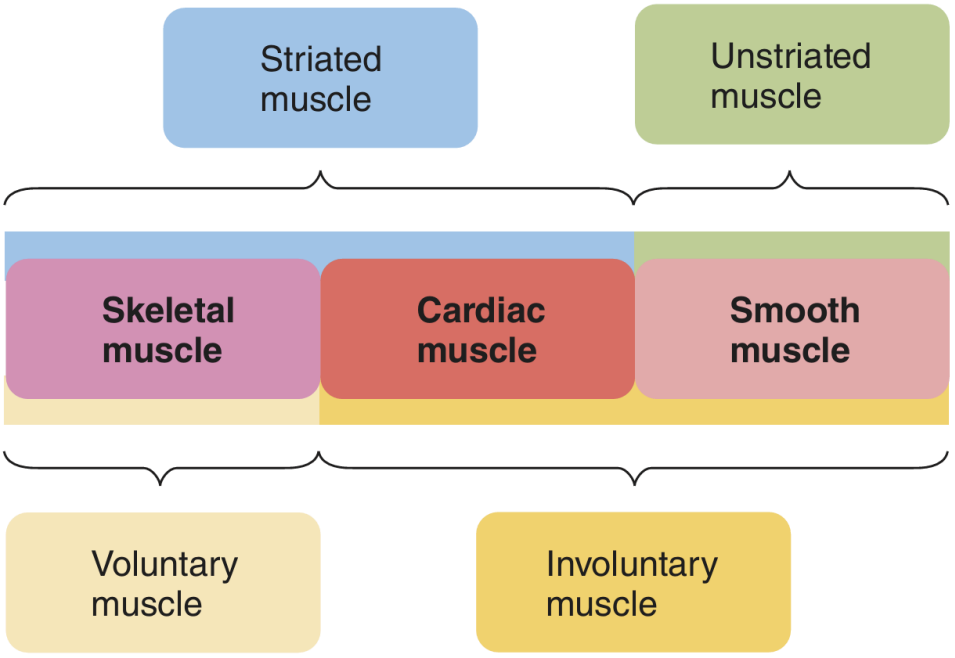
Muscles can develop tension and shorten (contract).

Three types of muscles:

- Skeletal
- Cardiac
- Smooth

Groups of muscles can produce movement. Skeletal muscle makes up 40% body weight in men, 32% in women.

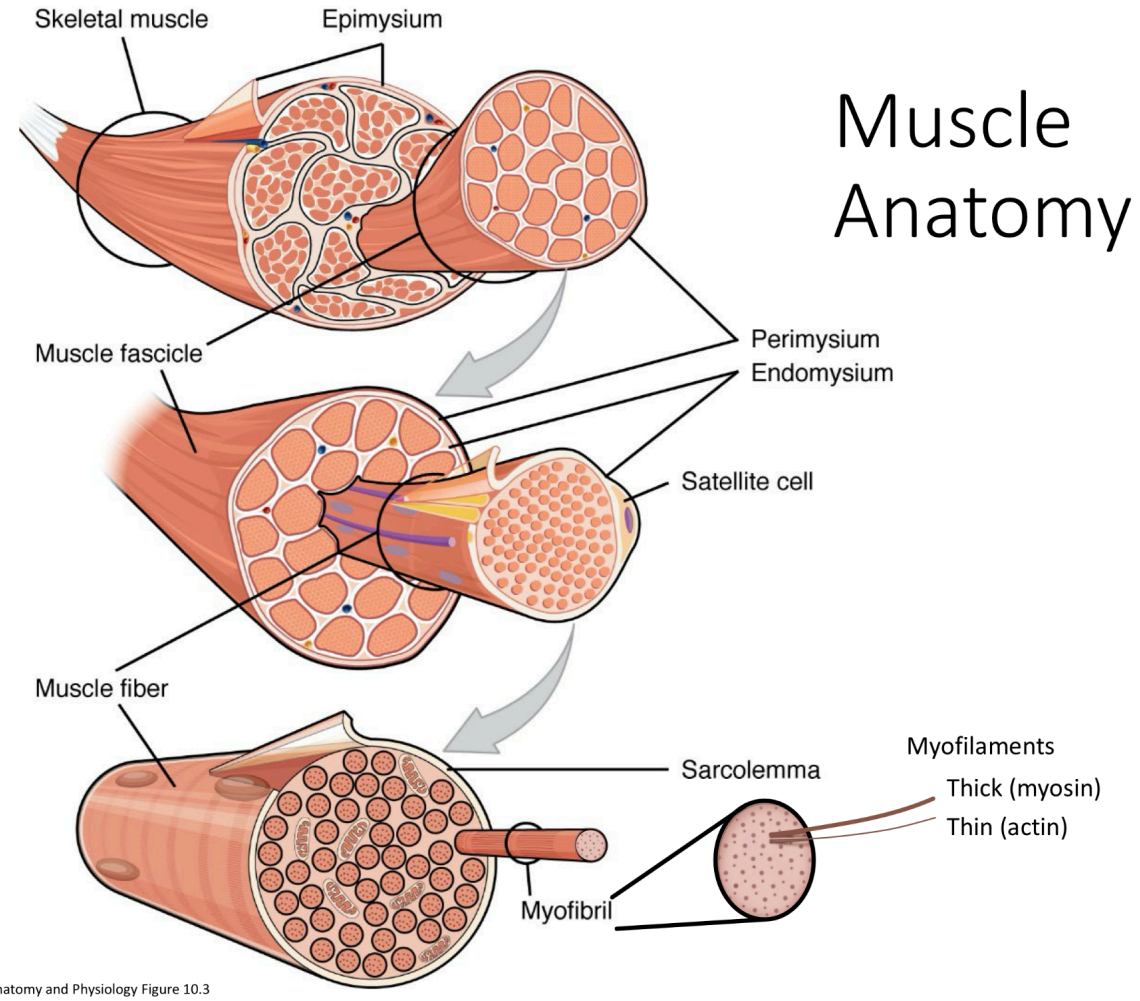
1. **Striated (skeletal, cardiac) vs. unstriated (smooth):** whether alternating dark and light bands can be seen when viewed under microscope
2. **Voluntary (skeletal) vs. involuntary (cardiac, smooth):** whether they are innervated by somatic (voluntary) or autonomic (involuntary)
    a. Skeletal muscle can be voluntarily controlled, but still many activity is subject to involuntary control (e.g. balance)



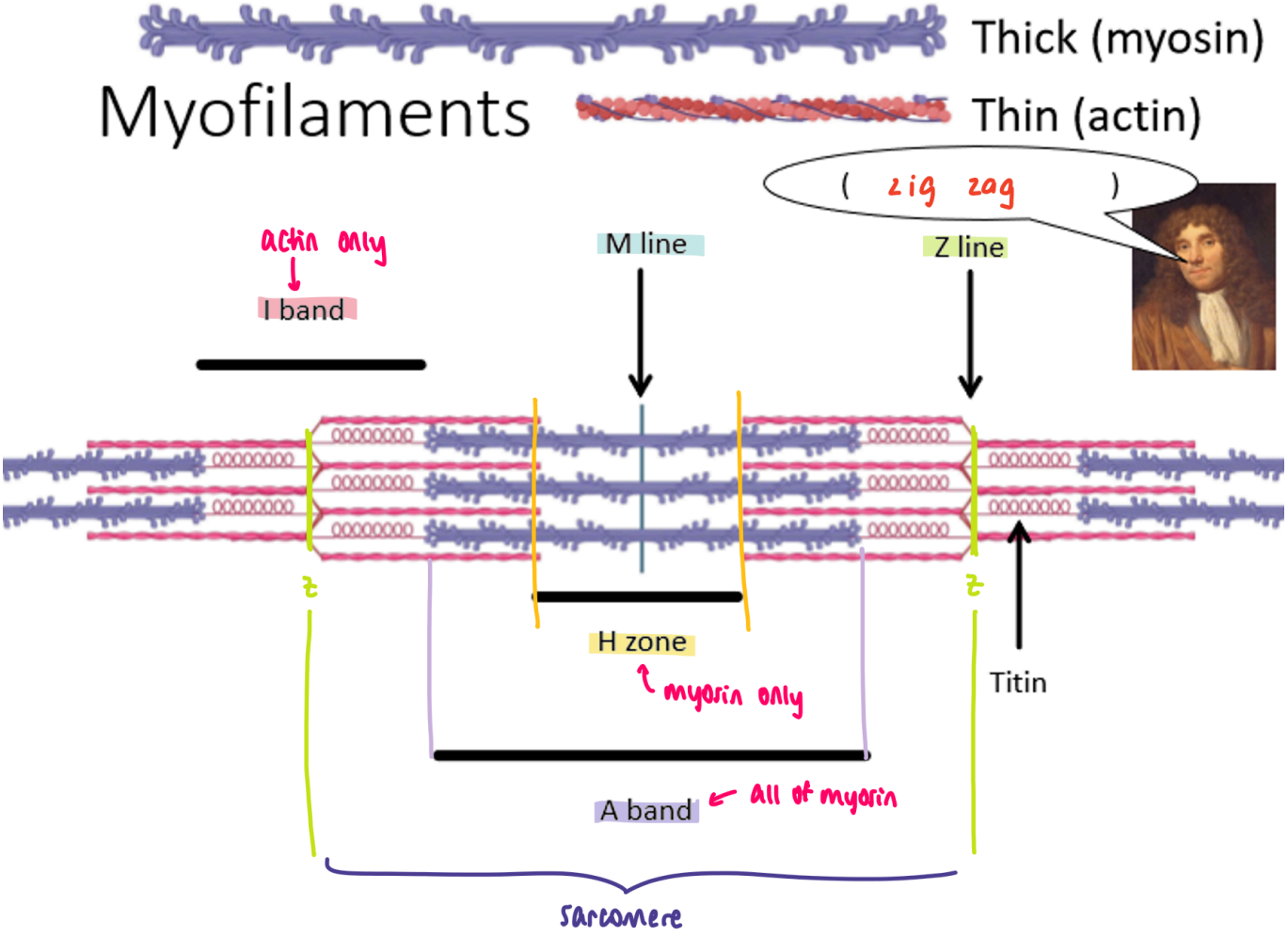
8.2 Structure of Skeletal Muscle

- Skeletal muscle cell is cylinder shaped, diameter 10-100 micrometre, up to 0.76m in length.
- Consists of muscle fibres parallel, bundled with connective tissue

Whole muscle	→	muscle fibre	→	myofibril	→	thick and thin filaments	→	myosin and actin
(an organ)		(cell)		(a specialized intracellular structure)		(cytoskeletal elements)		(protein molecules)



OpenStax Anatomy and Physiology Figure 10.3



Thick:

- **A-band:** All of thick filament, including overlap with thin
- **H-zone:** Lighter area in middle of A band, thick only
- **M-line:** Middle of A band, proteins that hold filaments vertically

Thin:

- **I-band:** Remaining portion of thin filament, not overlapping with thick
- **Z-line:** Middle of I band

Functional unit:

- **Sarcomere:** Between two Z-lines, **functional unit of skeletal muscle**
    - One full A + two halves of I on both sides

Titin:

- Elastic protein that ends both directions from M line to Z line
- Largest protein in body, 30000 amino acids
- (1) stabilizes thick filament relative to thin, (2) spring that augments muscle's elasticity

Cross bridges

- AKA “myosin heads”
- Extend from thick filament toward surrounding thin filament in areas where they overlap

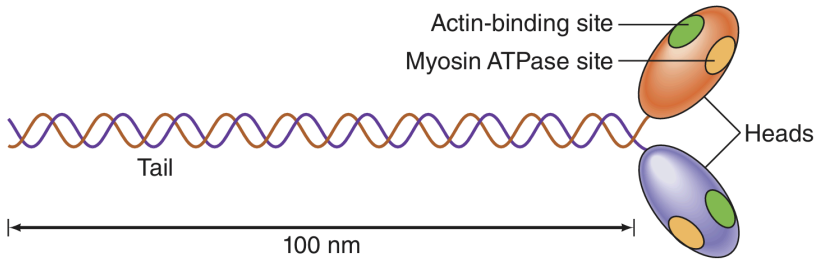
Each cross bridge has two important sites:

(1) Actin-binding site

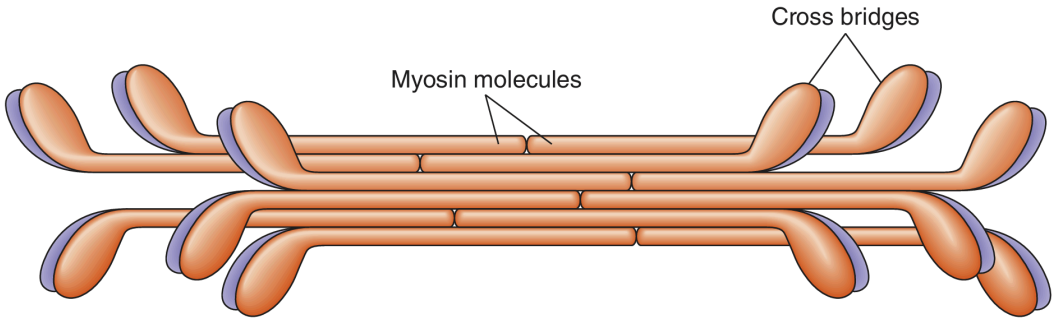
(2) Myosin ATPase: rate of ATPase activity on head of myosin that provides method of muscle fibre typing

Myosin

- Each thick filament has several hundred
- Motor protein, action-based mobility
- Two identical subunits, tails intertwined, staggered array with tails at center, heads protruding outwards at regular intervals



(a) Myosin molecule



(b) Thick filament

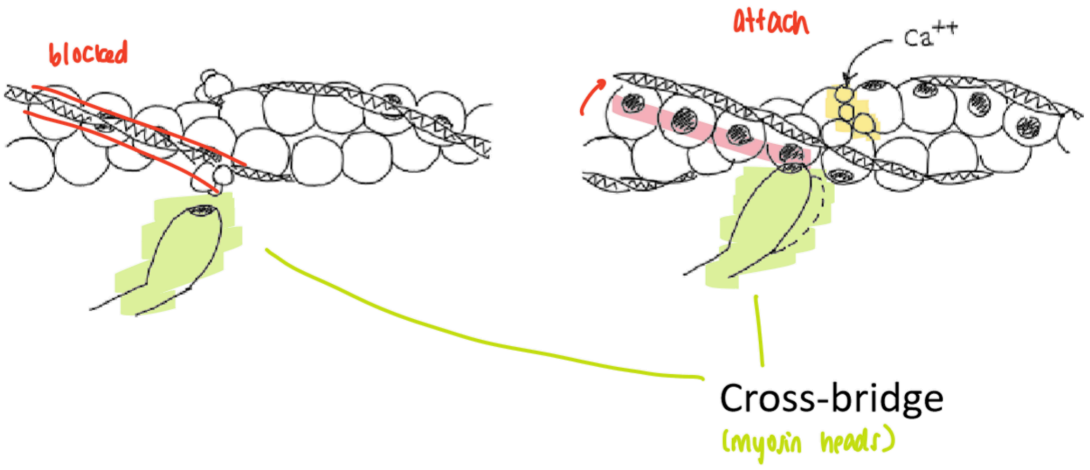
Actin, tropomyosin, troponin

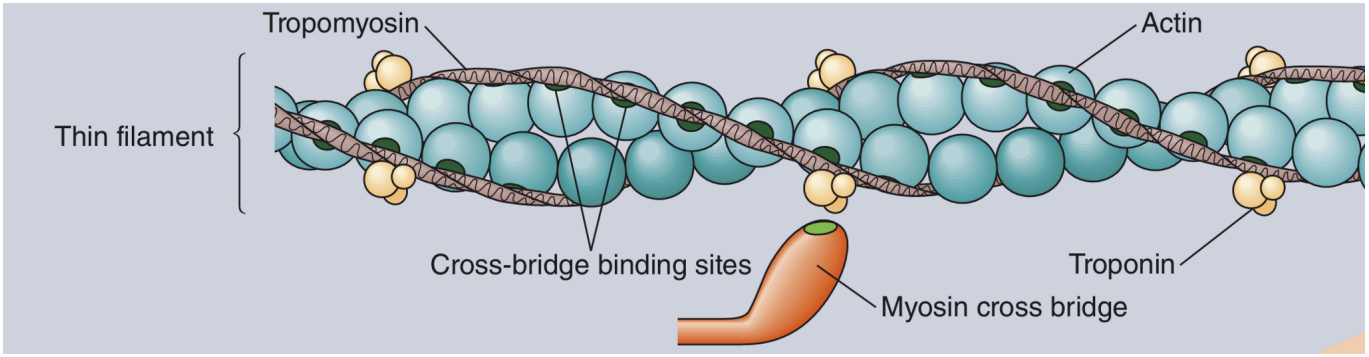
- Makes up thin filament
- **Actin:** primary spherical component
    - Two strands of actin twisted together
    - Contains binding site with myosin cross bridge
- **Regulatory proteins: tropomyosin & troponin**
    - When relaxed: **tropomyosin** lie along actin spiral, cover actin binding site (block contraction)
    - **Troponin:** protein complex with 3 units: bind to myosin, bind to actin, bind to Ca2+

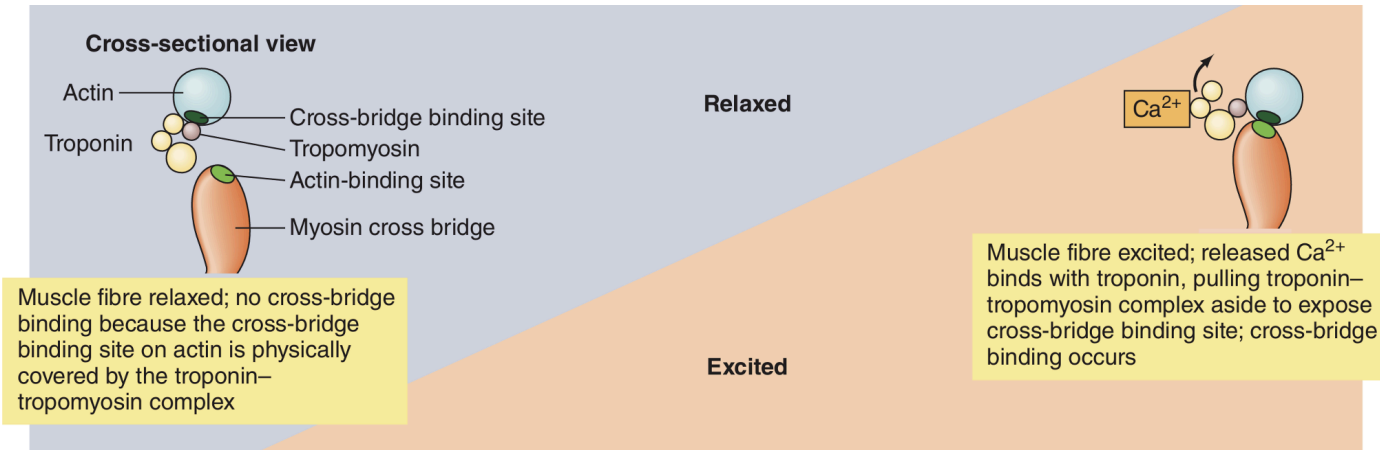
- When not bound to Ca2+, stabilizes tropomyosin in blocking position
- When bound to Ca2+, shape changes so that tropomyosin slips from blocking position

○ When tropomyosin out of way, actin and myosin can bind and interact at cross bridges

Myosin and actin are called **conctractile proteins**.



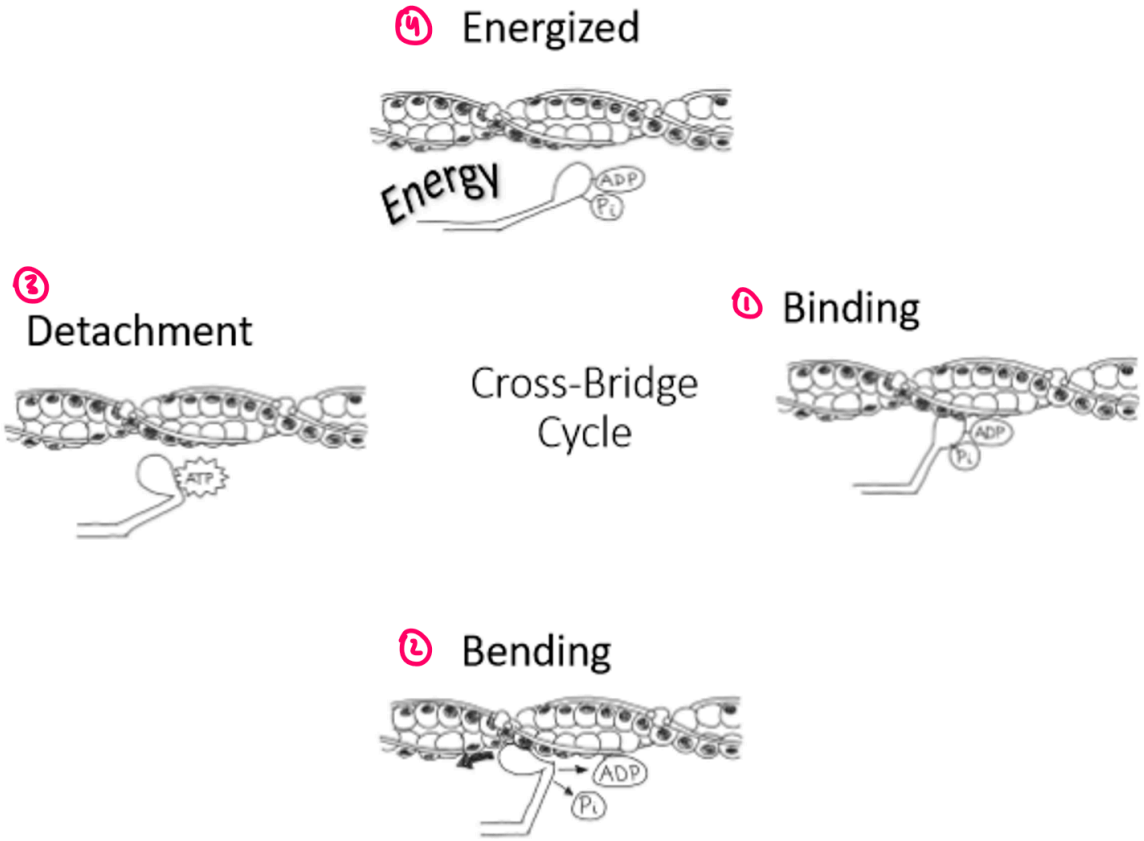




During contraction, the **thick and thin filaments themselves do not change length**. Instead, thin filaments from opposite sides of each sarcomere slides closer together between the thick filaments.

Cross bridge cycle:

1. **Binding:** The myosin cross-bridge (on the thick filament) binds to an actin molecule (on the thin filament). This can only happen if Calcium has already exposed the binding sites.
2. **Bending (Power Stroke):** The cross-bridge bends inward toward the center of the sarcomere. This "stroke" pulls the attached thin filament inward. ADP is released.
3. **Detachment:** At the end of the power stroke, the cross-bridge breaks its link with the actin molecule. New ATP is received.
4. **Energized:** ATP hydrolyzes to ADP + Pi, returns myosin to original shape



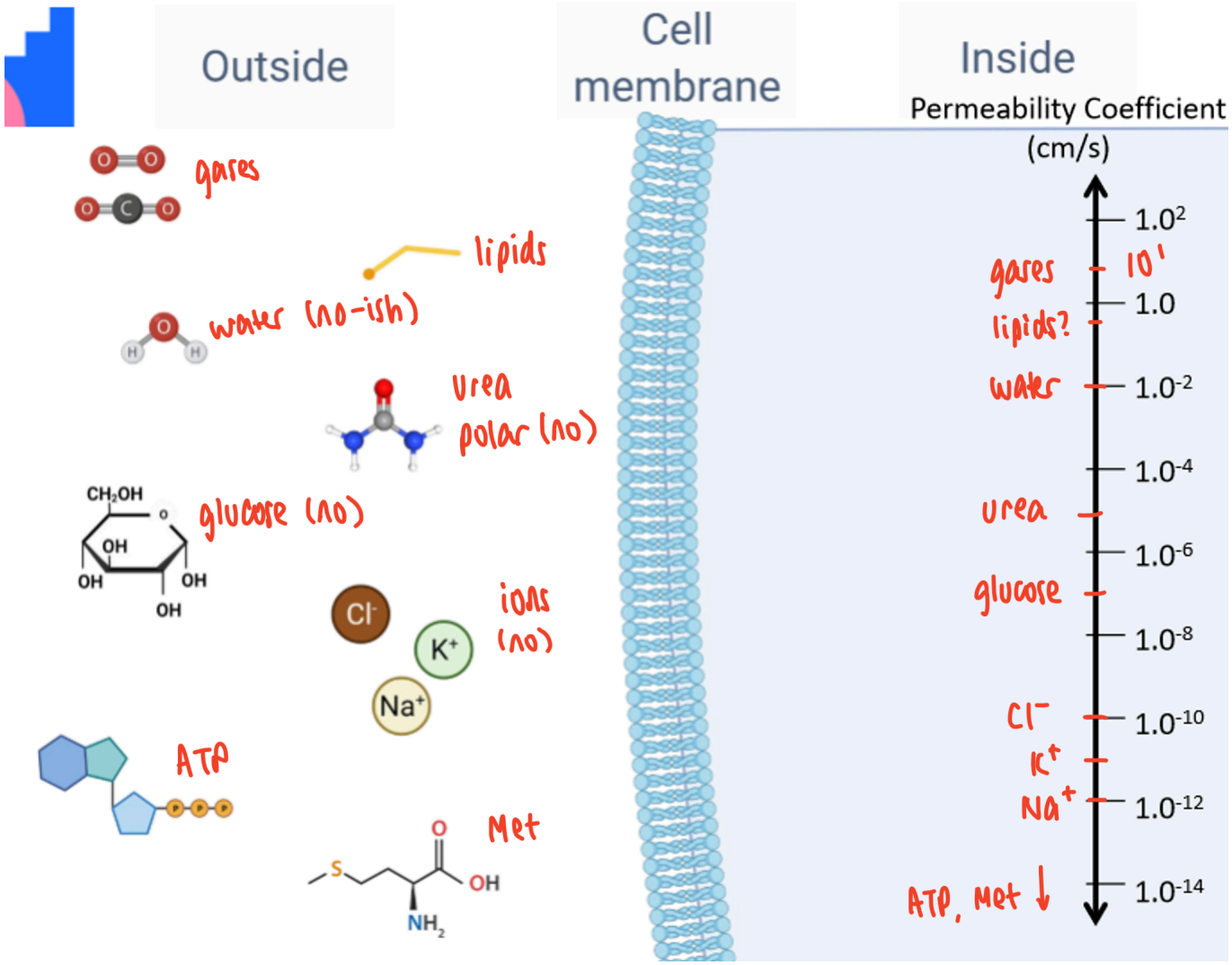
Miscellaneous stuff for midterm

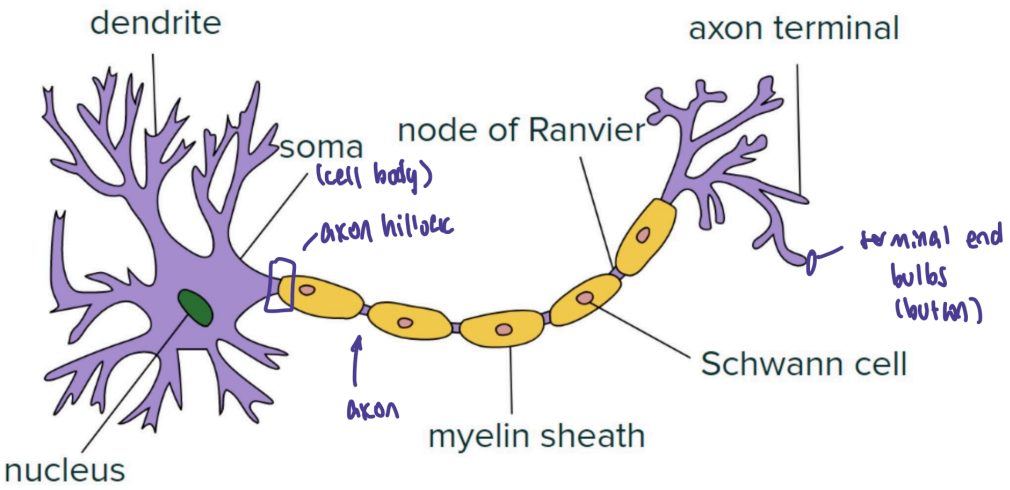
DNA

The DNA strand being copied (complement) is known as the **template** strand (3' to 5').

The other DNA strand (not being copied, same) is the **coding** strand (5' to 3').

Membrane





GABA

- Inhibitory neurotransmitter
- When GABA binds to **ligand-gated receptor**, it opens channel for Cl^-
- Primary function: regulate spiking of neurons by changing **membrane potential**
    - Allow chloride into cell -> inside is more negative (prevents neuron from firing signal)
- KCC2 is a **symporter** and carries **potassium** and **chloride** ions out of cell
    - Symporter = two ions moves same direction

Young neurons: KCC2 missing, **high internal chloride conc.** -> GABA is **excitatory** (chloride flows out of cell) -> neuron spikes when GABA added

Old neurons: KCC2 is present, **low internal chloride conc.** -> GABA is **inhibitory** (chloride flows into cell) -> neuron inhibited when GABA added

Excitatory neurotransmitters depolarize neurons (move more positive)

Inhibitory neurotransmitters hyperpolarize neurons (move more negative)

Somatic and Autonomic

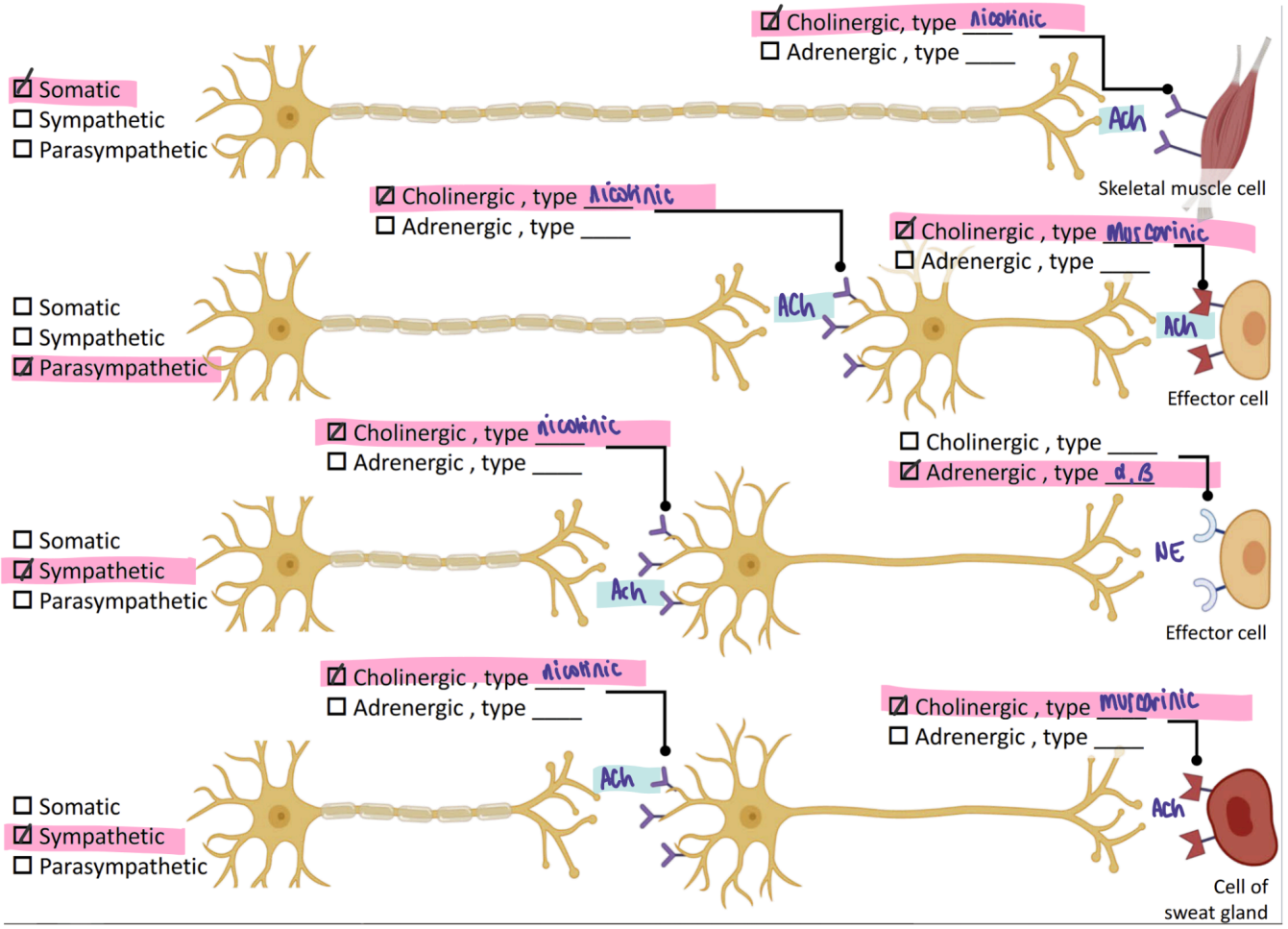


TABLE 5-10 Comparison of the Autonomic Nervous System and the Somatic Nervous System

Feature	Autonomic Nervous System	Somatic Nervous System
Site of Origin	Brain or lateral horn of spinal cord	Ventral horn of spinal cord for most; those supplying muscles in head originate in brain
Number of Neurons from Origin in CNS to Effector Organ	Two-neuron chain (preganglionic and postganglionic)	Single neuron (motor neuron)
Organs Innervated	Cardiac muscle, smooth muscle, exocrine and some endocrine glands	Skeletal muscle
Type of Innervation	Most effector organs dually innervated by the two antagonistic branches of this system (sympathetic and parasympathetic)	Effector organs innervated only by motor neurons
Neurotransmitter at Effector Organs	May be acetylcholine (parasympathetic terminals) or norepinephrine (sympathetic terminals)	Only acetylcholine
Effects on Effector Organs	Either stimulation or inhibition (antagonistic actions of two branches)	Stimulation only (inhibition possible only centrally through IPSPs on dendrites and cell body of motor neuron)
Types of Control	Under involuntary control	Subject to voluntary control; much activity subconsciously coordinated
Higher Centres Involved in Control	Spinal cord, medulla, hypothalamus, prefrontal association cortex	Spinal cord, motor cortex, basal nuclei, cerebellum, brain stem

Property	Somatic nervous system	Autonomic nervous system
Regulation	Voluntary/conscious	Involuntary/unconscious
Origin	Motor cortex, medulla, or dorsal horns of the spinal cord	Brain stem nuclei or varying regions of the spinal cord
Myelination	Yes	Preganglionic myelinated; postganglionic are not. (need to state postganglionic are not for full mark)
Neurotransmitter	ACh	ACh, NorEpi, Epi [need to state all three; otherwise half mark]
Target tissue	Skeletal	Any one or all of the following: Smooth and/or cardiac muscle; kidneys; lungs; blood vessels; GI tract; liver; bladder; eyes; salivary and sweat glands
Receptor subtype	Nicotinic	Nicotinic, muscarinic, alpha and beta adrenergic (or just adrenergic is acceptable) [Need to state all types of receptors for full mark; otherwise half mark]

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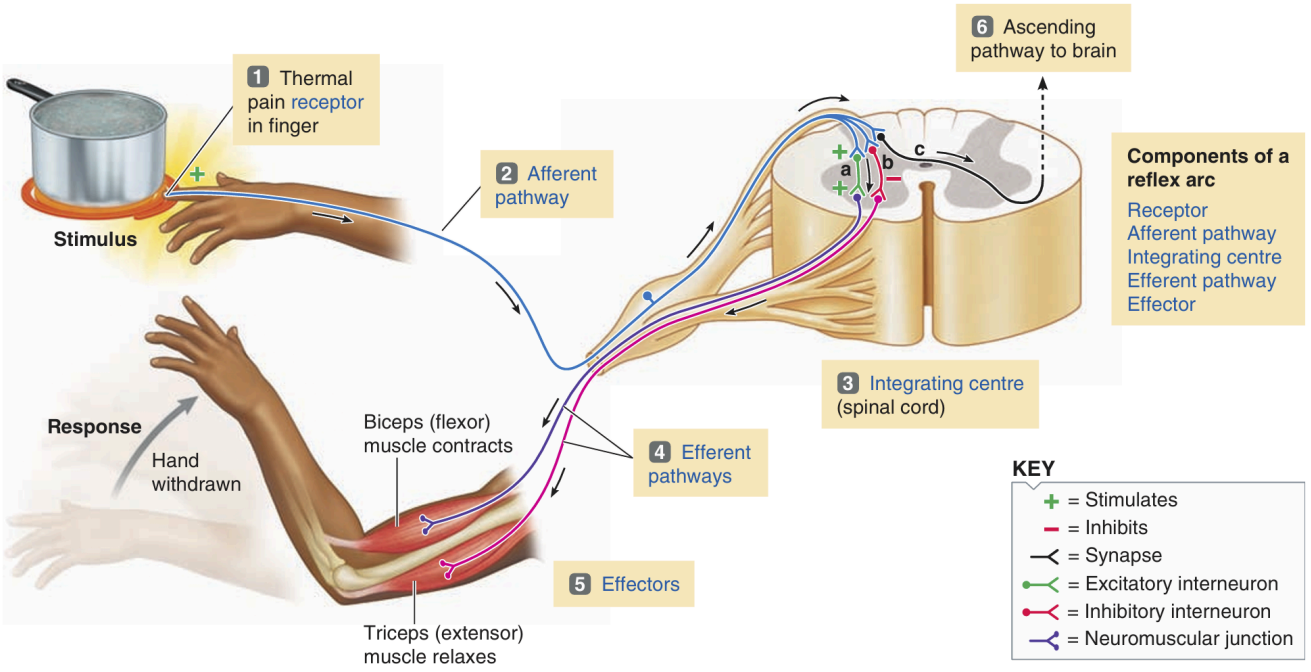
Reflex



1 Painful heat stimulus activates thermal pain receptor in finger.

2 Action potentials are generated in afferent pathway, which propagates impulses to the spinal cord.

3 Spinal cord serves as the integrating centre. Here afferent neuron stimulates;
3a excitatory interneurons, which stimulate motor neurons to biceps;
3b inhibitory interneurons, which inhibit motor neurons to triceps;
3c interneurons that are part of ascending pathway to brain.

4 One efferent pathway stimulates biceps to contract. Other efferent pathway leads to relaxation of triceps by preventing counterproductive excitation and contraction of this antagonistic muscle.

5 The biceps and triceps are effectors. Resultant flexion of elbow joint pulls hand away from painful stimulus. This response completes the withdrawal reflex.

6 Events that occur in brain on arrival of signal via ascending pathway, such as awareness of pain, memory storage, and so on, are above and beyond reflex arc.

Synapses

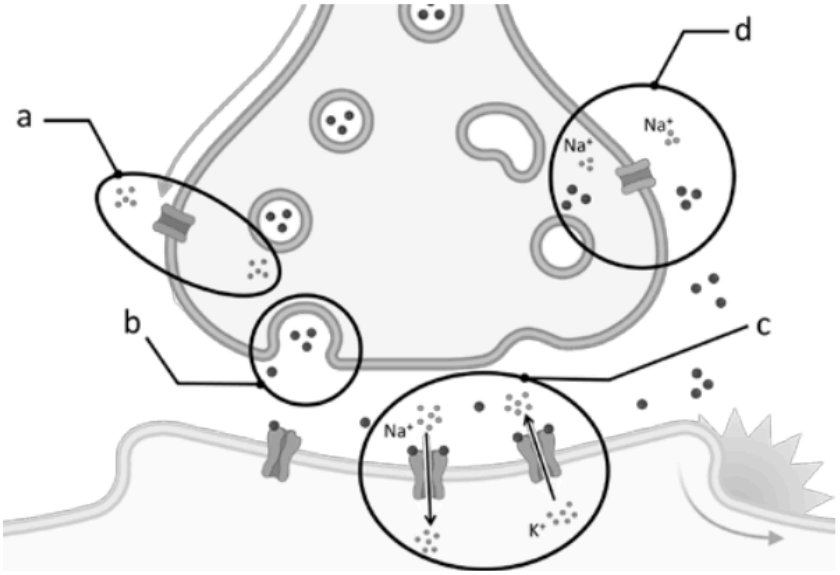


Figure 2

(a) Facilitated diffusion, channel mediated; voltage-gated Ca2+ channel

(b) Exocytosis

(c) **Non-specific cation channel**, ligand-gated channel; facilitated diffusion, channel mediated
Receptor: **ionotropic receptor**, cholinergric receptor
Ligand: **neurotransmitters**

(d) Secondary active transport, carrier mediated

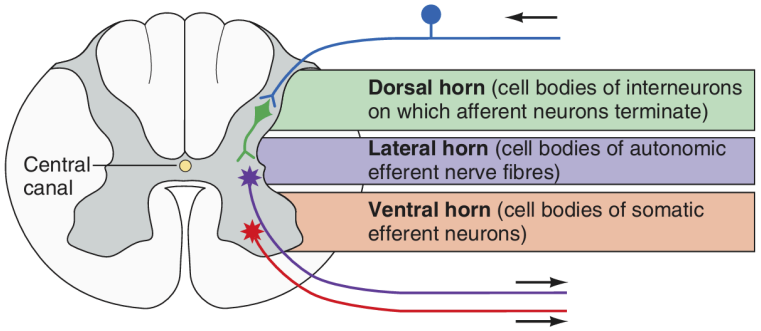
(e) (1) Diffusion
(2) Enzymatic inactivation (destruction by enzymes)
(3) Reuptake by glial cells

Grey matter and White Matter

Grey Matter (The "Butterfly")

Contains cell bodies, dendrites, and glial cells. It is divided into "horns":

- **Dorsal (posterior) Horn:** Receives incoming **afferent** (sensory) signals.
- **Lateral Horn:** Contains cell bodies for autonomic (involuntary) efferent fibers.
- **Ventral (anterior) Horn:** Contains cell bodies for **efferent** (voluntary) motor neurons.

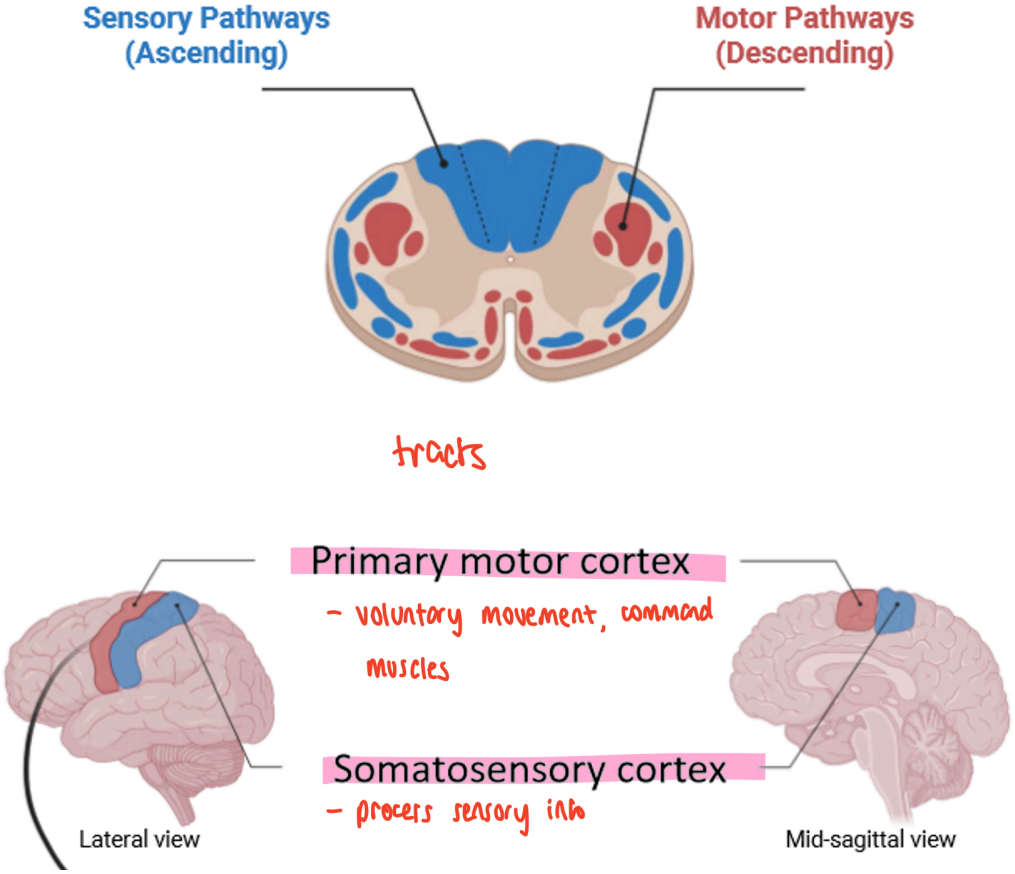


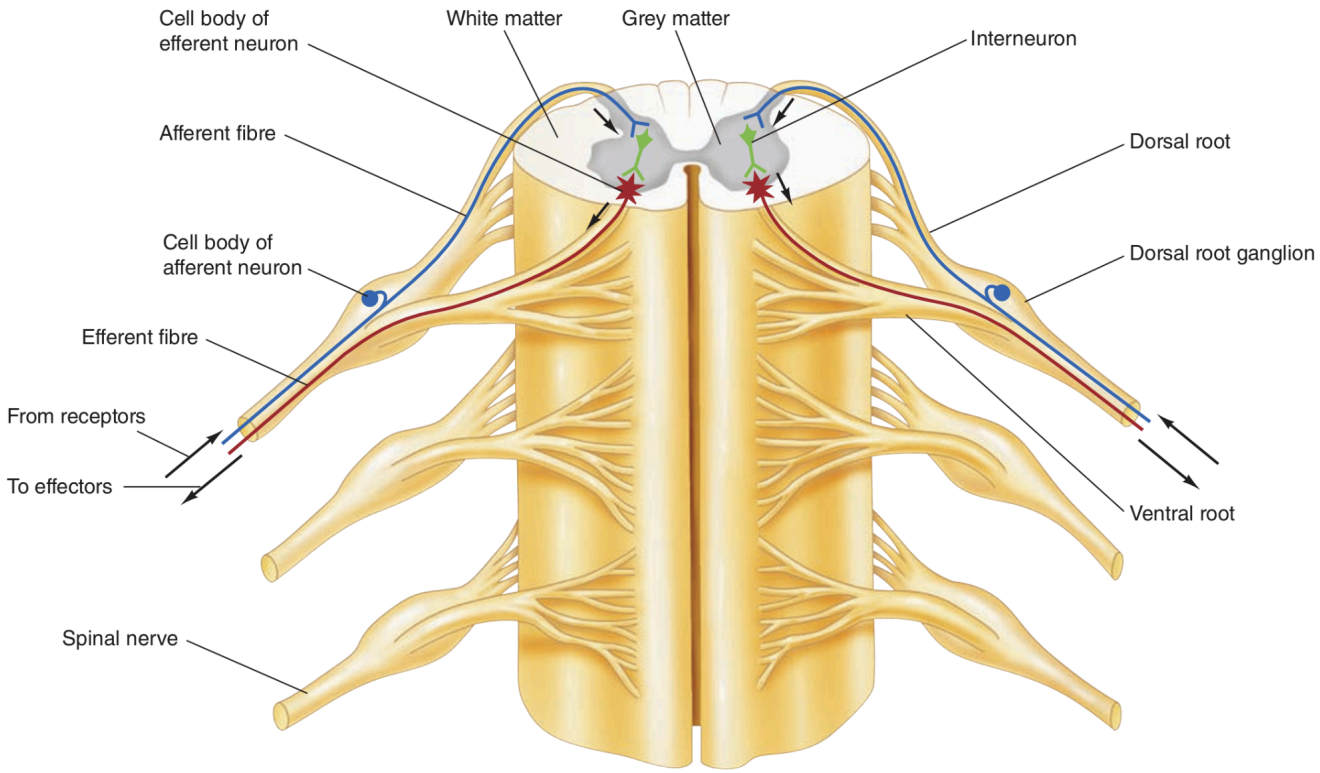
› FIGURE 4-12 Regions of the grey matter

White Matter (The "Tracts")

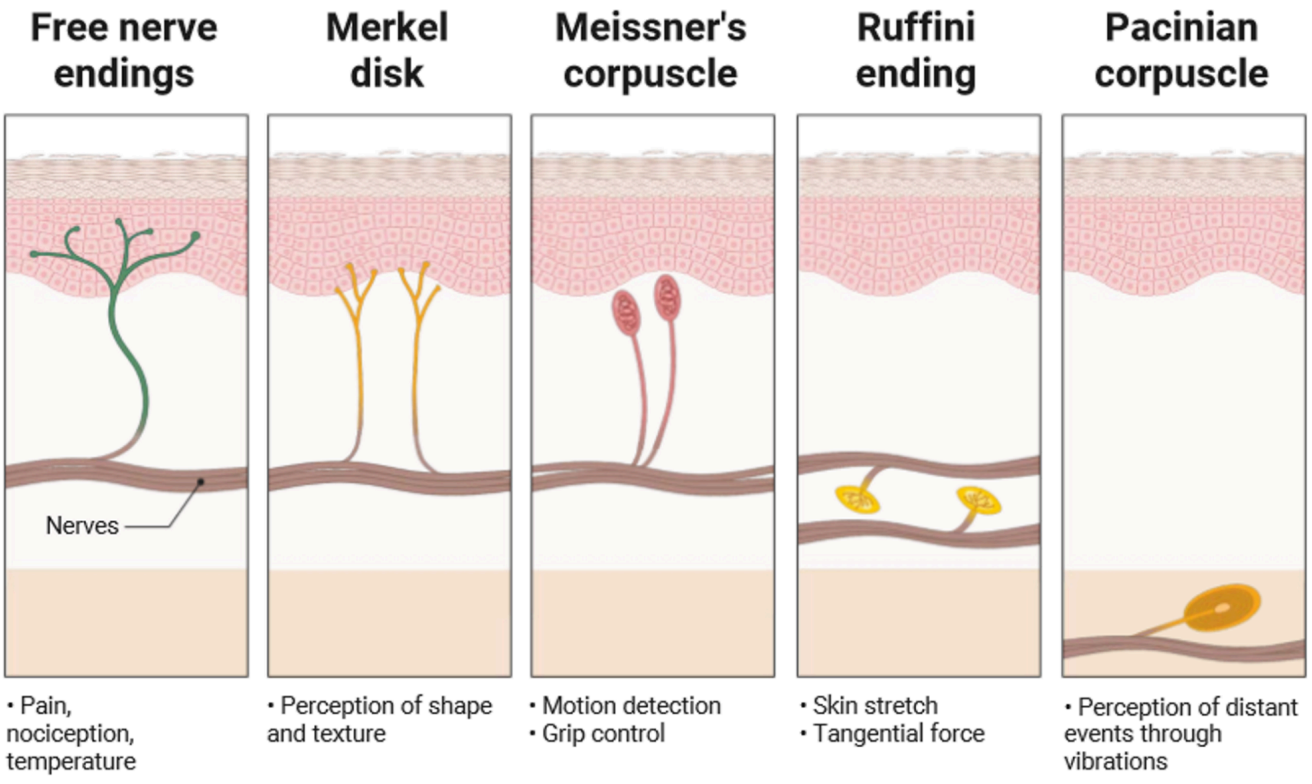
Contains bundles of axons (tracts) that carry signals up and down.

- **Ascending Tracts:** Carry sensory info **to the brain** (e.g., *spinocerebellar tract*).
- **Descending Tracts:** Carry motor commands **from the brain** (e.g., *corticospinal tract*).

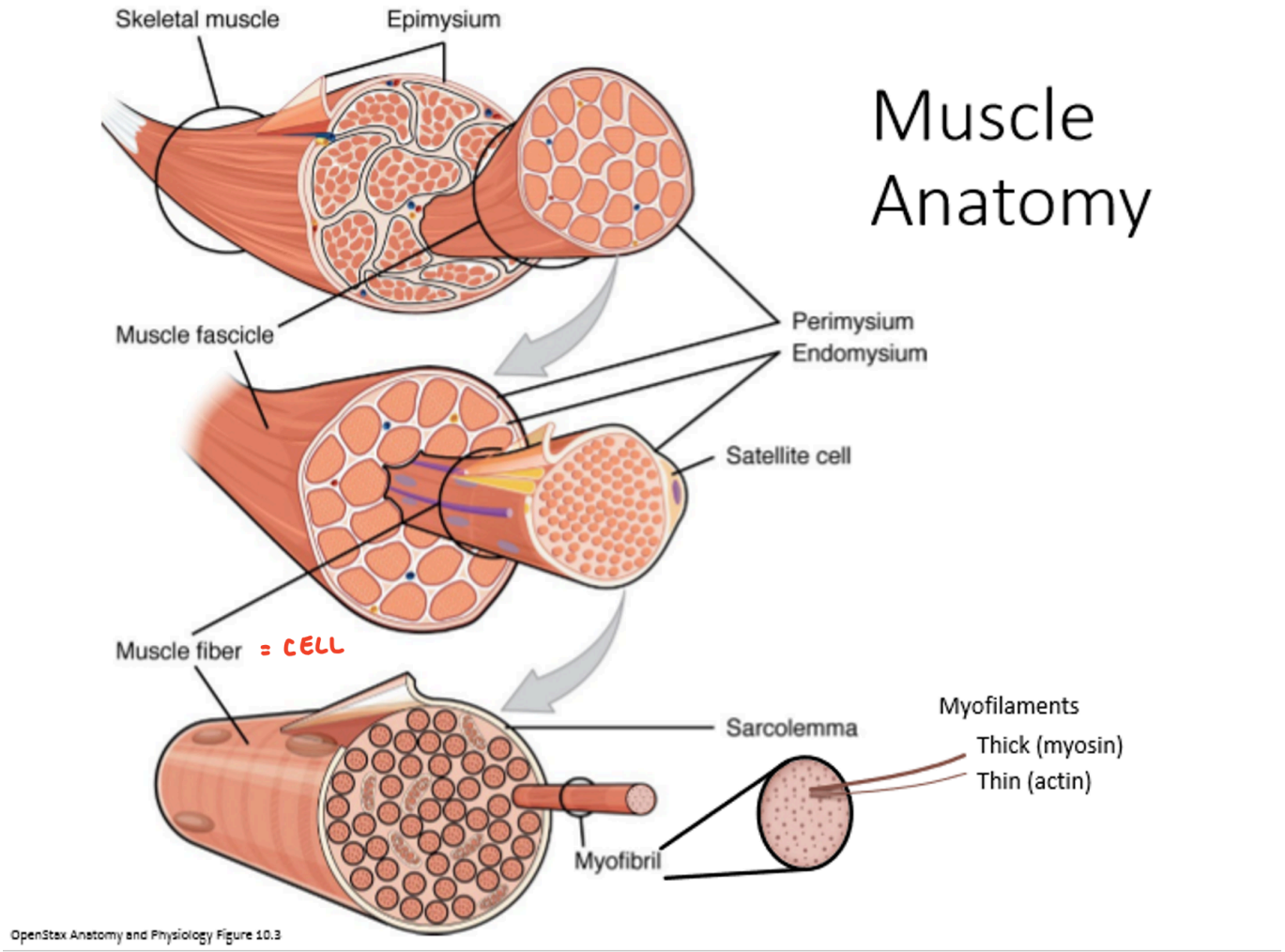




Location and Structure of Somatosensory Receptors



L12: Skeletal Muscle Structure



Myofilaments inside a Myofibril